A Broker-Realistic Backtesting Framework for Systematic Long/Short Equity Strategies

A Rigorous Approach

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Overview

This document describes the design and implementation of a broker-realistic backtesting framework for systematic long/short equity strategies, tested on the historical members of the S&P 500, where a *systematic* strategy refers to trading decisions made according to predefined rules. The framework is built around the following design goals:

- **A realistic brokerage ledger:** We track our brokerage account's (book) cash, margin debit, the long and short market values of our positions, the per-ticker (FIFO) short-proceeds lots, and posted collateral (if any). The ledger models overnight financing (debit interest, cash interest, stock-borrow fees, and short rebate), dividends on both long and short positions, trading costs (fees, half-spread, and price slippage), and Reg-T / FINRA margin-requirement mechanics. On top of this, the universe of investable stocks at any given date is the historical (point-in-time) membership of the S&P 500.
- **A pluggable, extensible interface for systematic strategies:** We include three example strategies with the framework: top-k/bottom-k selection from a momentum signal, top-k/bottom-k selection from a cross-sectional factor-model signal, and a constrained mean-variance portfolio optimisation. Given the framework's modular implementation, adding a new strategy is straightforward.

Each of the three example strategies currently implemented works by determining, at any chosen rebalance date, the optimal portfolio composition to hold over the next (configurable) period. To run such strategies (backtesting), we implemented:

- **Flexible rebalancing and termination:** The portfolio rebalances either on a scheduled calendar or dynamically (rebalancing early when the inter-rebalance return exceeds a threshold). The backtest terminates for one of three reasons: reaching the end of the scheduled backtest calendar, hitting a cumulative return target, or triggering a stop-loss.

Finally, to gauge and compare strategy performance, we implemented:

- **Diagnostics and KPIs:** Every backtest produces:
  - customizable plots of equity, LMV, SMV, cash, debit, total short proceeds, leverage, drawdown, and the margin-requirement series with its violation events;
  - a table of daily values of the book (the brokerage-account ledger), recorded at the open and the close;
  - an audit log of trades, position changes, and changes in the short-proceeds lots;
  - a structured KPI report covering returns, drawdowns, trading turnover, commissions and the execution cost paid on the gross notional traded, the largest single-trade share of daily market volume across all stocks during the backtest, margin-requirement violations and cures, the impact of financing and dividends on our P&L, a decomposition of the daily equity P&L into its price, dividend, financing, commission and execution-cost streams, and the ex-post market exposure of the realised equity curve.
- **Cross-strategy performance comparison metrics:** The framework compares strategies and configurations, reporting:
  - target-hit statistics (frequency of return-target hits, and average time-to-target conditional on a hit) for strategies run with matched configurations (same start date, same return target, dynamic rebalancing enabled or disabled), holding all other settings equal;

- Sharpe ratios (daily and annualized) for each strategy at specific historical periods, with the dynamic rebalancing rule enabled versus disabled (to gauge its effect on performance).
- ex-post market-exposure analytics for each realised equity curve: the beta and the alpha of its daily excess returns against the market's, estimated with heteroskedasticity- and autocorrelation-consistent standard errors, so that returns driven by exposure to the market can be distinguished from those that are not.

The codebase is organized as follows:

- **Core implementation**
  - 22 Python modules located in `modules/`, containing the backend implementation of the backtesting framework.
  - `download_data.py`, a standalone utility script used to pre-download the datasets required for backtesting. It can fetch data up to the present day.
- **Main notebooks**
  - `single_backtest_runner.ipynb`, used to run a complete end-to-end backtest with configurable parameters.
  - `strategy_comparison.ipynb`, used to compare the performance of different strategies across multiple parameter settings and backtest runs.
- **Auxiliary notebooks**
  - `mvo_solver.ipynb`, used to test and validate the portfolio optimizer used by the MVO strategy.
  - `factors_model.ipynb`, used to test and validate the factors model implementation used by the corresponding strategy.
  - `backtest_calendar.ipynb`, used to get a visual representation of the backtest calendar construction process.
  - `sp500_historical_members.ipynb`, used to test the retrieval of historical S&P 500 constituents from Wikipedia revision histories.

1 Introduction

1.1 What this framework does

The framework simulates the daily evolution of a brokerage account (or "book") that supports trading of a long/short equity portfolio (one that holds long positions in stocks expected to rise and short positions in stocks expected to fall, aiming to profit from moves in either direction). At each rebalance date, a strategy decides a target portfolio. Between rebalance dates, the book is marked to market, accrues financing costs and dividends, and is checked against margin requirements. Trades are simulated both to rebalance the portfolio and, when a daily maintenance-margin violation occurs, as part of one of the two supported procedures for curing it.

1.2 Scope and out-of-scope content

In scope:

- Daily-frequency backtesting of long/short equity strategies on the historical S&P 500 constituent stocks.
- Attempt to realistically simulate the P&L of our brokerage book, including financing, dividends, trading costs, and margin mechanics.
- A pluggable strategy interface (currently three concrete strategies are implemented).
- Two margin-cure variants applied globally to both rebalance infeasibility and maintenance-margin violations: Variant A posts cash collateral; Variant B uniformly deleverages positions.
- Three backtest termination causes: reaching the end of the schedule, hitting the return target, or triggering a stop-loss liquidation.
- Bad-data filtering and point-in-time construction of the investable universe for each date at which we rebalance our portfolio.
- Diagnostic plots, KPIs, audit logs, and comparison metrics.

Out of scope:

- Intraday execution: The framework does not model intraday execution. Trades are simulated at close prices, with a somewhat simple price-slippage modelling. Mark-to-market is performed twice daily: at the open (using open prices) and at the close (using close prices). Margin-requirement checks are performed once daily, using close prices, before any trades are possibly executed that day.
- Live trading: The framework does not connect to a broker.
- Tax accounting: The framework reports pre-tax performance.

1.3 Overview of the content

Sections §2–§5 describe the framework components common to all strategies: the brokerage book (ledger), margin-requirement mechanics, the backtest calendar, the backtest termination rules, and the optional dynamic portfolio-rebalancing rule applicable to any of the strategies. Section §6 explains how the investable universe of tickers (equities) is constructed at each rebalance date.

The framework currently includes three systematic strategies, which are described in sections §7 (momentum), §8 (cross-sectional factors model), and §9 (mean–variance portfolio optimisation). Section §10 discusses the caveats that apply to all three strategies. Section §11 justifies the choice of the risk-free rate we use to compute excess returns. Sections §12 and §13 cover performance diagnostics: §12 defines the key performance indicators (KPIs), and §13 walks through the complete output of a single backtest example. Section §14 presents the strategy-comparison methodology and results, in three parts: §14.1 compares risk-adjusted performance through Sharpe ratios computed over fixed historical windows, §14.2 compares strategies deployed at specific dates against specific return targets, and §14.3 measures, ex post from the realised equity curve, how much of each strategy's performance is attributable to exposure to the market rather than to the strategy itself. Building on these results, section §15 discusses a way to assess whether a strategy is suitable for deployment at the present day, taking into account both the comparison results and the caveats outlined in section §10. Section §16 lists known limitations of the current framework and discusses natural directions for future extension. Finally, the Appendix explains the bad-ticker pre-filter we apply to our price-data feed, introduced in §6.5. It presents two examples of data corruption in free price feeds (such as those obtained through the `yfinance` Python module) and explains the two filter rules used to mitigate these issues.

2 Brokerage Book Mechanics

2.1 The book as a state object

We model the brokerage account as a **book** (a ledger): a state object that holds cash, debit, long and short market values, posted collateral, per-ticker FIFO short-proceeds lots, and a vector of integer share positions. Two snapshots of the book are kept for each backtest date, corresponding to the market open and close:

- **open**: the state after marking to market the positions at the open (using the open prices of the stocks), after any overnight financing accruals and dividend bookings.
- **close**: the state after marking to market the positions at the close (using the close prices of the stocks), after any rebalance trades and/or any margin cures executed since the open.

2.2 Equity: definition and decomposition

Let Ω_t be the set of stock names (tickers) available for investing (the "universe of tickers") at time t, and let $q_{t,i} \in \mathbb{Z}$ denote the number of shares we hold in stock i at time t.

Value	Meaning
$q_{t,i} \geq 1$	Long position of $q_{t,i}$ shares in ticker i
$q_{t,i} \leq -1$	Short position of $\lvert q_{t,i}\rvert$ shares in ticker i
$q_{t,i} = 0$	No open position (flat) in ticker i

The **equity** of our brokerage account is its net asset value (residual value after deducting all liabilities):

$$\boxed{\text{Equity}_t := \text{LMV}_t(q_t) - \text{SMV}_t(q_t) + \text{ShortProceeds}_t + \text{Cash}_t - \text{Debit}_t + \text{MarginCollateral}_t} \quad (1)$$

The components are defined as follows:

- LMV_t (*Long Market Value*): aggregate mark-to-market value of all long positions:

$$\text{LMV}_t(q_t) := \sum_{i \in \Omega_t} p_{t,i}\, q_{t,i}^{+}$$

Notation

$x^+ := \max(x, 0)$, $x^- := \max(-x, 0)$. In particular, we have $x = x^+ - x^-$ and $|x| = x^+ + x^-$.

- SMV_t (*Short Market Value*): aggregate mark-to-market value of all short positions (non-negative):

$$\text{SMV}_t(q_t) := \sum_{i \in \Omega_t} p_{t,i}\, q_{t,i}^{-}$$

- Cash_t: free (unrestricted) cash available for purchasing shares; excludes short-sale proceeds. By convention $\text{Cash}_t \geq 0$; any cash deficit (margin loan) is reclassified as positive Debit_t (see §2.3).

- Debit_t: outstanding margin loan balance (non-negative liability); the broker charges interest at the margin debit rate.

- ShortProceeds_t: restricted cash credited by the broker from prior short sales that remain open at time t. These proceeds are held as collateral against open short positions. Under standard retail brokerage practice, they are not freely withdrawable and cannot be used to finance long positions. Although restricted, they remain assets of the account and enter the equity formula with a positive sign:

$$\text{ShortProceeds}_t := \sum_{i \in \Omega_t} \text{ShortProceeds}_{t,i}, \qquad \text{ShortProceeds}_{t,i} \geq 0$$

where $\text{ShortProceeds}_{t,i}$ is the aggregate restricted cash balance associated with open short-sale lots for ticker i.

- **Short-proceeds lot accounting:** Each short sale creates a lot entry in a per-ticker FIFO queue, recording the number of shares sold short and the execution price.

Opening / increasing a short position

A short sale of n shares each sold at execution price p^{exec} appends a new lot (n, p^{exec}) to the FIFO queue. Aggregate short proceeds for the ticker increase by $n \times p^{\text{exec}}$, and the net share position decreases by n (becomes more negative).
Note: in practice, individual shares may execute at different prices as a market order consumes order-book depth, thereby appending a new lot $(1, p^{\text{exec}})$ for each share sold at the unique execution price p^{exec}. Throughout this project we adopt the simplifying assumption that all shares of a given ticker traded on a given day execute at the same price (see §2.5.2).

Reducing / closing a short position

When buying to cover, restricted short-sale proceeds are released in FIFO order. For each lot partially or fully consumed, realised P&L is:

$$\text{Realised P\&L} = \left(p^{\text{short}} - p^{\text{cover}}\right) \times n^{\text{closed}}$$

where p^{short} is the lot's original short-sale price, p^{cover} is the cover price, and n^{closed} is the number of shares closed from that lot. Fully consumed lots are removed from the queue; partially consumed lots remain with their residual share count.

Remark

FIFO short-lot tracking provides a realistic method for computing realised P&L when short positions are partially covered.

- $\text{MarginCollateral}_t$: external capital posted to comply with broker margin requirements (see §3). It increases Equity_t one-for-one but represents no P&L generated by the strategy. For performance reporting it is stripped out:

$$\widetilde{\text{Equity}}_t = \text{Equity}_t - \text{MarginCollateral}_t$$

KPIs (return, drawdown) and the stop-loss trigger (see §5.2) and Sharpe ratios (see §14.1) all use $\widetilde{\text{Equity}}_t$; margin requirement checks use Equity_t (see §12).

2.3 Cash-to-debit reclassification and cash sweep

The following two rules are applied together, as a single subroutine, immediately after *every* ledger update that moves cash: namely after accounting for financing costs (§2.6.1), after accounting for dividends (§2.6.2), or after simulating trades (§2.8). They therefore may fire multiple times within a single backtest date.

First, if any ledger update would cause the free cash balance to become negative (such as increasing long exposure beyond available cash by drawing a margin loan (exemplified in §3.3), paying dividends owed on short positions, or accruing financing costs against a zero cash balance), the shortfall is reclassified immediately as borrowing:

$$\text{Debit}_t \leftarrow \text{Debit}_t + |\text{Cash}_t|, \qquad \text{Cash}_t \leftarrow 0.$$

Second, if $\mathrm{Cash}_t > 0$ and $\mathrm{Debit}_t > 0$ simultaneously after the update, we sweep cash against the debit balance to minimise interest expense on the margin loan, which exceeds the interest received on cash held in our broker account (in fact, that spread is a major source of broker revenue; see example values at §2.6.1):

$$x := \min(\mathrm{Cash}_t, \mathrm{Debit}_t), \qquad \mathrm{Cash}_t \leftarrow \mathrm{Cash}_t - x, \qquad \mathrm{Debit}_t \leftarrow \mathrm{Debit}_t - x.$$

2.4 Leverage ratios

The long, short, and gross leverage ratios are defined as:

$$l_t^L = \frac{\mathrm{LMV}_t}{\mathrm{Equity}_t}, \qquad l_t^S = \frac{\mathrm{SMV}_t}{\mathrm{Equity}_t}, \qquad l_t^G = l_t^L + l_t^S = \frac{G_t}{\mathrm{Equity}_t}$$

where $G_t = \mathrm{LMV}_t + \mathrm{SMV}_t$ is the gross exposure.

We can interpret these variables as follows. Given the definition of equity (§2.2):

- a change by r in the total long market value of our book from dates t_1 to t_2 causes an equity change from dates t_1 to t_2 equal to $r\, l_{t_1}^L$:

$$\mathrm{LMV}_{t_2} = (1+r)\,\mathrm{LMV}_{t_1} \implies \mathrm{Equity}_{t_2} = (1 + r\, l_{t_1}^L)\,\mathrm{Equity}_{t_1}$$

- a change by r in the total short market value of our book from dates t_1 to t_2 causes an equity change from dates t_1 to t_2 equal to $-r\, l_{t_1}^S$:

$$\mathrm{SMV}_{t_2} = (1+r)\,\mathrm{SMV}_{t_1} \implies \mathrm{Equity}_{t_2} = (1 - r\, l_{t_1}^S)\,\mathrm{Equity}_{t_1}$$

Therefore, leverage is a factor amplification of losses or gains, and must be handled carefully.

Remark

These leverage ratios are computed using the value of equity including any margin collateral we may post to comply with MR requirements. If we indeed do post margin collateral, then the leverage with respect to the strategy-generated equity (i.e., excluding the value of margin collateral), would be greater.

2.5 Valuation and trade execution prices

2.5.1 Marking to market

Marking to market ("MtM") at time t re-values existing positions using prices observed at time t, without changing the share holdings:

$$\mathrm{LMV}_t = \sum_i p_{i,t}\, q_{i,t}^+, \qquad \mathrm{SMV}_t = \sum_i p_{i,t}\, q_{i,t}^-$$

Cash, debit, posted margin collateral, shares, and short-proceeds lots are unchanged by MtM. Equity is recomputed via the formula in §2.2, and the leverage ratios via the formulae in §2.4.

Daily mark-to-market: At any given backtest date, positions are marked to market at the start of day using opening prices and at the end of day using close prices, both sourced from the `yfinance`

Python module [3] , which retrieves daily OHLC data from Yahoo Finance's publicly available API. For U.S.-listed stocks, the underlying prices reflect activity on the ticker's primary listing exchange: NYSE for NYSE-listed stocks (e.g., JPMorgan, Coca-Cola, ExxonMobil) and Nasdaq for Nasdaq-listed stocks (e.g., Apple, Microsoft, Tesla, Nvidia). S&P 500 constituents are split between NYSE and Nasdaq listings, with NYSE holding the larger share.

Adjustment convention: These open and close prices are adjusted for stock splits but not for distributions, so a dividend appears in them as a genuine price decline on the ex-date. That is the behaviour the ledger needs: positions are marked at the observed price, and the dividend is credited to the long leg or owed by the short leg separately on the same date (§2.6.2), so that the two together reproduce the total return without double-counting it. The distribution-adjusted `adj_close` series is consequently never used to mark the book; its only role in the framework is as the input to the extreme-return filter of A1.4.

Caveat

`yfinance` is intended for research and educational use and does not guarantee that its open/close values correspond to the official opening/closing auction prints from the primary exchange.

Mark-to-market after rebalancing or deleveraging trades: We simulate the trades of stocks (tickers) as market orders (thus executing instantly), filling at the execution price modelled in §2.5.2, derived from the day's close price for each ticker. **Trades therefore do not execute at the close**: every trade pays the half-spread and slippage embedded in that execution price, while the resulting position is marked at the close. That difference between the execution price and the close is a genuine reduction in equity, it is paid on every trade, and it is booked in no cash flow of its own; §2.6.4 defines it and §2.6.6 carries it as a named term of the daily P&L identity.

Deleveraging trades, triggered by a margin requirement violation identified from the portfolio marked-to-market at the day's close prices (Variant B in §3.5.2), are also simulated to execute at those same execution prices for each date. After the trades are applied, the updated portfolio is marked-to-market using the close prices, and the resulting portfolio value is recorded.

See §2.7 for the complete sequence of operations performed at each backtest date.

2.5.2 Execution price model

When executing a market order, the execution price reflects the friction of crossing the bid–ask half-spread (starting from the mid-price) and consuming liquidity from the order book. As market orders become larger relative to available liquidity, filling them walks further up (or down) the order book (temporary impact), consuming worse prices, and a market order of a certain size may also leave a lasting footprint: it shifts price persistently (permanent impact).

Consider a metaorder to trade $\Delta q_{t,i}$ shares, where $\Delta q_{t,i} = q_{t,i}^{\text{post}} - q_{t,i}^{\text{pre}}$ is the change in our share holding. A buy increases our share holding, yielding $\Delta q_{t,i} > 0$, and a sell decreases it ($\Delta q_{t,i} < 0$).

Define the VWAP ("Volume-Weighted Average Price") as:

$$p_{t,i}^{\text{VWAP}} := \frac{\sum_{k=1}^{m} \Delta q_{t,i}^{(k)} \cdot p_{t,i}^{(k)}}{\sum_{k=1}^{m} \Delta q_{t,i}^{(k)}}$$

where $\left(p_{t,i}^{(k)}, \Delta q_{t,i}^{(k)}\right)_{k=1,\ldots,m}$ are the order book price levels touched and the shares filled at each level, with $\sum_{k=1}^{m} \Delta q_{t,i}^{(k)} = \Delta q_{t,i}$. (Note that $p_{t,i}^{\text{VWAP}} > 0$).

The total cost incurred from the filling of the $\Delta q_{t,i}$ shares at the actual realized prices compared to filling them all at the pre-trade mid-price is equal to

$$\begin{aligned}
\sum_{k=1}^{m} (p_{t,i}^{(k)} - p_{t,i}^{\text{mid}})\, \Delta q_{t,i}^{(k)} &= \left(\sum_{k=1}^{m} p_{t,i}^{(k)}\, \Delta q_{t,i}^{(k)}\right) - p_{t,i}^{\text{mid}} \sum_{k=1}^{m} \Delta q_{t,i}^{(k)} \\
&= p_{t,i}^{\text{VWAP}} \sum_{k=1}^{m} \Delta q_{t,i}^{(k)} - p_{t,i}^{\text{mid}} \sum_{k=1}^{m} \Delta q_{t,i}^{(k)} \\
&= (p_{t,i}^{\text{VWAP}} - p_{t,i}^{\text{mid}})\, \Delta q_{t,i}.
\end{aligned}$$

In a static order-book snapshot (no price drift during execution), this total cost is a non-negative quantity representing a loss. For a buy ($\Delta q_{t,i} > 0$), every price level touched $p_{t,i}^{(k)}$ is greater than or equal to the best ask, hence strictly greater than $p_{t,i}^{\text{mid}}$, so $p_{t,i}^{\text{VWAP}} > p_{t,i}^{\text{mid}}$ (as a volume-weighted average); therefore $(p_{t,i}^{\text{VWAP}} - p_{t,i}^{\text{mid}})\, \Delta q_{t,i} > 0$. Similarly, for a sell, $\Delta q_{t,i} < 0$ and $p_{t,i}^{\text{VWAP}} < p_{t,i}^{\text{mid}}$, so again $(p_{t,i}^{\text{VWAP}} - p_{t,i}^{\text{mid}})\, \Delta q_{t,i} > 0$. (The inequality is strict whenever at least one level is strictly worse than the mid.)

This total cost is driven by two components:

- **Half-spread:** the cost of crossing the bid–ask spread, i.e. the mid-to-inside step: a buy pays the best ask rather than the mid; a sell receives the best bid rather than the mid. This component is always non-negative, representative of a cost.

- **Slippage:** the inside-to-VWAP step, i.e. the difference between the pre-trade best bid or ask and the realised VWAP, capturing both the depth consumed by our own order and any price drift during execution (if execution is not instantaneous):

$$\text{slippage} = \begin{cases} p_{t,i}^{\text{VWAP}} - \text{best ask}_{\text{pre}}, & \text{if we buy,} \\ \text{best bid}_{\text{pre}} - p_{t,i}^{\text{VWAP}}, & \text{if we sell.} \end{cases}$$

In a static book-walk, slippage is non-negative (the book is consumed at prices no better than the best bid/best ask). Once we allow price drift during a non-instantaneous execution, however, slippage becomes a *signed* quantity: the market may move favourably (negative slippage, a gain) or unfavourably (positive slippage, an additional cost) over the execution horizon.

We can model the execution price per share when trading $\Delta q_{t,i}$ shares of ticker i at t in a simple way as the **VWAP**, using

$$p_{t,i}^{\text{exec}}(\Delta q_{t,i}) := p_{t,i}\left(1 + \operatorname{sgn}(\Delta q_{t,i}) \cdot \frac{\text{halfspread\_bps}_{t,i} + \text{slip\_bps}_{t,i}}{10{,}000}\right) \tag{2}$$

where $p_{t,i}$ denotes the pre-trade mid-price of ticker i at date t, which we proxy by the close price of date t.

For both the half-spread and slippage, we can choose to fix them constant across all tickers i and backtest dates t, expressed in basis points relative to the pre-trade mid-price (proxied by the close price of the trade date t).

Close price as a proxy for the pre-trade mid

A fill at the official closing auction would not itself cross a bid–ask spread, since the auction fills buy and sell orders at a single price irrespective of order size. Our cost model is not meant to represent the closing auction. Instead, we use the close as a proxy for the pre-trade mid one would face trading in the continuous session shortly before the close, and it is that continuous-session execution (carrying the half-spread and slippage described above) that the model represents. We use the close price because intraday data is outside the framework's scope.

Numerical example

For a stock i with a pre-trade mid-price of \$50, the choice

$$\text{halfspread\_bps}_{t,i} = 1, \qquad \text{slip\_bps}_{t,i} = 3$$

gives a combined 4 bps cost, implying an execution price for a buy (of any number of shares) of $\$50 \times (1 + 0.0004) = \$50.02$ and for a sell (of any number of shares) of $\$50 \times (1 - 0.0004) = \$49.98$.

Acknowledged limitation

For S&P 500 stocks, which are for the most part highly liquid, a small order (such as 10 shares) almost never incurs price slippage. The displayed size of the best bid/best ask (top of the order book) is typically hundreds to thousands of shares, so a 10-share market order normally fills entirely at the best price with zero slippage. Slippage only occurs if our order exceeds the available size at the best price and "walks the book." For that stock i at that trade moment t we would set $\text{slip\_bps}_{t,i} = 0$.

In reality, execution costs depend on **order size** relative to available order-book depth. A more realistic model of the execution price would let slippage scale with trade size, according to the empirical square-root impact law [1]

$$\frac{\Delta p_{t,i}}{p_{t,i}} = Y_i \operatorname{sgn}(\Delta q_{t,i})\, \sigma_{t,i} \sqrt{\frac{|\Delta q_{t,i}|}{V_{t,i}}}$$

where $\Delta p_{t,i} = p_{t,i}^{\text{VWAP}} - p_{t,i}$, and where $\sigma_{t,i}$ is the estimated stock's volatility (standard deviation of daily price log-returns) over a recent window (e.g. the 60 consecutive trading days preceding t), and $V_{t,i}$ is the rolling average of the daily volume of shares traded for stock i (computed from the same chosen rolling window used for computing $\sigma_{t,i}$, and Y_i is some proportionality constant.

Therefore, if our trade sizes during the backtest are relatively large compared to

$V_{t,i}$, then, as a possible improvement in realism, we could replace the execution price (2) for

$$p^{\text{exec}}_{t,i}(\Delta q_{t,i}) := p_{t,i}\left(1 + Y_i\ \mathrm{sgn}(\Delta q_{t,i})\,\sigma_{t,i}\sqrt{\frac{|\Delta q_{t,i}|}{V_{t,i}}}\right)$$

with $Y_i = 1$.

2.6 Daily equity P&L decomposition

The daily change in equity decomposes into four P&L streams: price P&L, dividend P&L, financing P&L, and trading costs (plus, on days when a margin call is cured by posting cash, an external capital injection that is not strategy P&L, see §2.6.5 and §3.5.1). This subsection defines each contribution (§2.6.1–§2.6.5) and then combines them (§2.6.6).

2.6.1 Daily financing P&L

Financing P&L captures the interest income and expense that accrues on the book's cash balances and short positions over one trading day, for these reasons:

1. interest on the cash we borrow to fund our long positions (we pay)
2. interest on the cash we deposit in our account (we receive)
3. borrow fee on the stocks we short (we pay)
4. short rebate / interest on short-sale proceeds (we typically receive, although for hard-to-borrow names we may pay)

The bases used for the interest accruals are fixed at the close of trading day t, and accrue over the period $t \to t+1$:

$$\begin{aligned}
\mathrm{SMV}_{\text{base},t,i} &:= p_{t,i}\, q^{-}_{t,i} \\
\mathrm{SP}_{\text{base},t,i} &:= \mathrm{ShortProceeds}_{t,i} \\
\mathrm{Debit}_{\text{base},t} &:= \mathrm{Debit}_t \\
\mathrm{Cash}_{\text{base},t} &:= \mathrm{Cash}_t,
\end{aligned}$$

where $p_{t,i}$ is the close price on day t, $q^{-}_{t,i} \geq 0$ is the (absolute) short quantity in ticker i, and $\mathrm{ShortProceeds}_{t,i}$, Debit_t, Cash_t are the book values struck at that close (defined in §2.2).

Annualised rates at fixing time t:

- $b^{\text{ann}}_{t,i} \geq 0$: stock borrow fee for ticker i.
- $r^{\text{reb,ann}}_{t,i} \in \mathbb{R}$: short rebate rate for ticker i (negative for hard-to-borrow names).
- $i^{\text{deb,ann}}_{t} \geq 0$: margin debit rate.
- $r^{\text{cash,ann}}_{t} \geq 0$: cash deposit rate.

Let Δt be the accrual time fraction between the fixings at t and $t+1$, in years under the ACT/360 day-count convention (1/360 per calendar day). Then:

$$\text{Financing P\&L}_{t\to t+1} = -\sum_i b_{t,i}^{\text{ann}}\,\Delta t\,\text{SMV}_{\text{base},t,i} + \sum_i r_{t,i}^{\text{reb,ann}}\,\Delta t\,\text{SP}_{\text{base},t,i}$$
$$- i_t^{\text{deb,ann}}\,\Delta t\,\text{Debit}_{\text{base},t} + r_t^{\text{cash,ann}}\,\Delta t\,\text{Cash}_{\text{base},t}.$$

Default proxies we use:

Rate	Proxy	Backtest Value
Cash deposit rate $r_t^{\text{cash,ann}}$	EFFR minus broker spread, floored at zero	$\max(\text{EFFR}_t - 42\text{ bps}, 0)$
Margin debit rate $i_t^{\text{deb,ann}}$	EFFR plus broker spread	$\text{EFFR}_t + 300\text{ bps}$
Stock borrow fee $b_{t,i}^{\text{ann}}$	Flat rate (S&P 500 names are easy-to-borrow)	25 bps
Short rebate rate $r_{t,i}^{\text{reb,ann}}$	Zero (retail broker retains yield on proceeds)	0

The EFFR time series can be downloaded online (see §11.1). Note that for most of 2010–2015 and 2020–2022, EFFR was below 42 bps, so the cash rate is floored at zero for those periods.

Remark

$\text{MarginCollateral}_t$ earns no interest, consistent with typical retail broker practice.

2.6.2 Daily dividends P&L

Holders of equity shares receive dividends; short sellers owe dividends to the share lender.

Key dates (U.S., under T+1 settlement):

- *Ex-dividend date:* the first day shares trade without the right to the upcoming dividend. A position must be held at the open of the ex-date to be entitled to the dividend.
- *Record date:* under the current U.S. T+1 cycle, this coincides with the ex-dividend date.
- *Payment date:* cash is physically transferred, typically a few weeks after the ex-dividend date.

Entitlement logic. Dividend entitlement at ex-date t is based on the start-of-day position $q_{t,i}^{\text{open}}$. Selling during day t does not forfeit the dividend; buying during day t does not earn it.

Let $D_{t,i}$ be the declared cash dividend per share for ticker i with ex-date t (sourced from the `yfinance` Python module). As a simplification, dividends are booked directly to Cash on the ex-date:

$$\Delta\text{Cash}_t = q_{t,i}^{\text{open}}\,D_{t,i}$$

Remark

This is positive for long positions $q_{t,i}^{\text{open}} > 0$ (cash inflow) and negative for short positions $q_{t,i}^{\text{open}} < 0$ (cash outflow, as the dividend is owed to the lender).
The timing error introduced by booking on the ex-date rather than the payment date is negligible: for example, with a 1,000-share long position, a \$0.50 quarterly dividend, a 5% cash rate, and a 30-day lag, the timing error is approximately \$2.05 on a \$500 cash flow (roughly 0.4%).

2.6.3 Daily price P&L

We hold positions constant throughout $[t, t+1)$, subject to trades executed at the close of $t+1$ to cure a maintenance margin violation at $t+1$ under Variant B (see §3.5), or to rebalance our portfolio if $t+1$ is a rebalance date. Under Variant A a maintenance violation is cured by posting collateral rather than trading, so positions (and hence the price P&L below) are unaffected by the cure.

The price P&L for this period is:

$$\sum_{i \in \Omega_t} q_{t,i} \big(p_{t+1,i} - p_{t,i} \big)$$

2.6.4 Daily trading costs

While commission-free trading is now common for U.S. retail brokerage accounts, explicit per-share commissions remain relevant for some retail and institutional brokerage accounts. Therefore, we implement explicit broker commissions, and refer to the resulting per-trade charge as the **trading fee**. Trading fees are modelled as

$$C_{\text{fee}}(t, i, |\Delta q_{t,i}|) = \lambda_{t,i} \, |\Delta q_{t,i}|$$

where $\lambda_{t,i} \geq 0$ is the per-share commission for ticker i at date t. We assume the same rate for buys and sells, including the opening and closing trades of short positions. We use a time-invariant specification, setting $\lambda_{t,i} = \lambda = \$0.001$ per share for all tickers and dates.

Any trades executed (either under Variant B to cure a maintenance margin violation detected at the close of $t+1$ (using close prices of $t+1$), or to rebalance our portfolio at close $t+1$) are modelled as occurring at the close of $t+1$, filling at the execution price of §2.5.2 derived from that close. Trading fees reduce equity directly:

$$\Delta \text{Equity}_{t+1}^{\text{fees}}(\Delta q_{t+1}) = -\sum_{i} \lambda_{t+1,i} \, |\Delta q_{t+1,i}| < 0$$

where $\lambda_{t+1,i} \geq 0$ is the per-share commission for ticker i, and $\Delta q_{t+1,i} = q_{t+1,i} - q_{t,i}$ is the number of shares traded at date $t+1$.

Execution costs: The trading fee above is the only cost booked as an explicit cash flow. The half-spread and the slippage are applied *inside* the execution price (2), so they never appear as a separate debit, yet they reduce equity on every trade: we buy above the close and sell below it, and the resulting position is then marked at the close. Per the execution-price model, this difference is:

$$\Delta\text{Equity}^{\text{exec}}_{t+1}(\Delta q_{t+1}) = -\sum_i p_{t+1,i}\,\frac{\text{halfspread\_bps}_{t+1,i} + \text{slip\_bps}_{t+1,i}}{10{,}000}\,\left|\Delta q_{t+1,i}\right| \;\le\; 0$$

holding for each of the four trade cases of §2.8, since every share traded pays it exactly once. We report this quantity alongside the trading-fee total in the *Trading* section of the KPI report (§12.4); in the documented run of §13.1 it is roughly 55× the trading-fee line.

2.6.5 Daily posted collateral

If at date $t+1$ our book's equity drops below the maintenance margin requirement, the broker issues a margin call. We can either trade to reduce our long/short exposures (thereby incurring the daily trading costs of §2.6.4, which is Variant B of the backtest) or post the necessary margin collateral to our account, which increases equity one-for-one (§3.5.1), which is Variant A of the backtest.

In the latter case no trade occurs, and the collateral posted is

$$\Delta\text{Equity}^{\text{posting collateral}}_{t+1} = \text{MR}^{\text{maint}}(t+1, q_t) - \text{Equity}_{t+1} > 0$$

(see §3.5.1).

2.6.6 Daily equity P&L

Let $q_{t,i}$ denote the shares position of ticker i at the close of t.

Combining all contributions, the daily equity P&L decomposition for the period between consecutive closes $[t, t+1)$ (throughout which positions are held constant) is:

$$\begin{aligned}\text{Equity}_{t+1} - \text{Equity}_t = {} & \underbrace{\sum_{i\in\Omega_t} q_{t,i}\big(p_{t+1,i} - p_{t,i}\big)}_{\text{price P\&L}} + \underbrace{\sum_{i\in\Omega_t} q_{t,i}\, D_{t+1,i}}_{\text{dividend P\&L}} + \underbrace{\text{Financing P\&L}_{t\to t+1}}_{\text{financing P\&L}} \\ & + \underbrace{\Delta\text{Equity}^{\text{fees}}_{t+1}(\Delta q_{t+1})}_{\text{trading fees}} + \underbrace{\Delta\text{Equity}^{\text{exec}}_{t+1}(\Delta q_{t+1})}_{\text{execution costs}} + \underbrace{\Delta\text{Equity}^{\text{posting collateral}}_{t+1}}_{\text{posted collateral}}\end{aligned}$$

For the dividends attribution, we recall (§2.6.4) that $D_{t+1,i} = 0$ if $t+1$ is not an ex-dividend date for ticker i, and that $q^{\text{open}}_{t+1,i} = q_{t,i}$.

The last three terms depend on the actions taken at the close of $t+1$:

- *Trading fees* $\Delta\text{Equity}^{\text{fees}}_{t+1}(\Delta q_{t+1}) = -\sum_i \lambda_{t+1,i}\,|\Delta q_{t+1,i}|$ arise from any trades executed at $t+1$, whether a scheduled rebalance (§2.7, step 7) or a Variant B deleveraging cure (§3.5.2), and apply under *both* variants; the term is zero on days with no trades.

- *Execution costs* $\Delta\text{Equity}^{\text{exec}}_{t+1}(\Delta q_{t+1})$ (§2.6.4) arise from the same trades and are likewise zero on days with no trades. Unlike every other term they are booked in no cash flow: they are the gap between the execution price actually paid and the close at which the resulting position is then marked.

- *Posted collateral* $\Delta\text{Equity}_{t+1}^{\text{posting collateral}} \geq 0$ is non-zero only under Variant A, when a margin shortfall is cured by posting cash: either a maintenance violation (§3.5.1) or an infeasible rebalance (§3.4.1); it is zero otherwise.

In particular, on a Variant A rebalance day on which a margin call is also cured, both terms are non-zero simultaneously.

This identity is verified

The equation above explaining the equity change is checked in code. For each backtest date the implementation computes the left-hand side directly from the ledger, as $\text{Equity}_{t+1} - \text{Equity}_t$ taken from the two book snapshots, and the right-hand side independently, as the sum of the six streams. The *residual* is the difference between the two.
(Note: Every quantity entering the comparison is carried at full precision, so the only irreducible error is double-precision arithmetic.)

2.7 The intra-day sequence of operations

The backtest runs a fixed seven-step sequence at each backtest date $t > t_0$, where t_0 is the first backtest date (and also the first rebalance date). The cash-to-debit reclassification and cash sweep (§2.3) are *not* a separate step: they are invoked as a subroutine immediately after each cash-moving operation below (that is, after the financing and dividend accruals in step 1 and after every trade simulation in steps 4, 6, and 7)so that cash and debit are never both positive between operations.

1. **Apply overnight P&L:** Financing on cash, debit, and posted collateral; borrow fees and rebates on shorts; account dividends on long and short positions where t is a dividend *ex-date* (credited for longs, debited for shorts). The cash sweep (§2.3) is applied after the financing accrual and again after the dividend accrual.

2. **Mark to market:** Recompute LMV, SMV, equity, and leverage ratios using the open prices at t.

3. **Snapshot the book at this moment (at the open); bifurcate to a working "close" book:** The snapshot at the open is frozen for diagnostic display; subsequent steps mutate the "close" snapshot.

4. **Maintenance MR check (using close prices at t), and cure (Variant A or B) if MR violated:** Cure trades execute at close prices at t, consistent with the prices used to detect the violation; the cash sweep (§2.3) is applied after the cure trades.

5. **Termination tests (using close prices at t):** stop-loss, strategy return target, last backtest date reached (see §5). If any test triggers, the backtest terminates and steps 6–7 are skipped.

6. **Dynamic rebalancing rule (either enabled or disabled for the backtest), checked using close prices at t:** if triggered, an early rebalance is performed, with trades simulated

at close prices at t (see §5), followed by the cash sweep (§2.3). If this is triggered, step 7 is skipped.

7. **Scheduled rebalance?** If t is a scheduled rebalance date, trades are executed at the close prices on t, followed by the cash sweep (§2.3). The updated portfolio is then marked to market using the same close prices. Any simulated market impact from trading is reflected through the execution-price slippage model (see §2.5.2). Otherwise, the portfolio is marked to market using the close prices on t.

Steps 4, 6, and 7 may simulate trades. Step 5 may terminate the backtest. The **first rebalance date** t_0 is special: there is no overnight P&L (no prior day), and the book is initialised with the user's input for initial cash $\text{Equity}_{t_0} = \text{Cash}_{t_0} = \text{initial cash}$ and zero positions (recorded within the book's snapshot at the open); then the strategy is invoked to set the initial portfolio, with trades simulated at close prices at t_0 and the cash sweep (§2.3) applied afterward; the resulting book's snapshot at close is recorded.

2.8 Trades simulation: update of book values

This subsection specifies the exact ledger update rules for each trade case. These rules are applied at every trading event: at the initial rebalance date and at subsequent rebalance dates (§2.7, step 7), and whenever positions are reduced to cure a maintenance margin violation under Variant B (§3.5.2).

Let $q^{\text{pre}}_{t,i}$ denote the current position in ticker i, $q^{\text{post}}_{t,i}$ the intended new position, and $\Delta q_{t,i} = q^{\text{post}}_{t,i} - q^{\text{pre}}_{t,i}$ the number of shares traded. For every ticker i with $\Delta q_{t,i} \neq 0$, we update Cash_t, Debit_t, and the short-proceeds lots.

The following two operations (§2.3) are applied after every individual trade:

- **Cash-to-debit reclassification:** if $\text{Cash}^{\text{post}}_t < 0$: $\text{Debit}^{\text{post}}_t \leftarrow \text{Debit}^{\text{post}}_t + |\text{Cash}^{\text{post}}_t|$, then $\text{Cash}^{\text{post}}_t \leftarrow 0$.
- **Cash sweep:** if $\text{Cash}^{\text{post}}_t > 0$ and $\text{Debit}^{\text{post}}_t > 0$: set $x := \min(\text{Cash}^{\text{post}}_t, \text{Debit}^{\text{post}}_t)$, then $\text{Cash}^{\text{post}}_t \leftarrow \text{Cash}^{\text{post}}_t - x$ and $\text{Debit}^{\text{post}}_t \leftarrow \text{Debit}^{\text{post}}_t - x$.

2.8.1 Initialisation

Before looping over tickers:

$$\text{Cash}^{\text{post}}_t \leftarrow \text{Cash}^{\text{pre}}_t, \qquad \text{Debit}^{\text{post}}_t \leftarrow \text{Debit}^{\text{pre}}_t$$

2.8.2 Case 1: covering a short (possibly flipping to long)

Applies when $q^{pre}_{t,i} < 0$ and $\Delta q_{t,i} > 0$.

Define:

$$\text{cover}_{t,i} := \min(\Delta q_{t,i},\, |q^{\text{pre}}_{t,i}|) \geq 0, \qquad \text{flip}_{t,i} := \max(\Delta q_{t,i} - |q^{\text{pre}}_{t,i}|,\, 0) \geq 0$$

- **Step 1:** Consume lots from the FIFO short-lot queue of ticker i until $\text{cover}_{t,i}$ shares have been matched. Fully consumed lots are removed; partially consumed lots retain their residual share count.

- **Step 2:** Update cash from the covering trade:

$$\text{Cash}_t^{\text{post}} \leftarrow \text{Cash}_t^{\text{post}} - (p_{t,i}^{\text{exec}} + \lambda_{t,i}) \cdot \text{cover}_{t,i} + \sum_{\text{lot } k \text{ consumed}} p_i^{(k)} \cdot n_i^{(k)}$$

- **Step 3:** If $\text{flip}_{t,i} > 0$, open a long position:

$$\text{Cash}_t^{\text{post}} \leftarrow \text{Cash}_t^{\text{post}} - (p_{t,i}^{\text{exec}} + \lambda_{t,i}) \cdot \text{flip}_{t,i}$$

- **Step 4:** Apply cash-to-debit reclassification or cash sweep.

2.8.3 Case 2: increasing a short

Applies when $q_{t,i}^{pre} \leq 0$ and $\Delta q_{t,i} < 0$.

- **Step 1:** Append a new lot $(|\Delta q_{t,i}|, p_{t,i}^{\text{exec}})$ to the FIFO short-lot queue of ticker i. Also, update $\text{ShortProceeds}_{t,i} \leftarrow \text{ShortProceeds}_{t,i} + p_{t,i}^{\text{exec}} \cdot |\Delta q_{t,i}|$.

- **Step 2:** Update cash (trading fee only):

$$\text{Cash}_t^{\text{post}} \leftarrow \text{Cash}_t^{\text{post}} - \lambda_{t,i}\, |\Delta q_{t,i}|$$

- **Step 3:** Apply cash-to-debit reclassification or cash sweep.

2.8.4 Case 3: reducing a long (possibly flipping to short)

Applies when $q_{t,i}^{pre} > 0$ and $\Delta q_{t,i} < 0$.

Define:

$$\text{sell}_{t,i} := \min(|\Delta q_{t,i}|,\, q_{t,i}^{\text{pre}}) \geq 0, \qquad \text{short}_{t,i} := \max(|\Delta q_{t,i}| - q_{t,i}^{\text{pre}},\, 0) \geq 0$$

- **Step 1:** Update cash from selling the long:

$$\text{Cash}_t^{\text{post}} \leftarrow \text{Cash}_t^{\text{post}} + (p_{t,i}^{\text{exec}} - \lambda_{t,i}) \cdot \text{sell}_{t,i}$$

- **Step 2:** If $\text{short}_{t,i} > 0$, open a short position: Append a new lot $(\text{short}_{t,i}, p_{t,i}^{\text{exec}})$ to the FIFO queue. Also, update $\text{ShortProceeds}_{t,i} \leftarrow \text{ShortProceeds}_{t,i} + p_{t,i}^{\text{exec}} \cdot \text{short}_{t,i}$.

- **Step 3:** Update cash (trading fee on the short sale):

$$\text{Cash}_t^{\text{post}} \leftarrow \text{Cash}_t^{\text{post}} - \lambda_{t,i} \cdot \text{short}_{t,i}$$

- **Step 4:** Apply cash-to-debit reclassification or cash sweep.

2.8.5 Case 4: increasing a long

Applies when $q^{pre}_{t,i} \geq 0$ *and* $\Delta q_{t,i} > 0$.

- **Step 1:** Update cash:

$$\text{Cash}^{\text{post}}_t \leftarrow \text{Cash}^{\text{post}}_t - (p^{\text{exec}}_{t,i} + \lambda_{t,i})\, \Delta q_{t,i}$$

- **Step 2:** Apply cash-to-debit reclassification or cash sweep.

2.8.6 Final mark-to-market

After looping over all tickers with $\Delta q_{t,i} \neq 0$, mark the updated portfolio to market using the close prices (§2.5.1):

$$\text{LMV}^{\text{post}}_t = \sum_i p_{t,i}\, (q^{\text{post}}_{t,i})^+, \qquad \text{SMV}^{\text{post}}_t = \sum_i p_{t,i}\, (q^{\text{post}}_{t,i})^-$$

The short proceeds balance is recomputed from the updated lot queues:

$$\text{ShortProceeds}^{\text{post}}_{t,i} = \sum_k p^{(k)}_i\, n^{(k)}_i$$

Post-trade equity is:

$$\text{Equity}^{\text{post}}_t = \text{LMV}^{\text{post}}_t - \text{SMV}^{\text{post}}_t + \text{Cash}^{\text{post}}_t + \sum_i \text{ShortProceeds}^{\text{post}}_{t,i} - \text{Debit}^{\text{post}}_t + \text{MarginCollateral}_t$$

Remark

Under the assumption that post-trade prices equal the pre-trade prices, we have:

$$\text{Equity}^{\text{post}}_t = \text{Equity}^{\text{pre}}_t - \sum_{i,\, \Delta q_{t,i} \neq 0} \lambda_{t,i}\, |\Delta q_{t,i}|$$

i.e. the only equity impact of the trades (at constant prices) are the trading fees paid.

3 Margin Requirements

3.1 Purpose

A long/short portfolio with gross exposure greater than the deposited equity is leveraged (recall §2.4). Brokers require collateral against this leverage, in the form of equity that must remain above a per-position-weighted threshold. The requirement comes in two forms:

- **Initial margin** is checked at trade time, when a position is opened or increased.
- **Maintenance margin** is checked continuously, whenever the book is marked to market.

The margin requirement is checked against our equity (§2.2). **The broker requires that equity remain sufficient to absorb simultaneous adverse moves across all positions.** Specifically, that for a decline of $h^L_{t,i}$ in every long and a rise of $h^S_{t,j}$ in every short, equity remains non-negative:

$$\text{Equity}_t - \sum_i h^L_{t,i}\,\text{LMV}_{t,i} \;-\; \sum_j h^S_{t,j}\,\text{SMV}_{t,j} \;\geq\; 0$$

The haircuts $h^L_{t,i}$, $h^S_{t,j}$ are set by a combination of regulation and broker in-house rules (§3.2). They act as risk-based buffers: the larger the haircut, the more equity must be held.

The rationale is as follows. Given the equity formula of §2.2:

- **For long positions:** a price decline of $h^L_{t,i}$ reduces $\text{LMV}_{t,i}$ by $h^L_{t,i}\,\text{LMV}_{t,i}$, reducing equity by the same amount.

- **For short positions:** a price rise of $h^S_{t,j}$ increases $\text{SMV}_{t,j}$ by $h^S_{t,j}\,\text{SMV}_{t,j}$. Since SMV enters the equity definition with a negative sign, equity falls by the same amount.

Reg-T sets the standard initial requirement at 50% for both long and short positions; FINRA Rule 4210 sets maintenance floors of 25% for longs and 30% for shorts. Real broker requirements may be higher (especially for volatile names); we use these regulatory floors as defaults. Our code supports per-(date, ticker) overrides if a user wants different margin haircuts.

Variant A vs. Variant B: Throughout §3, the backtest runs in one of two modes, fixed for the entire run. *Variant A* always cures a margin shortfall by posting external cash collateral; *Variant B* always cures it by deleveraging through uniform position shrinkage. This single choice governs both situations in which a margin shortfall can arise: an infeasible scheduled rebalance (§3.4.1) and a maintenance-margin violation detected post-MtM (§3.5). In our implementation, the two situations are never handled by different variants within the same backtest.

3.2 Initial and maintenance margin

- The **maintenance margin requirement** at date t for the current portfolio q_t is

$$\text{MR}^{\text{maint}}(t, q_t) = \sum_i \Big[h_i^{L,\text{maint}}\, p_{i,t}\, q_{i,t}^+ + h_i^{S,\text{maint}}\, p_{i,t}\, q_{i,t}^- \Big] \tag{3}$$

where $h_i^{L,\text{maint}}$ and $h_i^{S,\text{maint}}$ are the **maintenance haircuts** for ticker i (defaults 25% and 30% respectively).

Equity must cover the maintenance MR:

$$\text{Equity}_t \geq \text{MR}^{\text{maint}}(t, q_t)$$

If the inequality is violated post-MTM, the cure procedure described in §3.5 (with q_t' set to q_t) kicks in.

- The **initial margin requirement** at date t for opening/increasing positions by Δq_t is

$$\mathrm{MR}^{\mathrm{init}}(t, q_t) \;=\; \sum_i \Big[h_i^{L,\mathrm{init}}\, p_{i,t}\,(\Delta q_{i,t})^+ + h_i^{S,\mathrm{init}}\, p_{i,t}\,(\Delta q_{i,t})^-\Big]$$

where $h_i^{L,\mathrm{init}}$ and $h_i^{S,\mathrm{init}}$ are the **initial haircuts** for ticker i (defaults 50% and 50% respectively).

The equity requirement for changing positions (opening/increasing, or decreasing/closing) is covered by the margin requirement for rebalancing the portfolio (§3.4).

3.3 Simulated margin loans drawn during the backtest

An initial haircut $h_{t,i}^{L,\mathrm{init}} < 100\%$ permits a margin loan: equity need only cover the fraction $h_{t,i}^{L,\mathrm{init}}$ of any new long market value, so the remainder can be financed by debit. Per §2.3, any cash shortfall created by the trade is reclassified as positive Debit_t.

Example

Suppose the book has no open positions and $\mathrm{Cash}_t = \mathrm{Equity}_t = \$10{,}000$, $\mathrm{Debit}_t = 0$. We consider opening a long position of $q_{t,i}$ shares of an S&P 500 constituent i (e.g. AAPL) at the pre-trade price $p_{t,i}$, which the broker subjects to a haircut of $h_{t,i}^{L,\mathrm{init}} = 50\%$ (the default Regulation T initial margin for long positions). The initial margin requirement is

$$\mathrm{Equity}_t \;\ge\; \mathrm{MR}_i^{\mathrm{init}}(t, q_{t,i}), \qquad \mathrm{MR}_i^{\mathrm{init}}(t, q_{t,i}) := h_{t,i}^{L,\mathrm{init}}\, p_{t,i}\, q_{t,i}.$$

This is equivalent to

$$q_{t,i} \;\le\; \frac{\mathrm{Equity}_t}{h_{t,i}^{L,\mathrm{init}}\, p_{t,i}},$$

and since $q_{t,i}$ must be a whole number of shares, the largest admissible position is

$$q_{t,i}^{\max} \;=\; \left\lfloor \frac{\mathrm{Equity}_t}{h_{t,i}^{L,\mathrm{init}}\, p_{t,i}} \right\rfloor$$

with corresponding long market value $\mathrm{LMV}_{t,i} := p_{t,i}\, q_{t,i}^{\max} \approx \mathrm{Equity}_t / h_{t,i}^{L,\mathrm{init}} = \$20{,}000$.

Assuming the account is permitted to draw a margin loan to fund admissible positions, booking $q_{t,i}^{\max}$ shares at $p_{t,i}$ reduces cash by $\mathrm{LMV}_{t,i}$, so the post-booking cash balance is

$$\mathrm{Cash}_t - \mathrm{LMV}_{t,i} \;\approx\; -\$10{,}000.$$

By §2.3 any such deficit is reclassified as positive debit, resulting in a debit balance of

$$\mathrm{Debit}_t \;=\; \big|\mathrm{Cash}_t - \mathrm{LMV}_{t,i}\big| \;\approx\; \$10{,}000.$$

with $\mathrm{Cash}_t = 0$ after the sweep.

Remark

Equity itself is unchanged by the trade, up to trading costs and market impact (§2.5.2).

Therefore, provided the margin requirement $\mathrm{Equity}_t \ge \mathrm{MR}_i^{\mathrm{init}}(t, q_{t,i})$ is satisfied, the book can open or grow long exposure. A margin loan is drawn exactly when the trade exhausts available cash, i.e. when $p_{t,i}\, q_{t,i} > \mathrm{Cash}_t$; in that case the rebalance strictly increases Debit_t.

3.4 Margin requirement for rebalancing

At any rebalance date t, the portfolio holds positions carried over from the previous rebalance date (or is flat if this is the first rebalance date), and we wish to move to a new target portfolio built from tickers in the current investable universe (see §6 for the construction of the investable universe).

We assume the rebalance is fully executed on the rebalance date and that prices are constant during execution (no market impact).

Notation

Let Ω_t^{prev} be the investable universe at the previous rebalance date (empty set if this is the first rebalance date), and Ω_t^{new} the universe at the current rebalance date. Define $\bar{\Omega}_t := \Omega_t^{\mathrm{prev}} \cup \Omega_t^{\mathrm{new}}$. Current positions are $q_t = (q_{t,i})_{i\in\bar{\Omega}_t}$ and intended new positions are $q'_t = (q'_{t,i})_{i\in\bar{\Omega}_t}$, with $q_{t,i} = 0$ for $i \notin \Omega_t^{\mathrm{prev}}$ and $q'_{t,i} = 0$ for $i \notin \Omega_t^{\mathrm{new}}$.

Execution order: All reductions and closures of existing positions are executed first (releasing maintenance margin); all increases and openings are executed afterward (requiring initial margin).

We define the following partition of $\bar{\Omega}_t$:

Symbol	Description	Condition
$\mathcal{R}^+(q_t, q'_t)$	Reduction of a long	$0 < q'_{t,i} < q_{t,i}$
$\mathcal{R}^-(q_t, q'_t)$	Reduction of a short	$q_{t,i} < q'_{t,i} < 0$
$\mathcal{I}^+(q_t, q'_t)$	Increase of a long	$0 < q_{t,i} < q'_{t,i}$
$\mathcal{I}^-(q_t, q'_t)$	Increase of a short	$q'_{t,i} < q_{t,i} < 0$
$\mathcal{S}^+(q_t, q'_t)$	Switch from short to long	$q_{t,i} < 0 < q'_{t,i}$
$\mathcal{S}^-(q_t, q'_t)$	Switch from long to short	$q'_{t,i} < 0 < q_{t,i}$
$\mathcal{C}^+(q_t, q'_t)$	Close a long	$q_{t,i} > 0,\ q'_{t,i} = 0$
$\mathcal{C}^-(q_t, q'_t)$	Close a short	$q_{t,i} < 0,\ q'_{t,i} = 0$
$\mathcal{O}^+(q_t, q'_t)$	Open a long	$q_{t,i} = 0,\ q'_{t,i} > 0$
$\mathcal{O}^-(q_t, q'_t)$	Open a short	$q_{t,i} = 0,\ q'_{t,i} < 0$
$\mathcal{U}(q_t, q'_t)$	Unchanged	$q_{t,i} = q'_{t,i}$

The minimum equity necessary and sufficient to execute the full rebalance without violating any margin constraint at any intermediate step is $\mathrm{MR}(t, q_t, q'_t)$, with:

$$
\begin{aligned}
\mathrm{MR}(t, q_t, q'_t) = \mathrm{MR}^{\mathrm{maint}}(t, q_t) & \\
&+ \sum_{i\in\mathcal{C}^+} \Big(\lambda_{t,i} - h_{t,i}^{L,\mathrm{maint}} p_{t,i}\Big) q_{t,i} \\
&+ \sum_{i\in\mathcal{C}^-} \Big(\lambda_{t,i} - h_{t,i}^{S,\mathrm{maint}} p_{t,i}\Big) |q_{t,i}| \\
&+ \sum_{i\in\mathcal{O}^+} \Big(\lambda_{t,i} + h_{t,i}^{L,\mathrm{init}} p_{t,i}\Big) q_{t,i} \\
&+ \sum_{i\in\mathcal{O}^-} \Big(\lambda_{t,i} + h_{t,i}^{S,\mathrm{init}} p_{t,i}\Big) |q_{t,i}| \\
&+ \sum_{i\in\mathcal{R}^+} \Big(\lambda_{t,i} - h_{t,i}^{L,\mathrm{maint}} p_{t,i}\Big) (q_{t,i} - q'_{t,i}) \\
&+ \sum_{i\in\mathcal{R}^-} \Big(\lambda_{t,i} - h_{t,i}^{S,\mathrm{maint}} p_{t,i}\Big) (q'_{t,i} - q_{t,i}) \\
&+ \sum_{i\in\mathcal{I}^+} \Big(\lambda_{t,i} + h_{t,i}^{L,\mathrm{init}} p_{t,i}\Big) (q'_{t,i} - q_{t,i}) \\
&+ \sum_{i\in\mathcal{I}^-} \Big(\lambda_{t,i} + h_{t,i}^{S,\mathrm{init}} p_{t,i}\Big) (q_{t,i} - q'_{t,i}) \\
&+ \sum_{i\in\mathcal{S}^+} \Big[\Big(\lambda_{t,i} - h_{t,i}^{S,\mathrm{maint}} p_{t,i}\Big) |q_{t,i}| + \Big(\lambda_{t,i} + h_{t,i}^{L,\mathrm{init}} p_{t,i}\Big) q'_{t,i}\Big] \\
&+ \sum_{i\in\mathcal{S}^-} \Big[\Big(\lambda_{t,i} - h_{t,i}^{L,\mathrm{maint}} p_{t,i}\Big) q_{t,i} + \Big(\lambda_{t,i} + h_{t,i}^{S,\mathrm{init}} p_{t,i}\Big) |q'_{t,i}|\Big].
\end{aligned}
$$

Interpretation: The portfolio begins with its maintenance requirement $\mathrm{MR}^{\mathrm{maint}}(t, q_t)$. Each subsequent term reflects the net equity impact of one class of trade:

- For reductions and closures, required equity changes by the trading fee paid minus the maintenance margin released (a negative contribution, since the haircut multiplied by the price always exceeds the fee for realistic values: $p_{t,i} > 1$, $h_{t,i}^{L,\mathrm{maint}}, h_{t,i}^{S,\mathrm{maint}} > 25\%$, $\lambda_{t,i} \le 10^{-1}$).
- For increases and openings, it changes by the trading fee paid plus the initial margin required (a positive contribution).
- For switches, both effects occur sequentially: the old side is closed first, then the new side is opened.

3.4.1 Algorithm for rebalancing the portfolio

At each rebalance date t:

- If $\mathrm{Equity}_t \ge \mathrm{MR}(t, q_t, q'_t)$: rebalance as intended.
- Otherwise:
  - **Variant A (posting collateral):** Post external cash $F_t = \mathrm{MR}(t, q_t, q'_t) - \mathrm{Equity}_t > 0$ into the MarginCollateral account, increasing Equity_t by F_t without altering positions. Record the posted collateral.

– **Variant B (deleveraging):** Compute $\alpha^\star$ via bisection (§3.4.4) and replace the intended target q'_t by $\mathrm{trunc}(\alpha^\star q'_t)$. Record the shrinkage factor $\alpha^\star$.

3.4.2 Variant B: feasibility test and uniform shrinkage

If $\mathrm{Equity}_t < \mathrm{MR}(t, q_t, q'_t)$, the intended rebalance is infeasible. We replace q'_t by a uniformly shrunk version $q_t'^{(\alpha)} := \mathrm{trunc}(\alpha q'_t)$ for $\alpha \in [0,1]$, where truncation toward zero is defined as:

$$\mathrm{trunc}(x) := \begin{cases} \lfloor x \rfloor, & x \geq 0 \\ \lceil x \rceil, & x < 0. \end{cases}$$

We assume only integer share amounts can be traded: Uniform shrinkage is chosen because it preserves (i) the sign of every position, (ii) relative proportions between positions (before integer truncation), (iii) the Sharpe ratio of the intended investment (before integer truncation), and (iv) any beta-neutrality target, as verified by the following identities:

- *Sharpe ratio preservation:* $\dfrac{\mu_t^\top (\alpha w'_t)}{\sqrt{(\alpha w'_t)^\top \Sigma_t (\alpha w'_t)}} = \dfrac{\mu_t^\top w'_t}{\sqrt{{w'_t}^\top \Sigma_t w'_t}}$ (see §9.1 for the definition of these variables).

- *Beta neutrality preservation:* $|\beta_t^\top (\alpha w_t)| \leq |\beta_t^\top w_t| \leq \varepsilon$ whenever the pre-shrinkage portfolio satisfies the beta neutrality constraint $|\beta_t^\top w_t| \leq \varepsilon$, where ε is small (see §9.3.2).

3.4.3 Optimal shrinkage factor

Define $f(\alpha) := \mathrm{MR}(t, q_t, q_t'^{(\alpha)})$. Infeasibility means $\mathrm{Equity}_t < f(1)$. We seek the largest feasible α:

$$\alpha^\star := \sup S, \qquad S := \{\alpha \in [0,1] : \mathrm{Equity}_t \geq f(\alpha)\}$$

Assumption (A1):

$$\min\left(h_{t,i}^{L,\mathrm{maint}}\, p_{t,i}\,,\, h_{t,i}^{S,\mathrm{maint}}\, p_{t,i}\right) \geq \lambda_{t,i}$$

for all t and $i \in \bar{\Omega}_t$. This holds for our backtest values: $\lambda = 10^{-3}$, $h^{\mathrm{maint}} \geq 25\%$, $p_{t,i} \geq 1$.

Remark

Under (A1), we have that full liquidation to zero is feasible:

$$\mathrm{Equity}_t \geq \mathrm{MR}(t, q_t, 0) = \sum_{i \in \Omega_t,\, q_{t,i} \neq 0} \lambda_{t,i} |q_{t,i}|$$

Indeed, this is implied by the satisfaction of the maintenance margin requirement:

$$
\begin{aligned}
\text{Equity}_t &\geq \sum_{i\in\Omega_t,\, q_{t,i}\neq 0} h_{t,i}^{L,\text{maint}}\, p_{t,i}(q_{t,i})^+ + h_{t,i}^{S,\text{maint}}\, p_{t,i}(q_{t,i})^- \\
&\geq \sum_{i\in\Omega_t,\, q_{t,i}\neq 0} \min\left(h_{t,i}^{L,\text{maint}}\, p_{t,i}\,,\, h_{t,i}^{S,\text{maint}}\, p_{t,i}\right) |q_{t,i}| \\
&\overset{(A1)}{\geq} \sum_{i\in\Omega_t,\, q_{t,i}\neq 0} \lambda_{t,i}|q_{t,i}|
\end{aligned}
$$

Under (A1), f is nondecreasing (each term's contribution is nondecreasing in α under the sign conditions imposed by the respective haircut coefficient).

Furthermore, $0 \in S$, and $1 \notin S$. Therefore, this implies:

$$S = [0, \alpha^\star] \quad \text{or} \quad S = [0, \alpha^\star)$$

3.4.4 Approximation of the shrinkage factor via bisection

Since f is nondecreasing, $\alpha^\star$ can be approximated by bisection to any desired precision $P > 0$ using the standard bisection algorithm:

```python
low, hi = 0.0, 1.0
while hi - low > P:
    mid = (low + hi) / 2.0
    if f(mid) <= equity:
        low = mid
    else:
        hi = mid
alpha = low
```

Python Script 1: Bisection algorithm approximating the optimal shrinkage factor with arbitrary precision P

The invariant $f(\text{low}) \leq \text{Equity}_t < f(\text{hi})$ is maintained at every iteration, and the interval $[\text{low}, \text{hi}]$ halves at each iteration. Therefore, at termination, the algorithm returns a feasible α (*i.e.* $\alpha \in S$) satisfying $0 \leq \alpha^\star - \alpha \leq P$.

In the backtest implementation we use the precision $P = 10^{-6}$.

3.5 Curing maintenance margin violations

If on any backtest date equity drops below the maintenance margin requirement:

$$\text{Equity}_t < \text{MR}^{\text{maint}}(t, q_t)$$

the broker issues a margin call and we must restore compliance.

3.5.1 Variant A: posting collateral

Post the amount of external cash

$$F_t = \mathrm{MR}^{\mathrm{maint}}(t, q_t) - \mathrm{Equity}_t > 0$$

into the MarginCollateral account. This increases Equity_t by F_t without altering positions. Record the posted collateral.

Under this procedure in our backtesting framework, there is no limit on the exposure implied by the intended new positions q_t: we may acquire them as long as we post the required margin collateral F_t, which may therefore be several times greater than Equity_t, depending on q_t. As illustrated in the example of §13.11.2, we run a backtest of the momentum strategy under this MR-compliance procedure (Variant A). At the first rebalance, starting from an initial cash of \$100,000 (which constitutes our initial equity), the total collateral posted to reach the strategy's intended positions throughout the backtest is \$49,139.59 (see Figure 27), roughly half of our initial equity.

3.5.2 Variant B: deleveraging via uniform position rescaling

Apply the uniform shrinkage procedure of §3.4.2 with intended portfolio equal to the current portfolio ($q'_t = q_t$). With $\Omega_t^{\mathrm{prev}} = \Omega_t^{\mathrm{new}} = \Omega_t$, the rebalance margin requirement simplifies to:

$$\mathrm{MR}\left(t, q_t, \mathrm{trunc}(\alpha q_t)\right) = \mathrm{MR}^{\mathrm{maint}}\left(t, \mathrm{trunc}(\alpha q_t)\right) + \sum_{i \in \Omega_t,\, q_{t,i} \neq 0} \lambda_{t,i} \left| q_{t,i} - \mathrm{trunc}(\alpha q_{t,i}) \right|$$

Apply the bisection algorithm of §3.4.4 to find the largest feasible $\alpha \in [0, 1]$, execute the resulting position reductions via §2.8, and record the shrinkage factor and trading costs.

4 Backtest Calendar and Inter-Rebalance Periods

4.1 The three calendars

The trading calendar is the NYSE.

A backtest is anchored on three related date sequences, all derived from a single user-specified start date t_0 and a rebalance frequency rule:

1. **Rebalance dates** $t_0 < t_1 < \cdots < t_P$: the dates on which a new target portfolio is computed and traded into. The number of *inter-rebalance periods* is P. The user specifies P and the frequency rule (see §4.2).

2. **Training rebalance dates** $t_{-P_{\mathrm{train}}} < \cdots < t_{-1} < t_0 < t_1 < \cdots < t_P$: the rebalance grid **extended** backwards by P_{train} periods, used to build rolling-window estimators (covariance matrices, betas, factor regressions, etc.). Strategy-input estimators that need W periods of history at t_k require $P_{\mathrm{train}} \geq W$.

3. **Backtest dates:** every NYSE trading day in $[t_0, t_{P+1}]$, where t_{P+1} is one further rebalance period after t_P under the same frequency rule, included to mark-to-market the book from the last rebalance date to the last backtest date (*i.e.*, the period $(t_P, t_{P+1}]$.

All dates are NYSE trading days (no weekends, no exchange holidays). If the user-supplied start date t_0 doesn't fall on a trading day, it is snapped backward to the closest preceding trading day.

4.2 Frequency rules

Two frequency rules are supported:

- `monthly`: rebalance on the **first NYSE trading day of each calendar month**. The first rebalance after t_0 is the first trading day of the *next* calendar month (so the rebalance never falls in t_0's own month).
- `ndays`: rebalance every N trading days, where N is user-specified. For $N = 21$, this approximates monthly rebalancing (since a calendar month contains roughly 21 NYSE trading days on average) but produces *uniform-length* inter-rebalance periods.

4.3 Inter-rebalance periods

An **inter-rebalance period** is the half-open interval $(t_{k-1}, t_k]$. Its **period return** is

$$R_k^{\text{period}} = \prod_{t \in (t_{k-1}, t_k]} (1 + r_t) - 1$$

where r_t is the daily return from the previous trading date to t. This is the compounded return realised over the period.

Period returns are computed for three quantities:

- **Equity** of the portfolio (used for KPI reporting in §12).
- **Per-ticker close prices** (used as the regression target for the factors model in §8, and as the input to the rolling covariance matrix and beta estimators in §9.7).
- **Risk-free rate** (used to compute period excess returns).

4.4 Sizing the data window

Given the rebalance and training calendars, we compute the **data window** $[t_{\text{fetch}}, t_{P+1}]$ that must be covered by price data, with

$$t_{\text{fetch}} = t_{-P_{\text{train}}} - W^{\text{daily}} \text{ (trading days)}$$

where W^{daily} is the **largest** of the daily rolling windows used by the feature engineering layer:

- Volatility window (default 60 trading days);
- Momentum lookback + buffer (default $W_{mom} + b_{mom} = 92 + 23 = 115$ trading days);
- ADV window when capping universes by trailing volume (see §6), (default 60 trading days).

As of our current code implementation, t_{fetch} cannot be earlier than the earliest date our code supports for fetching Wikipedia's S&P 500 list (2007-03-06; see §6).

5 Termination Rules and Dynamic Rebalancing

5.1 Overview: early termination and early rebalancing

A backtest will run to its scheduled end, unless two situations require ending the backtest early:

1. The book has **lost too much equity** ("stop-loss" rule)
2. The strategy has **already met its return target**.

We also implement **dynamic rebalancing**: rebalance early (ahead of the next scheduled rebalance date) when the inter-rebalance return rises above a threshold, taking advantage of the favourable move rather than waiting for the next scheduled rebalance.

These two termination triggers and the early-rebalance trigger are evaluated in a fixed order at every backtest date, after the maintenance-MR cure step (step 4 of §2.7). Each is described below.

5.2 Stop-loss

The stop-loss rule fires when post-MTM equity (excluding posted collateral) drops to a critical fraction of initial equity with which we started the backtest, also accounting for the cost of liquidating the current portfolio.

Threshold: Let $\rho \in (0, 1)$ be the fraction (default 0.5, a 50% drawdown) and Equity_{t_0} be the initial equity. The stop-loss threshold at date t is

$$\theta_t \;=\; \rho\,\mathrm{Equity}_{t_0} + \sum_i \left[\lambda_{i,t} + p_{i,t} \cdot \frac{\mathrm{halfspread\_bps}_{t,i} + \mathrm{slip\_bps}_{t,i}}{10\,000}\right] |q_{i,t}| \tag{4}$$

where Equity_{t_0} is the value of the equity of our book after marking-to-market our positions at the open prices of date t_0, and the second term is the trading cost of fully liquidating the current book at t: half-spread and slippage from our execution price model (§2.5.2), and trading commission, summed across all positions.

The intent is that **after liquidation**, the residual equity covers $\rho\,\mathrm{Equity}_{t_0}$.

Trigger condition: If $\widetilde{\mathrm{Equity}}_t \le \theta_t$, the stop-loss is triggered. The current book snapshot is recorded as that date's close snapshot (it is *not* automatically liquidated, the backtest simply terminates; a real-world stop-loss would close out positions).

5.3 Strategy return target

The strategy return target rule fires when the cumulative return of the portfolio (excluding collateral) reaches a user-specified threshold R^{strategy} relative to initial equity:

$$\frac{\widetilde{\mathrm{Equity}}_t - \widetilde{\mathrm{Equity}}_{t_0}}{\widetilde{\mathrm{Equity}}_{t_0}} \;\ge\; R^{\mathrm{strategy}}$$

Where $\widetilde{\text{Equity}}_{t_0}$ is the initial capital deployed, that is equity at the open of the first backtest date, before any trade has executed, and $\widetilde{\text{Equity}}_t$ is equity after marking-to-market our positions at the *close* prices of date t, which is where this rule is evaluated in the daily sequence (§2.7). The stop-loss (§5.2) and the return reported in the KPI report (§12.1) use the same baseline, so all three measure from the capital deployed.

When triggered, the backtest terminates without further trades.

Typical values are $R^{\text{strategy}} \in [5\%, 30\%]$ for an annual horizon. The strategy-comparison methodology in §14 sweeps this parameter explicitly.

Distinction from the inter-rebalance target

The strategy target is measured from the *initial* equity and *terminates* the backtest; the inter-rebalance target (§5.4) is measured from the *previous rebalance* equity and only *triggers an early rebalance*. The two rules can be active simultaneously; for example, a strategy with $R^{\text{strategy}} = 15\%$ and $R^{\text{inter}} = \frac{3\%}{12}$ will rebalance early on inter-rebalance gains and also terminate if and when the cumulative gain hits 15%.

5.4 Inter-rebalance return target (dynamic rebalance)

The inter-rebalance target rule rebalances *early*, before the next scheduled rebalance date, when the return since the *previous* rebalance reaches a user-specified threshold R^{inter}.

Trigger condition: Let t_k be the most recent rebalance date (from the dynamic schedule). At any backtest date t with $t > t_k$:

$$\frac{\widetilde{\text{Equity}}_t - \widetilde{\text{Equity}}_{t_k}}{\widetilde{\text{Equity}}_{t_k}} \geq R^{\text{inter}}$$

where the values of equity at the respective dates are those after marking-to-market our positions at the open prices of the respective dates.

When triggered, t is treated as a rebalance date: the strategy is invoked, trades are executed, and the future rebalance schedule is **rebuilt** from t forward using the same frequency rule (rebalances every N trading days from t, or the first trading day of the next month, etc.).

Rationale: The fixed-calendar mode commits to rebalancing only at scheduled dates, regardless of intervening market moves. If a strategy has already realised most of the period's expected return, holding the portfolio passively until the next scheduled rebalance exposes the book to adverse reversion. **Rebalancing early locks in the profit and refreshes the optimal portfolio under the strategy's view at t.**

When this is useful: The dynamic-rebalance rule is particularly relevant for momentum and factors-model strategies, whose signals can decay rapidly. For mean-variance optimization with moderate variance targets, the rule has less effect (the optimizer's portfolio doesn't change dramatically between rebalances unless the input estimates do).

5.5 Universe at a dynamic rebalance date

When the dynamic rule fires at a date t that is not a pre-schedule rebalance date, the backtest constructs a point-in-time universe at t on the fly (see §6). Pre-built universes are reused when the dynamic rebalance date happens to coincide with a pre-scheduled rebalance date.

5.6 Last scheduled date

If neither the stop-loss early-termination rule nor the strategy return target early-termination rule is triggered, the backtest continues until reaching the last backtest date. No special trades are executed at the last date: the final snapshot (at close) is the result from the mark-to-market of our positions at the close prices of t_{P+1}, possibly after a maintenance-MR cure on that date.

5.7 Order of evaluation

The three termination tests are evaluated **in this order** at each backtest date, immediately after the MTM cure step:

1. Stop-loss: if fired, terminate backtest.
2. Strategy return target: if fired, terminate backtest.
3. Last scheduled date: if reached, terminate backtest.

If none of the three terminate the backtest, the dynamic rebalancing rule is evaluated next; if it fires, the date becomes a dynamic rebalance date. Otherwise, the backtest orchestrator checks whether t is on the (possibly updated) scheduled rebalance dates grid and rebalances if so.

6 Universe Construction

6.1 Historical members of the S&P 500 index

A realistic backtest must use **only information available at the time of the trade**. For S&P 500 strategies, this means the universe at rebalance date t_k is the set of S&P 500 constituents *as of* t_k, not the constituents as of today.

Using today's constituents (a common backtest mistake known as **survivorship bias**) systematically excludes failed companies (which were dropped from the index) and includes successful companies before they joined the index. Both effects improve the backtest performance unrealistically.

6.2 Source of historical membership

We source the S&P 500 historical membership from the revision history of Wikipedia's *List of S&P 500 companies* page [8]. The page is updated whenever the index composition changes (additions, deletions, ticker changes). For any past date t, we identify the latest revision strictly before t, fetch the rendered page at that revision, and parse the constituents table.

Earliest supported date: 2007-03-06: Before this date, Wikipedia's table was present in a differently structured format from later dates, which we do not handle.

Caveats: Wikipedia is community-maintained; corrections sometimes lag actual index changes by days. For most research purposes, the list is good enough; for production trading, a paid data vendor (e.g. Compustat, CRSP) would be preferable. We log a warning if the latest revision before a target date is more than 60 days old, and a warning if the constituent count falls outside $[400, 600]$.

6.3 Anchoring membership to the rebalance grid

We sample membership at the **first NYSE trading day of each month** in the data window (§4.4). Every (training) rebalance date t_k then takes its candidate universe from the membership at the first trading day of t_k's calendar month.

Why first-of-month: The S&P 500 membership is formally reviewed quarterly (March, June, September, December), but constituent changes can occur at any time due to mergers, acquisitions, bankruptcies, spin-offs, or other corporate events. In practice, most membership changes are event-driven rather than tied to scheduled quarterly reviews. For daily-frequency backtests, a monthly membership snapshot is already a sufficiently accurate approximation, while avoiding the need to reconstruct and store roughly 252 daily membership snapshots per year instead of just 12 monthly ones.

6.4 Eligibility filters

From the candidate set at t_k (the historical S&P 500 members), we apply two filters in sequence:

1. **Forward price data available:** The ticker has non-NaN open and close prices for *every* NYSE trading day in the upcoming inter-rebalance period $[t_k, t_{k+1}]$. This prevents the strategy from entering positions whose future daily mark-to-market series cannot be computed, either because of missing price data or because the ticker disappears before the next rebalance (e.g. delisting). We acknowledge that this introduces a mild survivorship bias.

2. **ADV available** *(optional)*: When the universe is capped to a fixed number of members (e.g. 200), tickers are ranked by rolling-average daily dollar volume (ADV), a liquidity measure, and only the top-ranked names are retained. In this case, the ticker must have a non-NaN ADV value at t_k.

6.5 Bad-ticker pre-filter

We first screen the entire price panel (all historical S&P 500 members since the earliest date used) for data corruption, then build the point-in-time universe of investable tickers for each rebalance date from the cleaned panel.

We use the two rules explained in Appendix to detect and discard corrupted data.

6.6 The eligibility report

In the code implementation, the bulk universe-construction function returns both the per-date universe dictionary and an **eligibility DataFrame** indexed by `(rebalance_date, ticker)` with one boolean column per filter (`forward_close_prices_available`, `forward_open_prices_available`, and, if capping the number of tickers per universe, `adv_available`). The DataFrame is useful for diagnosing exclusion decisions, most commonly forward-period data gaps near delistings.

7 Strategy 1: Top-k/Bottom-k Selection from A Momentum Signal

7.1 The signal

Momentum is the empirical tendency of stocks that have performed well or badly over a recent time window to continue doing so in the near future.

Our momentum signal at date t_k for ticker i is the log return over a rolling window, with a buffer at the recent end:

$$M_{i,t_k} \;=\; \ln \frac{p_{i,t_k-b}}{p_{i,t_k-(b_{mom}+W_{mom})}}$$

where W_{mom} is the rolling window in trading days (default $W_{mom} = 92 \approx 4$ trading months) and b is the buffer (default $b_{mom} = 23 \approx 1$ trading month). The buffer is meant to skip the most recent month, which empirically exhibits short-term reversal that contaminates the medium-term momentum signal.

7.2 The selection rule

The idea is to form a long-short portfolio of recent winners and losers, with the expectation that they will continue to be so in the near future (at least until our next scheduled rebalance date).

At each rebalance date t_k:

1. Compute M_{i,t_k} for every ticker i from the investable universe constructed at this date.
2. Rank tickers by M.
3. **Long the top** k tickers (highest momentum) with shares $+Q$ each.
4. **Short the bottom** k tickers (lowest momentum) with shares $-Q$ each.

The number k (default 10) and per-position size Q (default 100 shares) are user-specified.

In practice, the position size Q is reduced if opening the corresponding long or short positions would exceed:

- the maximum margin borrowing capacity available for long positions, or
- the maximum permitted short exposure for short positions,

as determined by the rebalance margin requirements (see §3.4). Specifically, the reduction is done under Variant B, where we shrink the intended positions as little as possible until the intended portfolio becomes feasible (see §3.4.2, §3.4.3).

Sign filter: We additionally require positive momentum on the long side and negative momentum on the short side. If fewer than k tickers have $M > 0$, only those with $M > 0$ are bought (and similarly for the short side). This prevents the strategy from going long names with $M < 0$ when the entire universe has trended down, or going short with $M > 0$ when the entire universe has trended up.

7.3 Per-position size and gross exposure

The strategy uses a **fixed share count per position** (not a fixed dollar amount). The dollar value of each leg therefore varies with the prices of the selected names. With $Q = 100$ shares and S&P 500 prices in the \$50˘\$500 range, each long position is roughly $\$5K$˘$\$50K$, and the long book is roughly k times that. Recall that, per the rebalance margin requirement (§3.4), our equity may not allow us to draw a sufficient margin loan to fund the intended long positions. In that case, the intended positions are either uniformly shrunk to feasibility (Variant B) or we post the required extra collateral (Variant A).

Why fixed shares rather than fixed dollars: Fixed-share position sizing makes per-trade volume predictable. The downside is that gross exposure varies as a function of the selected names' prices (a basket of high-priced stocks produces more leverage than a basket of low-priced stocks).

7.4 Static vs dynamic mode

In **static (fixed-calendar) mode**, momentum can be precomputed for all scheduled rebalance dates in one batch before running the backtest. In **dynamic mode**, momentum at a dynamic-rebalance date t is computed on the fly for that date's universe of tickers.

8 Strategy 2: Top-k/Bottom-k Selection From A Cross-Sectional Factors Model Signal

8.1 Idea

Strategy 2 uses the same long-top-k / short-bottom-k selection rule as the momentum strategy (§7.2), but ranks tickers by a **predicted next-period return** rather than by the momentum signal of §7.1. The prediction is the output of a rolling cross-sectional linear regression of realized period returns on factor values. Where momentum uses one signal directly, the factors model lets the data decide how to weight (linearly) several signals to best fit the **realized returns of the past period(s)**, and re-estimates those weights at every rebalance. **Notation:** Recall from §4.1 that rebalance dates are $t_0 < t_1 < \cdots < t_P$. At date t_k the investable universe contains n_{t_k} tickers (below we write n instead of n_{t_k} when the date is clear from context). For the inter-rebalance period $[t_k, t_{k+1}]$, each ticker i has a realized simple return equal to

$$R_i(t_k, t_{k+1}) \;=\; \frac{p_{i,t_{k+1}} - p_{i,t_k}}{p_{i,t_k}}$$

computed on close prices. This is the regression target (at date t_k, this is a random variable).

8.2 Factors

$p = 2$ factors are computed at each rebalance date t_k for every ticker in the candidate universe.

Factor 1: Momentum M_{i,t_k}, the buffered log-return from §7.1:

$$M_{i,t_k} \;=\; \ln \frac{p_{i,\,t_k - b_{mom}}}{p_{i,\,t_k-(b_{mom}+W_{mom})}}$$

with rolling window $W_{mom} = 92$ trading days (≈ 4 trading months) and buffer $b_{mom} = 23$ trading days (≈ 1 month); all offsets are in *trading days*. The buffer skips the most recent month, which empirically exhibits short-term reversal that contaminates the medium-term momentum signal.

Factor 2: Volatility V_{i,t_k}, the annualized standard deviation of **daily log returns** over the trailing W_{vol} trading days of returns (default $W_{vol} = 60$). Writing the daily log return as $r_{i,\tau} = \ln \frac{p_{i,\tau}}{p_{i,\tau-1}}$ for consecutive trading days τ:

$$V_{i,t_k} \;=\; \sqrt{252}\,\sqrt{\frac{1}{W_{vol}-1}\sum_{\tau=t_k-W_{vol}+1}^{t_k}\left(r_{i,\tau}-\bar{r}_{i,t_k}\right)^2}, \qquad \bar{r}_{i,t_k} \;=\; \frac{1}{W_{vol}}\sum_{\tau=t_k-W_{vol}+1}^{t_k} r_{i,\tau}$$

The sum ranges over W_{vol} log returns; forming them requires the $W_{vol}+1$ prices $p_{i,t_k-W_{vol}}, \ldots, p_{i,t_k}$.

We use log returns instead of simple returns because they are additive across time. The sum of consecutive daily log returns equals the log return for the period:

$$\sum_{\tau=t_k-W_{vol}+1}^{t_k} r_{i,\tau} \;=\; \sum_{\tau=t_k-W_{vol}+1}^{t_k}\left(\ln p_{i,\tau}-\ln p_{i,\tau-1}\right) \;=\; \ln\frac{p_{i,t_k}}{p_{i,t_k-W_{vol}}}$$

Under the assumption that daily log returns are uncorrelated with common (finite) variance σ^2, variance scales linearly with time:

$$\mathrm{Var}\left(\sum_{\tau=1}^{T} r_{i,\tau}\right) = T\sigma^2$$

Therefore volatility (the standard deviation) scales with the square root of time, giving annualized volatility

$$\sigma_{\text{annual}} = \sqrt{252}\,\sigma_{\text{daily}}$$

where we have used $T = 252$ trading days in one year.

Remark

The estimator uses $1/(W_{vol}-1)$ so that, under the assumption that daily log returns are uncorrelated with common mean and common (finite) daily variance σ^2, the estimator

$$S^2 := \frac{1}{W_{vol}-1}\sum_{\tau=t_k-W_{vol}+1}^{t_k}\left(r_{i,\tau}-\bar{r}_{i,t_k}\right)^2$$

is an unbiased estimator of σ^2. Note, however, that $\mathbb{E}[\sqrt{S^2}] < \sqrt{\mathbb{E}[S^2]} = \sigma$ (Jensen's inequality, applicable because $\sqrt{\cdot}$ is concave), therefore V_{i,t_k} is **not** an unbiased estimator of annualized volatility (it carries a downward bias).

In order of increasing sophistication, more refined volatility estimates would: use a longer window to reduce estimation variance so a single outlier cannot dominate (though under time-varying volatility this trades estimation variance for staleness bias); use EWMA so older observations decay smoothly (used for expected-return estimation, §9.7.1.2); or model returns explicitly (GARCH to capture volatility clustering and persistence, or jump-diffusion to separate normal (diffusive) volatility from the jump component). These are noted as possible extensions.

8.3 Cross-sectional preprocessing

Both factors are transformed before entering the regression. Every step is computed per rebalance date.

1. **Winsorization:** At each rebalance date independently, each factor is clipped to its cross-sectional $[\,1^{\text{st}}, 99^{\text{th}}\,]$ percentile range. This caps the influence of outliers (e.g. a stock fresh off a massive single-day move) without dropping the observation.

2. **Z-scoring:** At each rebalance date, each winsorized factor is standardized to have cross-sectional mean 0 and standard deviation 1. Specifically, we subtract the cross-sectional mean from the factor and divide by the cross-sectional standard deviation.

Z-scoring puts M and V on a common scale, so the fitted coefficients are directly comparable and stable across rebalance dates. Winsorizing beforehand prevents extreme observations from disproportionately affecting the estimated scale used in the z-scoring step.

From this point on, M_{i,t_k} and V_{i,t_k} denote the processed (winsorized, then z-scored) factor values.

8.4 The single-period regression model

Fix a rebalance date t_j. Let n_{t_j} denote the number of investable tickers (stocks) at t_j (the "universe" of tickers). We consider the following linear model, explaining the cross-section of realized period returns by the factors observed at the start of that period:

$$R(t_j, t_{j+1}) \;=\; \mathbf{1}\,\alpha_{t_j} \;+\; b_{t_j,M}\,M_{\cdot,t_j} \;+\; b_{t_j,V}\,V_{\cdot,t_j} \;+\; \varepsilon_{t_j}$$

i.e.:

$$R(t_j, t_{j+1}) \;=:\; X_{t_j}\,\beta_{t_j} \;+\; \varepsilon_{t_j}$$

where:

- $R(t_j, t_{j+1}) \in \mathbb{R}^{n_{t_j}}$ is the random vector of realized period returns;
- $X_{t_j} = \left[\,\mathbf{1}_n \mid M_{\cdot,t_j} \mid V_{\cdot,t_j}\,\right] \in \mathbb{R}^{n_{t_j}\times(p+1)}$ is the design matrix of the regression: a column of ones (intercept) prepended to the factor columns;
- $\beta_{t_j} = (\alpha_{t_j},\, b_{t_j,M},\, b_{t_j,V})^\top \in \mathbb{R}^{p+1}$ is the coefficient vector, with intercept $\beta_{t_j,0} = \alpha_{t_j}$ and the factor loadings $\beta_{t_j,1} = b_{t_j,M}, \quad \beta_{t_j,2} = b_{t_j,V}$;
- $\varepsilon_{t_j} \in \mathbb{R}^{n_{t_j}}$ is the random residual vector.

Here $p = 2$, so $X_{t_j} \in \mathbb{R}^{n_{t_j}\times 3}$ and $\beta_{t_j} \in \mathbb{R}^3$; the formulation holds for general p, where we use p factors.

Working hypotheses $(\mathcal{H})$:

$$(\mathcal{H}) \begin{cases} (H_1) : \operatorname{rank}(X_{t_j}) = p+1 \\ (H_2) : \varepsilon_{t_j,i} \text{ are i.i.d. with } \mathbb{E}[\varepsilon_{t_j,i} \mid X_{t_j}] = 0,\ \operatorname{Var}(\varepsilon_{t_j,i} \mid X_{t_j}) = \sigma^2. \end{cases}$$

(H_1) holds whenever the two processed factors are not perfectly collinear in the cross-section. (H_2) is an idealization: equity residuals are cross-sectionally correlated (common market and industry shocks) and heteroskedastic, so the variance and interval results below should be read as idealized diagnostics, not exact diagnostics. Point estimation (§8.5) and prediction (§8.8) do not require (H_2).

8.5 OLS estimation

The OLS estimator is the least-squares minimizer

$$\hat{\beta}_{t_j} \;:=\; \arg\min_{\beta \in \mathbb{R}^{p+1}} \left\| R(t_j, t_{j+1}) - X_{t_j}\,\beta \right\|^2$$

which, under (H_1), has the closed-form solution

$$\hat{\beta}_{t_j} \;=\; \left(X_{t_j}^\top X_{t_j}\right)^{-1} X_{t_j}^\top R(t_j, t_{j+1})$$

Under $(\mathcal{H})$ the estimator is **unbiased**, $\mathbb{E}[\hat{\beta}_{t_j}] = \beta_{t_j}$, with covariance

$$\operatorname{Cov}(\hat{\beta}_{t_j}) \;=\; \sigma^2 \left(X_{t_j}^\top X_{t_j}\right)^{-1}$$

Interval diagnostics

Adding the Gaussian assumption $\varepsilon_{t_j} \sim \mathcal{N}(0, \sigma^2 I_n)$, a $(1-\alpha)$-level confidence interval for $\beta_{t_j,\ell}$ is

$$\hat{\beta}_{t_j,\ell} \pm t_\alpha\, \hat{\sigma}\, \sqrt{[(X_{t_j}^\top X_{t_j})^{-1}]_{\ell\ell}},$$

with $\hat{\sigma}^2 = \|R - X_{t_j}\hat{\beta}_{t_j}\|^2/(n-(p+1))$, and where $t_\alpha := t_{n-(p+1)}(1-\alpha/2)$ is the quantile of order $1-\alpha/2$ of a Student distribution with $n-(p+1)$ degrees of freedom. This is reported only as a rough diagnostic, subject to the (H_2) caveat above.

8.6 Rolling-window estimation (the model actually used)

A single cross-section (n rows, $p+1$ parameters) may yield noisy coefficients. We can instead model β as **constant over a rolling window of** $W := W_{\textbf{factor}}$ **consecutive inter-rebalance periods** (*e.g.* $W = 6$) and pool the observations.

At rebalance date t_k, the training set is every (ticker, date) pair from the W most recently **completed** periods: factor dates $t_{k-W}, \ldots, t_{k-1}$ paired with their realized returns $R(t_{k-W}, t_{k-W+1}), \ldots, R(t_{k-1}, t_k)$. We index the estimator built from this window by its **last factor date**, t_{k-1}.

All of these returns are known by t_k, so there is no look-ahead. Stacking the cross-sections:

$$\underbrace{\begin{pmatrix} R(t_{k-1},t_k) \\ R(t_{k-2},t_{k-1}) \\ \vdots \\ R(t_{k-W},t_{k-W+1}) \end{pmatrix}}_{\mathcal{R}_{t_{k-1}}} = \underbrace{\begin{pmatrix} X_{t_{k-1}} \\ X_{t_{k-2}} \\ \vdots \\ X_{t_{k-W}} \end{pmatrix}}_{\mathcal{X}_{t_{k-1}}} \beta^{(W)}_{t_{k-1}} + \tilde{\varepsilon}_{t_{k-1}}$$

with $\mathcal{R}_{t_{k-1}} \in \mathbb{R}^N$, $\mathcal{X}_{t_{k-1}} \in \mathbb{R}^{N\times(p+1)}$, $\tilde{\varepsilon}_{t_{k-1}} \in \mathbb{R}^N$, where $N = \sum_{j=k-W}^{k-1} n_{t_j}$ is the total number of pooled (ticker, date) rows. Under hypothesis $(\mathcal{H})$, the OLS estimator of $\beta^{(W)}_{t_{k-1}}$ is

$$\hat{\beta}^{(W)}_{t_{k-1}} = \left(\mathcal{X}^\top_{t_{k-1}} \mathcal{X}_{t_{k-1}}\right)^{-1} \mathcal{X}^\top_{t_{k-1}} \mathcal{R}_{t_{k-1}}$$

The single-period model of §8.4–8.5 is the special case $W = 1$ (the window reduces to $\{t_{k-1}\}$ and $\hat{\beta}^{(1)}_{t_{k-1}} = \hat{\beta}_{t_{k-1}}$).

Relation to Fama–MacBeth

Stacking the cross-sections into a single OLS fit is an alternative to the Fama–MacBeth (1973) procedure, which runs a separate cross-sectional regression *per period* and takes the equal-weighted time-series average of the resulting slopes.

8.7 In-sample fitting

For any $j \in \{\, k-W, \ldots, k-1 \,\}$, the in-sample predictions on the training window are

$$\hat{R}(t_j, t_{j+1}) = X_{t_j} \hat{\beta}^{(W)}_{t_{k-1}} = \hat{\alpha}_{t_{k-1}} + \hat{b}_{M,t_{k-1}} M_{\cdot,t_j} + \hat{b}_{V,t_{k-1}} V_{\cdot,t_j}$$

8.8 Out-of-sample prediction and the ranking signal

Assuming the coefficient vector is stable from the training window into the upcoming period ($\beta^{(W)}_{t_k} \approx \beta^{(W)}_{t_{k-1}}$), at t_k we apply the fitted coefficients $\hat{\beta}^{(W)}_{t_{k-1}}$ to the **current** cross-section X_{t_k} (which is *not* in the training set) to forecast the next period's return:

$$\hat{R}(t_k, t_{k+1}) = X_{t_k} \hat{\beta}^{(W)}_{t_{k-1}} = \hat{\alpha}_{t_{k-1}} + \hat{b}_{M,t_{k-1}} M_{\cdot,t_k} + \hat{b}_{V,t_{k-1}} V_{\cdot,t_k}$$

This vector $\hat{R}(t_k, t_{k+1})$ is the **ranking signal** for the selection rule of §8.9. It may also serve as the expected-return input $\hat{\mu}_{t_k}$ to the portfolio optimizer of §9 (an alternative to the rolling historical means).

When the prediction is used for ranking, out-of-sample **rank skill** (§8.11), not in-sample fit quality, is the metric that matters most.

8.9 Selection rule

Identical to the momentum strategy (§7.2), with the predicted return $\hat{R}$ replacing M as the ranking variable. At each rebalance date t_k:

1. Compute the predictions $\hat{R}_i(t_k, t_{k+1})$ for every ticker i in the universe.
2. Rank tickers by $\hat{R}$.
3. **Long the top** m tickers ($+Q$ shares each).
4. **Short the bottom** m tickers ($-Q$ shares each).

Defaults are $m = 10$ and $Q = 100$ shares. The §7.2 sign filter carries over: long only names with $\hat{R} > 0$, short only names with $\hat{R} < 0$.

8.10 Multicollinearity and Ridge regression

(Subscripts are dropped for readability; $\mathcal{X}$ is the stacked design matrix, and $A := \mathcal{X}^\top \mathcal{X}$. All vector norms below are the Euclidean norm $\|v\|_2 = (v^\top v)^{1/2}$, and all matrix norms are the induced operator norm $\|M\|_2 = \sup_{v \neq 0} \frac{\|Mv\|_2}{\|v\|_2}$).

If the factors are linearly dependent, $\mathcal{X}$ is rank-deficient and the OLS estimator is not uniquely defined. Even when A has full rank, a **near-zero eigenvalue** inflates the estimator's variance. Indeed, since A is symmetric positive semidefinite (and positive definite under (H_1)) it admits the eigendecomposition $A = Q \Lambda Q^\top$ with $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_{p+1})$, $\lambda_\ell > 0$, so $A^{-1} = Q \Lambda^{-1} Q^\top$ and

$$\mathrm{Var}(\hat{\beta}_j) \;=\; \sigma^2 \, [A^{-1}]_{jj} \;=\; \sigma^2 \sum_{\ell=1}^{p+1} \frac{q_{j\ell}^2}{\lambda_\ell}$$

which blows up as some $\lambda_\ell \to 0$. The stability of the estimator is summarized by the **condition number**

$$\kappa(A) \;:=\; \|A\| \, \|A^{-1}\| \;=\; \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$$

(the second equality derived using that A is symmetric positive definite).

As shown above (§8.6), $\hat{\beta}$ solves $A\hat{\beta} = b$ with $b := \mathcal{X}^\top \mathcal{R}$. A large $\kappa(A)$ signals that small perturbations of b can produce large changes in $\hat{\beta}$. Indeed, perturbing $b \mapsto b + \Delta b$ shifts the solution by $\Delta\hat{\beta} = A^{-1} \Delta b$, so

$$\|\Delta\hat{\beta}\| \;\leq\; \|A^{-1}\| \, \|\Delta b\| \qquad \text{and} \qquad \|b\| \;=\; \|A\hat{\beta}\| \;\leq\; \|A\| \, \|\hat{\beta}\|$$

which combine to give the standard perturbation bound

$$\frac{\|\Delta\hat{\beta}\|}{\|\hat{\beta}\|} \;\leq\; \kappa(A) \, \frac{\|\Delta b\|}{\|b\|}. \tag{$\star$}$$

(This bound is realized: there exist directions of Δb for which it is attained.)

For example, if $\kappa(A) = 100$, a 1% relative perturbation of b can, in the worst case, shift $\hat{\beta}$ by as much as 100% in relative norm. In that case, $\hat{\beta}$ is potentially far too sensitive to the response $\mathcal{R}$ to be trustworthy ($b := \mathcal{X}^\top \mathcal{R}$).

We report κ per date; see `condition_numbers` in §8.11.

Ridge regression is used when the factors are linearly dependent, or also when A has near-zero eigenvalues (which inflate the estimator's variance):

$$\hat{\beta}^{\mathrm{ridge}} \;:=\; \arg\min_{\beta \in \mathbb{R}^{p+1}} \left\{ \|\mathcal{R} - \mathcal{X}\beta\|^2 + \lambda_{\mathrm{ridge}} \|\beta\|^2 \right\} \;=\; \left(A + \lambda_{\mathrm{ridge}} I\right)^{-1} \mathcal{X}^\top \mathcal{R}, \qquad \lambda_{\mathrm{ridge}} > 0$$

Since $A + \lambda_{\text{ridge}} I$ has eigenvalues $\lambda_\ell + \lambda_{\text{ridge}} > 0,\ l \in \{1, \ldots, p+1\}$, **it is always invertible**.

Ridge trades bias for variance:

$$\mathbb{E}[\hat{\beta}^{\text{ridge}}] \;=\; \beta \;-\; \lambda_{\text{ridge}}\,(A + \lambda_{\text{ridge}} I)^{-1}\beta$$

$$\operatorname{Var}(\hat{\beta}_j^{\text{ridge}}) \;=\; \sigma^2 \sum_{\ell=1}^{p+1} \frac{\lambda_\ell\, q_{j\ell}^2}{(\lambda_\ell + \lambda_{\text{ridge}})^2} \;<\; \sigma^2 \sum_{\ell=1}^{p+1} \frac{q_{j\ell}^2}{\lambda_\ell} \;=\; \operatorname{Var}(\hat{\beta}_j^{\text{OLS}})$$

the strict inequality holding for every $\lambda_{\text{ridge}} > 0$ because $\lambda_\ell/(\lambda_\ell + \lambda_{\text{ridge}})^2 < 1/\lambda_\ell$. With the mean-squared-error decomposition

$$\operatorname{MSE}(\hat{\beta}_j) \;=\; \mathbb{E}\big[(\hat{\beta}_j - \beta_j)^2\big] \;=\; \big(\mathbb{E}[\hat{\beta}_j] - \beta_j\big)^2 + \operatorname{Var}(\hat{\beta}_j)$$

Ridge improves MSE whenever the variance it saves outweighs the bias it introduces: typically the case under multicollinearity or ill-conditioning, but not guaranteed: it depends on the true β and on the choice of λ_{ridge}.

Conclusion

OLS and Ridge with a small penalty (e.g. $\lambda_{\text{ridge}} \approx 0.01$) produce similar predictions. Ridge is retained as the extension should more (and possibly linearly-dependent) factors be added later.

8.11 Model fit evaluation metrics and reported diagnostics

In this subsection, we keep the notation $R_i(t_j, t_{j+1})$ to refer to the observed realized values of the random variable $R_i(t_j, t_{j+1})$, and $\hat{R}_i(t_j, t_{j+1})$ to refer to the observed realized values of the predictor $\hat{R}_i(t_j, t_{j+1})$.

We define $\bar{R}(t_j, t_{j+1}) := n_{t_j}^{-1} \sum_{i=1}^{n_{t_j}} R_i(t_j, t_{j+1})$, the cross-sectional mean of realized return for the period $(t_j, t_{j+1}]$.

Order of presentation: We report several metrics, presented below in decreasing order of relevance to the strategy's profitability:

1. the **sign hit rate** on the traded names (primary): does each placed bet win money?
2. the **Spearman IC** (secondary): are the bets selected toward the strongest moves?
3. the **R^2 and MSE** level-fit metrics (tertiary): useful mainly to diagnose over-fitting, not to judge profitability directly.

The Pearson IC is reported alongside the Spearman IC as a level-correlation companion. We explain the reason for this hierarchy as we introduce each metric.

Sign hit rate is the primary profitability metric: The metric most directly tied to whether the strategy makes money is the *directional accuracy* of the predictions on the tickers actually traded. For a single traded position, the realized P&L is (up to position size) the realized return of the stock times the position direction: $+R_i$ for a long (which we would open if $\hat{R}_i > 0$) and $-R_i$ for a short (which we would open if $\hat{R}_i < 0$). The P&L contribution is therefore positive **when** $\operatorname{sign}(\hat{R}_i) = \operatorname{sign}(R_i)$. Thus a correct sign *is* the event "this position made money," and sign accuracy on the traded tickers is therefore the first-order determinant of profitability.

Sign hit rate on traded tickers

A correlation (Pearson or Spearman) is invariant to an additive shift of the predictions and is therefore *blind* to sign placement: it can be high while most predicted signs are wrong. We therefore measure sign agreement directly.

At rebalance date t_k, let $\mathcal{T}_{t_k} \subseteq \{1, \ldots, n_{t_k}\}$ be the set of *traded* tickers: the top-k (long) and bottom-k (short) tickers selected by $\hat{R}(t_k, t_{k+1})$, subject to the §8.9 sign filter (only $\hat{R}_i > 0$ eligible for the long bucket, $\hat{R}_i < 0$ for the short bucket). The **sign hit rate** is the fraction of traded tickers whose predicted and realized returns share the same sign:

$$\mathrm{HR}(t_k) \;=\; \frac{1}{|\mathcal{T}_{t_k}|} \sum_{i \in \mathcal{T}_{t_k}} \mathbf{1}\Big\{\mathrm{sign}\big(\hat{R}_i(t_k, t_{k+1})\big) = \mathrm{sign}\big(R_i(t_k, t_{k+1})\big)\Big\}$$

where $\mathrm{sign}(x) = +1$ if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$. By construction every traded ticker has $\hat{R}_i \neq 0$, so a hit requires the long tickers ($\hat{R}_i > 0$) to realize $R_i > 0$ and the short tickers ($\hat{R}_i < 0$) to realize $R_i < 0$. Realized returns exactly equal to zero count as misses. We also report the hit rate for the long ($\mathrm{HR}^+(t_k)$) and short ($\mathrm{HR}^-(t_k)$) sides separately, so that asymmetry between the two books is visible: the same formula applies with $\mathcal{T}_{t_k}$ replaced by $\mathcal{T}^+_{t_k}$ (tickers traded long) for the long bucket, and by $\mathcal{T}^-_{t_k}$ (tickers traded short) for the short bucket.

We have

$$\mathrm{HR}(t_k) \in [0, 1],$$

with $\mathrm{HR} = 1$ meaning every traded bet was directionally correct. The **no-information baseline** is 0.5: if predicted signs were assigned by fair coin flips (say positive for heads, negative for tails), then for every traded ticker i, fixing $s_i := \mathrm{sign}(R_i) \in \{+1, -1\}$ (assuming $P(R_i = 0) = 0$),

$$\mathbb{E}\big[\mathbf{1}\{\hat{s}_i = s_i\}\big] \;=\; P(\hat{s}_i = s_i) \;=\; \tfrac{1}{2}.$$

and thus by linearity of expectation $\mathbb{E}[\mathrm{HR}(t_k)] = \frac{1}{2}$.

Ranking is the secondary determinant of profitability. In the event that the predicted signs of the traded tickers are correct, total profit is larger when the largest predicted returns line up with the largest realized returns. Conditional on the traded set (determined by the top-k/bottom-k selection rule), correct ordering of the predictions against the realized returns *concentrates* exposure on the strongest moves, amplifying the profit of correctly-signed bets. Ranking with wrong signs still loses money, so ordering is relevant only on top of correct direction. We measure ordering skill with the Spearman IC (defined below via the Pearson IC), and read it as a secondary diagnostic; a window-averaged Spearman IC above 0 suggests out-of-sample ranking skill.

Pearson Information Coefficient (Pearson IC): the empirical correlation between the **out-of-sample** predicted and realized next-period returns. Specifically, at rebalance date t_k, treat each ticker $i = 1, \ldots, n_{t_k}$ as contributing a pair $(\hat{R}_i,\, R_i)$, with $\hat{R}_i := \hat{R}_i(t_k, t_{k+1})$ and $R_i := R_i(t_k, t_{k+1})$, and we define

$$\mathrm{IC}_{\mathrm{Pearson}}(t_k) \;=\; \widehat{\mathrm{corr}}(\hat{R}, R) \;=\; \frac{\displaystyle\sum_{i=1}^{n_{t_k}} (\hat{R}_i - \bar{\hat{R}})(R_i - \bar{R})}{\sqrt{\displaystyle\sum_{i=1}^{n_{t_k}} (\hat{R}_i - \bar{\hat{R}})^2}\;\sqrt{\displaystyle\sum_{i=1}^{n_{t_k}} (R_i - \bar{R})^2}}$$

where $\bar{\hat{R}} = n_{t_k}^{-1} \sum_{i=1}^{n_{t_k}} \hat{R}_i$ is the cross-sectional mean of predicted returns.

Spearman Information Coefficient (Spearman IC): the Spearman IC is the Pearson correlation (defined above) applied to the *ranks* of the predicted and realized returns rather than to their raw values. Specifically, let $y_1, \ldots, y_n$ be a vector of values (here either the predictions $\hat{R}_i$ or the realized returns R_i for $i = 1, \ldots, n_{t_k}$). Let $r_1 < r_2 < \cdots < r_K$ be the *distinct* values occurring among the y_i, so that $K \le n$. For each i there is a unique $j \in \{1, \ldots, K\}$ with $y_i = r_j$, and we define

$$\mathrm{rank}(y_i) \;:=\; j.$$

Equal values thus share a rank, and the next strictly larger value takes the immediately following integer (*dense ranking*). We write $\mathrm{rank}(\hat{R}), \mathrm{rank}(R) \in \mathbb{R}^{n_{t_k}}$ for the resulting rank vectors.

Then, treating each ticker $i = 1, \ldots, n_{t_k}$ as contributing the pair $\big(\mathrm{rank}(\hat{R}_i),\, \mathrm{rank}(R_i)\big)$, we define the Spearman IC as

$$\mathrm{IC}_{\mathrm{Spearman}}(t_k) \;=\; \widehat{\mathrm{corr}}\big(\mathrm{rank}(\hat{R}),\, \mathrm{rank}(R)\big).$$

Geometric interpretation of the sign

The sign of the Spearman IC equals the sign of the slope of the least-squares line fitting the rank pairs. Writing the rank pairs as $(x_i, z_i) := \big(\mathrm{rank}(\hat{R}_i), \mathrm{rank}(R_i)\big)$ and

$$\mathrm{sse}(a, b) \;=\; \sum_{i=1}^{n_{t_k}} \big(z_i - (a + b\, x_i)\big)^2,$$

the minimizer $(a^\star, b^\star)$ has slope

$$b^\star \;=\; \frac{\widehat{\mathrm{cov}}(x, z)}{\widehat{\mathrm{Var}}(x)} \;=\; \widehat{\mathrm{corr}}(x, z)\, \frac{\widehat{\sigma}(z)}{\widehat{\sigma}(x)}.$$

Since the ratio $\widehat{\sigma}(z)/\widehat{\sigma}(x)$ is strictly positive whenever both rank vectors are non-constant, $b^\star$ and $\widehat{\mathrm{corr}}(x, z)$ share the same sign. The correlation (hence the IC) is undefined only in the degenerate case where all predictions, or all realized returns, are equal (so that one rank vector is constant).

Level-fit metrics (R^2 and MSE): The remaining metrics measure how well the predictions reproduce the *levels* of realized returns. They do not map directly to profitability (a model can fit levels poorly yet still sign and rank the traded tickers correctly), and we report them mainly as *over-fitting diagnostics*: a positive in-sample R^2 alongside a negative out-of-sample R^2 signals a model fitting in-sample noise that does not generalize.

Cross-sectional R^2 : in-sample over the training window, out-of-sample at the realized next period:

$$R^2_{\text{in}} = 1 - \frac{\displaystyle\sum_{j=k-W}^{k-1} \sum_{i=1}^{n_{t_j}} \big(R_i(t_j, t_{j+1}) - \hat{R}_i(t_j, t_{j+1})\big)^2}{\displaystyle\sum_{j=k-W}^{k-1} \sum_{i=1}^{n_{t_j}} \big(R_i(t_j, t_{j+1}) - \bar{R}(t_j, t_{j+1})\big)^2}$$

$$R^2_{\text{oos}} = 1 - \frac{\displaystyle\sum_{i=1}^{n_{t_k}} \big(R_i(t_k, t_{k+1}) - \hat{R}_i(t_k, t_{k+1})\big)^2}{\displaystyle\sum_{i=1}^{n_{t_k}} \big(R_i(t_k, t_{k+1}) - \bar{R}(t_k, t_{k+1})\big)^2}$$

Interpretation of the value of R^2

In general, for a set of observed values $(y_i)_i$ with empirical mean $\bar{y}$, and a model that predicts the corresponding values $(\hat{y}_i)_i$, we define the quantity R^2 as

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

The denominator is fixed by the data, so R^2 grows as the residual sum of squares in the numerator shrinks. A value of 1 corresponds to a perfect fit: every prediction equals its observed value ($\hat{y}_i = y_i$ for all i), making the numerator zero.

Case of an OLS linear model fit with an intercept

When the predictor is the OLS predictor of a linear model **with an intercept** (our case in §8.4 and §8.6), we can show, by construction of the OLS predictor as an orthogonal projection onto the vector subspace generated by the columns of the design matrix, that the two following properties hold:

1. **Equal means of fitted values and observations:**

$$\bar{\hat{y}} = \bar{y}.$$

2. **Zero empirical covariance of fitted values and residuals:**

$$\widehat{\mathrm{Cov}}(\hat{y},\, y - \hat{y}) = 0.$$

Now, writing

$$y_i - \bar{y} \;=\; (y_i - \hat{y}_i) \;+\; (\hat{y}_i - \bar{y})$$

and computing the empirical variances using the inner product $\langle x, y \rangle = \sum_i x_i y_i$, and using the two properties above, we get:

$$\widehat{\mathrm{Var}}(y) = \widehat{\mathrm{Var}}(\hat{y}) + \widehat{\mathrm{Var}}(y - \hat{y})$$

and

$$R^2 = \frac{\widehat{\mathrm{Var}}(\hat{y})}{\widehat{\mathrm{Var}}(y)}, \qquad \text{with } R^2 \in [0,1].$$

This is why the (**in-sample**) R^2 is called the "fraction of the variance of the observations explained by the model". Furthermore, the following equality is also verified:

$$R^2 = \widehat{\mathrm{corr}}(y, \hat{y})^2.$$

Outside this OLS-with-intercept setting, these properties may fail. For an arbitrary predictor, or when R^2 is evaluated out of sample, the general definition of R^2 above can produce **negative** values.
This happens when the model does worse (higher sum of squared errors) than the constant baseline $\bar{y}$ (the model predicting the same value $\bar{y}$ for every stock's return), i.e. when

$$\sum_i (y_i - \hat{y}_i)^2 > \sum_i (y_i - \bar{y})^2$$

Error metrics: mean squared error on the in-sample and out-of-sample sets:

$$\mathrm{MSE}_{\mathrm{in}} = \frac{1}{N} \sum_{j=k-W}^{k-1} \sum_{i=1}^{n_{t_j}} \big(R_i(t_j, t_{j+1}) - \hat{R}_i(t_j, t_{j+1})\big)^2, \qquad \mathrm{MSE}_{\mathrm{oos}} = \frac{1}{n_{t_k}} \sum_{i=1}^{n_{t_k}} \big(R_i(t_k, t_{k+1}) - \hat{R}_i(t_k, t_{k+1})\big)^2$$

where $N = \sum_{j=k-W}^{k-1} n_{t_j}$ is the total number of pooled (ticker, date) rows (§8.6).

Implementation reference

In our implementation, the fit–predict routine returns a dataclass `FactorsModelResult` with five fields:

- `predictions`: the predicted returns $\hat{R}$ (the signal used for selection) at every rebalance date.
- `coefs`: the regression coefficients ("factors loadings") $(\hat{\alpha}, \hat{b}_M, \hat{b}_V)$ at every rebalance date.
- `in_sample_metrics`: MSE and R^2 on the pooled training set, for every rebalance date.
- `out_sample_metrics`: MSE, R^2, Spearman IC, Pearson IC, and the sign hit rates (overall, long side, and short side) between predicted and realized period returns for the next period, at every rebalance date.
- `condition_numbers`: the condition number $\kappa(A)$ (see §8.10) for the regression, at every rebalance date.

On choosing W: The helper `evaluate_rolling_periods_by_hit_rate(...)` from the module `returns_prediction_model.py` sweeps a grid of candidate training windows W and reports, for each one, the average out-of-sample sign hit rate on the traded tickers (top-k/bottom-k, §8.9) across a set of rebalance dates. Because the sign hit rate is the primary profitability proxy (§above), we may select the window that maximizes this average for use in our strategy. The notebook `factors_model.ipynb` runs this sweep.

9 Strategy 3: Mean-Variance Portfolio Optimisation

Mean-variance portfolio optimisation (MVO) selects, at each rebalance date, the portfolio that optimally balances expected return against return variance, subject to constraints on market-beta exposure, net dollar exposure, and gross dollar exposure. We implement two formulations of the same trade-off:

- a **return-maximising** formulation (`max_return`): maximise expected return subject to a cap on portfolio variance;
- a **variance-minimising** formulation (`min_variance`): minimise portfolio variance subject to a floor on expected return.

Both share the same constraint set; they differ only in which quantity is the objective and which is the constraint. The choice between them is a matter of which quantity the user prefers to specify (a volatility cap, or a return floor).

The optimiser operates in **weight space**. We therefore proceed in three logical stages, and the section is organised accordingly:

1. **§9.1** expresses the portfolio's return, expected return, and variance as functions of the vector of *equity weights* w (the change of variables from share holdings q to weights w).
2. **§9.2–§9.5** set up the optimisation problem in w, establish that it is well-posed, and solve it for the optimal weights $w^\star$.
3. **§9.6** maps $w^\star$ back to integer share counts $q^\star$ (the inverse of the change of variables of §9.1).

Finally, **§9.7** details how the optimiser's inputs (μ, Σ, β) are estimated.

9.1 Equity return, excess return, and equity weights

At any rebalance date t_k we rebalance to new share positions q_{t_k}, intending to hold them constant until the next rebalance date t_{k+1}.

For the purpose of formulating the optimisation problem, we make the following simplifying assumptions:

1. **Positions are constant (no trading)** throughout $[t_k, t_{k+1}]$.
2. **No posting collateral** to cure MR violations.
3. **No financing P&L** (interest income and expense are omitted).
4. **No dividends.**

We see that assumptions 1 and 2 are not realistic, since margin requirements cannot be ignored in practice, and thus, in the event of violation, we would be required to either deleverage our positions (Variant B) or posting margin collateral (Variant A), therebyu. Assumptions 3 and 4 may be reasonably close to reality over a monthly horizon, or over shorter rebalancing intervals, depending on the dividends paid/received based on the shares held in the portfolio and on the rates and position bases used to determine financing P&L (§2.6.1). Nevertheless, the discussion that follows should be regarded as a simplification rather than a fully accurate representation of reality.

Equity change

Under assumptions 1-4, the only source of P&L over the period is price movement (§2.6.6), so

$$\text{Equity}_{t_{k+1}} - \text{Equity}_{t_k} = \sum_{i \in \Omega_{t_k}} q_{t_k,i} \left(p_{t_{k+1},i} - p_{t_k,i} \right)$$

where Ω_{t_k} is the investable universe at t_k and $q_{t_k,i} > 0$ (< 0) for long (short) positions.

Equity weights

For every ticker $i \in \Omega_{t_k}$, define the **equity weight**

$$w_{t_k,i} := \frac{q_{t_k,i}\, p_{t_k,i}}{\text{Equity}_{t_k}}$$

Then $w_{t_k,i} > 0$ for long positions and $w_{t_k,i} < 0$ for short positions, and $\mathbf{1}^\top w_{t_k}$ is the **net dollar exposure as a fraction of equity**.

Equity return as a weighted average of stock returns

Let

$$R_i(t_k, t_{k+1}) := \frac{p_{t_{k+1},i} - p_{t_k,i}}{p_{t_k,i}}$$

be the simple return of ticker i over $[t_k, t_{k+1}]$. Dividing the equity-change identity by Equity_{t_k} and inserting the definition of $w_{t_k,i}$:

$$R^{\mathrm{Equity}}(t_k, t_{k+1}) = \sum_{i \in \Omega_{t_k}} \frac{q_{t_k,i}\, p_{t_k,i}}{\mathrm{Equity}_{t_k}} \cdot \frac{p_{t_{k+1},i} - p_{t_k,i}}{p_{t_k,i}} = \sum_{i \in \Omega_{t_k}} w_{t_k,i}\, R_i(t_k, t_{k+1})$$

that is,

$$\boxed{R^{\mathrm{Equity}}(t_k, t_{k+1}) = w_{t_k}^{\top} R(t_k, t_{k+1})}$$

Therefore, under the above assumptions 1-4, the equity return is equal to the weighted average of constituent returns, with the weights defined above. This identity is what justifies the change of variables from shares $q_{t_k,i}$ to weights $w_{t_k,i}$: the optimiser need only choose the weights $w_{t_k,i}$.

Expected return and variance

Let

$$\mu_{t_k} := \mathbb{E}\big[R(t_k, t_{k+1})\big]$$

$$\Sigma_{t_k} := \mathrm{Cov}\big(R(t_k, t_{k+1})\big)$$

Since w_{t_k} is $\mathcal{F}_{t_k}$-measurable (chosen using information available at the rebalance date) and $R^{\mathrm{Equity}} = w_{t_k}^{\top} R$, linearity of expectation and the bilinearity of covariance give immediately

$$\mathbb{E}\big[R^{\mathrm{Equity}}(t_k, t_{k+1})\big] = \mu_{t_k}^{\top} w_{t_k}$$

$$\mathrm{Var}\big(R^{\mathrm{Equity}}(t_k, t_{k+1})\big) = w_{t_k}^{\top} \Sigma_{t_k}\, w_{t_k}$$

Excess return

Let $R^f(t_k, t_{k+1})$ be the simple risk-free return over $[t_k, t_{k+1}]$ (see §11); it is $\mathcal{F}_{t_k}$-measurable. The portfolio's **excess return** is

$$R^{\mathrm{Equity,excess}}(t_k, t_{k+1}) := R^{\mathrm{Equity}}(t_k, t_{k+1}) - R^f(t_k, t_{k+1})$$

$$\mathbb{E}\big[R^{\mathrm{Equity,excess}}\big] = \mu_{t_k}^{\top} w_{t_k} - R^f(t_k, t_{k+1})$$

9.2 The two optimisation formulations

Let $n_{t_k} := |\Omega_{t_k}|$ be the number of investable tickers at t_k. Both formulations solve for an optimal weight vector $w_{t_k} \in \mathbb{R}^{n_{t_k}}$ under the common constraint set C2–C4 (defined in §9.3).

9.2.1 Return-maximising formulation (`max_return`)

Because $R^f(t_k, t_{k+1})$ does not depend on w_{t_k}, maximising expected *excess* return is equivalent to maximising $\mu_{t_k}^\top w_{t_k}$. The problem is:

$$
\begin{aligned}
\max_{w_{t_k} \in \mathbb{R}^{n_{t_k}}} \quad & \mu_{t_k}^\top w_{t_k} \\
\text{s.t.} \quad & w_{t_k}^\top \Sigma_{t_k} w_{t_k} \le \sigma_{\max}^2 && \text{(C1: variance cap)} \\
& |\beta_{t_k}^\top w_{t_k}| \le \varepsilon_\beta && \text{(C2: beta neutrality (near-zero beta cap))} \\
& |\mathbf{1}^\top w_{t_k}| \le \varepsilon_N && \text{(C3: net exposure cap)} \\
& \mathbf{1}^\top |w_{t_k}| \le G_{\max} && \text{(C4: gross exposure cap)}
\end{aligned}
$$

This formulation is **always feasible**: the null portfolio $w = 0$ satisfies every constraint (for $\sigma_{\max}, \varepsilon_\beta, \varepsilon_N, G_{\max} > 0$).

9.2.2 Variance-minimising formulation (`min_variance`)

The dual viewpoint minimises portfolio variance subject to a return lower bound $\mathcal{R}_{\min}$:

$$
\begin{aligned}
\min_{w_{t_k} \in \mathbb{R}^{n_{t_k}}} \quad & w_{t_k}^\top \Sigma_{t_k} w_{t_k} \\
\text{s.t.} \quad & \mu_{t_k}^\top w_{t_k} \ge \mathcal{R}_{\min} && \text{(C1}'\text{: return lower bound)} \\
& |\beta_{t_k}^\top w_{t_k}| \le \varepsilon_\beta && \text{(C2: beta neutrality (near-zero beta cap))} \\
& |\mathbf{1}^\top w_{t_k}| \le \varepsilon_N && \text{(C3: net exposure cap)} \\
& \mathbf{1}^\top |w_{t_k}| \le G_{\max} && \text{(C4: gross exposure cap)}
\end{aligned}
$$

Unlike `max_return`, this formulation **can be infeasible** (if $\mathcal{R}_{\min}$ is set above the maximum return achievable under C2–C4); the recovery procedure is given in §9.4.

9.3 The constraints

The two formulations swap their objective and their first constraint; constraints C2–C4 are identical across both. We describe each below.

9.3.1 Variance cap (C1) and return lower bound (C1′)

$\sigma_{\max}^2$ is the maximum permitted variance of the portfolio's simple return over $[t_k, t_{k+1}]$ (for example $\sigma_{\max} = 20\%$ per inter-rebalance period). $\mathcal{R}_{\min}$ is the minimum acceptable expected return (the lower bound for the return). Each formulation makes one of these quantities a hard bound and optimises the other.

9.3.2 Beta neutrality (C2)

The portfolio beta (estimated in §9.7.3) is constrained to a tolerance ε_β of zero:

$$|\beta_{t_k}^\top w_{t_k}| \leq \varepsilon_\beta$$

This removes most market-direction exposure (at least at the inception of the portfolio at t_k), so that portfolio returns are driven by idiosyncratic stock-price movements rather than by broad index (market) moves. We set $\varepsilon_\beta = 0.01$. The bound is on the absolute value, not an exact equality $\beta_{t_k}^\top w_{t_k} = 0$, as that would confine the portfolio to a hyperplane with empty interior, whereas the small slack $\varepsilon_\beta > 0$ keeps the feasible set with a non-empty interior.

We note, however, that beta neutrality is not guaranteed to persist over the holding period: price drift moves the weights w_t away from their inception values w_{t_k}, so the constraint is enforced only at t_k and may be violated for $t > t_k$.

9.3.3 Net exposure cap (C3)

The net dollar exposure (as a fraction of equity) is bounded by a small tolerance ε_N, targeting an approximately dollar-neutral book:

$$|\mathbf{1}^\top w_{t_k}| \leq \varepsilon_N$$

We set $\varepsilon_N = 0.1\%$ ($\varepsilon_N = 0.001$). As with C2, we write this constraint as an inequality rather than an equality ($\mathbf{1}^\top w_{t_k} = 0$) so that the feasible set has a non-empty interior.

Reason for including this constraint: From the definition of the weights (§9.1), $\mathbf{1}^\top w_{t_k} = 0$ is equivalent to

$$\mathrm{LMV}_{t_k} = \mathrm{SMV}_{t_k},$$

i.e. equal capital deployed long and short. We seek this (at least at inception of the portfolio, since prices subsequently drift between rebalances) for the following reason: if every stock in the book, across the long and short legs, undergoes a common price shock c over $[t_k, t_{k+1}]$, the long-leg gain offsets the short-leg loss and equity is unchanged. Indeed, under the assumptions of §9.1 (price movement as the sole source of P&L), the equity-change identity of §9.1 with $p_{t_{k+1},i} = (1+c)\, p_{t_k,i}$ gives

$$\Delta\mathrm{Equity}_{t_k} \;=\; c\left(\mathrm{LMV}_{t_k} - \mathrm{SMV}_{t_k}\right) \;=\; 0.$$

Without dollar neutrality the same identity gives $\Delta\mathrm{Equity}_{t_k} = c\,(\mathrm{LMV}_{t_k} - \mathrm{SMV}_{t_k}) \neq 0$: a market-wide drop ($c < 0$) produces a loss when the book is net long ($\mathrm{LMV}_{t_k} > \mathrm{SMV}_{t_k}$), and a market-wide rise ($c > 0$) produces a loss when the book is net short ($\mathrm{LMV}_{t_k} < \mathrm{SMV}_{t_k}$).

9.3.4 Gross exposure cap (C4)

The gross exposure (recall its definition from §2.4) is capped at $G_{\max}$:

$$\mathbf{1}^\top |w_{t_k}| \leq G_{\max}$$

This bounds total leverage and prevents the optimiser from taking excessively large positions. $G_{\max}$ is a configurable leverage parameter; for example $G_{\max} = 2$ (200% gross). As shown in §9.4, C4 is also what makes the feasible set bounded, and hence guarantees that an optimum to the constained problem exists.

9.3.5 Optional per-ticker weight box

We may, as an optional additional constraint (offered in the code implementation), set a global floor and cap for the weight of all the tickers of the investable universe of t_k, in order to cap or floor the concentration in each ticker:

$$\forall i \in \Omega_{t_k}, \quad w_{\min} \leq w_{t_k,i} \leq w_{\max}$$

This is complementary to the separate per-position-size cap (expressed in *shares*, not weights) which we apply after the solve (§9.6).

9.4 Discussion of the convexity, feasibility, existence, and uniqueness

Let $\mathcal{F}$ denote the *feasible set* of each formulation, defined as the intersection of its constraint sets: C1–C4 for `max_return` and C1(')–C4 for `min_variance`. Below we write $\mathcal{F}$ for both feasible sets and, and rely on context to disambiguate.

- **Convexity:** Σ_{t_k} is positive semidefinite by construction (§9.7.2). Hence the quadratic form $w \mapsto w^\top \Sigma_{t_k} w$ is convex, so the sublevel set C1 (an ellipsoid) is convex and the `min_variance` objective is convex. C1$'$ is linear. Constraints C2 and C3 are absolute-value constraints, and C4 is an ℓ_1-norm constraint; all three defining convex sets. The feasible set $\mathcal{F}$ is the intersection of these convex sets and is therefore convex. The `max_return` objective is linear. Thus `max_return` is a convex problem (linear objective over a convex set) and `min_variance` is a convex quadratic program (QP).

- **Feasibility:** The null vector $w = 0$ satisfies C2–C4 strictly (for $\varepsilon_\beta, \varepsilon_N, G_{\max} > 0$) and satisfies C1 strictly (for $\sigma_{\max} > 0$). Hence `max_return` is always feasible (i.e. the intersection of the constraint sets is non-empty). For `min_variance`, $w = 0$ gives $\mu_{t_k}^\top w = 0$, which satisfies C1$'$ if and only if $\mathcal{R}_{\min} \leq 0$. For $\mathcal{R}_{\min} > 0$, feasibility depends on whether the gross budget $G_{\max}$ and the expected-return vector μ_{t_k} admit a portfolio reaching $\mathcal{R}_{\min}$ within C2–C4.

  Feasibility recovery for `min_variance` (lower bound halving)

  When a user-specified $\mathcal{R}_{\min}$ is infeasible, we retry the solve with $\mathcal{R}_{\min}/2$, then $\mathcal{R}_{\min}/4$, and so on, down to a configurable floor (default 0). The lower bound $\mathcal{R}_{\min} = 0$ is always feasible, since $w = 0$ satisfies $\mu_{t_k}^\top w = 0 \geq 0$ together with C2–C4. The return lower bound actually used and the number of halving steps are logged (by the code implementation) for diagnostics.

- **Existence:** The gross constraint C4 bounds $\mathcal{F}$ (since $\|w\|_1 \leq G_{\max}$ on $\mathcal{F}$); each constraint set is closed; therefore $\mathcal{F}$ is compact. When $\mathcal{F}$ is non-empty, the Weierstrass theorem guarantees that the continuous objective (linear for `max_return`, quadratic for `min_variance`) attains its optimum on $\mathcal{F}$.

- **Uniqueness:**
  - *`max_return`:* The maximum of a linear function over a convex set is attained on the boundary but is **generally not unique**: the optimal set may be an entire face of $\mathcal{F}$. The optimal *value* is nonetheless unique, and the solver returns one particular optimiser.

– *min_variance:* After the shrinkage regularisation of §9.7.2, Σ_{t_k} is positive definite, so the objective $w \mapsto w^\top \Sigma_{t_k} w$ is strictly convex. A strictly convex objective over a non-empty convex compact set has a **unique** minimiser.

9.5 Solver

We formulate the problem with the Python module `cvxpy` and solve it with **ECOS** (an open-source interior-point solver) as the default, falling back to **SCS** (another solver) when ECOS fails to converge. In spite of the covariance matrix Σ_{t_k} being theoretically symmetric by construction, we additionally symmetrise Σ_{t_k} numerically (replacing it by $\frac{1}{2}(\Sigma_{t_k} + \Sigma_{t_k}^\top)$) to remove floating-point arithmetic asymmetry, which may cause the solver to fail.

9.6 From optimal weights to integer share counts

The solver returns an optimal weight vector $w^\star_{t_k}$. We convert it back to tradable, integer share counts in two steps:

Step 1: Weights to fractional shares. Inverting $w_{t_k,i} = q_{t_k,i}\, p_{t_k,i}/\text{Equity}_{t_k}$, given current book equity Equity_{t_k} and close prices p_{t_k}:

$$q^{\text{frac}}_{t_k,i} = \frac{w^\star_{t_k,i}\,\text{Equity}_{t_k}}{p_{t_k,i}}$$

Step 2: Per-position-size cap, then truncation. To enforce a maximum position size of $Q_{\max}$ shares per ticker (default $Q_{\max} = 10{,}000$), we scale **all** positions down uniformly by

$$\gamma = \min\left(1,\ \min_{i \in \Omega_{t_k}} \frac{Q_{\max}}{|q^{\text{frac}}_{t_k,i}|}\right)$$

and then truncate toward zero:

$$q^\star_{t_k,i} = \text{trunc}\big(\gamma\, q^{\text{frac}}_{t_k,i}\big)$$

where $\text{trunc}(\cdot)$ rounds toward zero (§3.4.2). Uniform scaling preserves the relative proportions of the optimal portfolio, and therefore its Sharpe ratio (§3.4.2): scaling every position by γ multiplies both equity return and equity volatility by γ, leaving their ratio unchanged.

Reason for capping shares per ticker: The cap is an upper bound applied to each optimal position (amount of shares), to comply with eventual broker concentration limits.

Margin compliance: If the rounded positions $q^\star_{t_k}$ violate the rebalance margin requirement, we apply Variant A or Variant B as described in §3.4.1 (Variant B shrinks the positions (see §3.4.2, §3.4.3)).

9.7 Estimation of the optimization inputs

All three optimiser inputs (the expected-return vector μ_{t_k}, the covariance matrix Σ_{t_k}, and the beta vector β_{t_k}) are unknown at t_k and are estimated by rolling-window methods on past data.

9.7.1 Expected returns

For each ticker $i \in \Omega_{t_k}$ we estimate the expected simple return over the next inter-rebalance period. We write, for conciseness,

$$r_{i,k} := r_i(t_k, t_{k+1})$$

for the one-period-ahead return of ticker i. In the code implementation, three estimators are supported, selected via the `next_period_expected_returns_estimation_method` setting.

9.7.1.2 Exponentially weighted moving average

To down-weight distant observations:

$$\hat{\mu}_{t_k,i} := \frac{\displaystyle\sum_{q=1}^{K_\mu} (1-\alpha)^{q-1}\, r_{i,k-q}}{\displaystyle\sum_{q=1}^{K_\mu} (1-\alpha)^{q-1}}, \qquad \alpha \in (0,1)$$

where a higher α reduces the weight $(1-\alpha)^{q-1}$ on past observations, making the weights decay faster and thus concentrating weight on the most recent periods.

Remark

The weights decrease as the lag q increases in the summation above. It is convenient to set α from an interpretable **half-life** h (in inter-rebalance periods), defined as the number of inter-rebalance periods over which the weight halves:

$$(1-\alpha)^h = \frac{1}{2} \iff \alpha = 1 - 2^{-1/h},$$

so that an observation h periods older than the most recent one carries half its weight. For example, $h = 6$.

9.7.1.3 Factors model prediction

Expected returns can instead be taken as the out-of-sample prediction of the cross-sectional linear factors model of §8:

$$\hat{\mu}_{t_k,i} = \hat{\alpha} + \hat{b}_M\, M_{i,t_k} + \hat{b}_V\, V_{i,t_k}$$

where M_{i,t_k} and V_{i,t_k} are the cross-sectionally winsorised and z-scored momentum and volatility of ticker i at rebalance date t_k, and the coefficients $(\hat{\alpha}, \hat{b}_M, \hat{b}_V)$ are fit by rolling cross-sectional OLS (or Ridge) regression over the previous W_{factor} rebalance dates. See §8.6 for the full specification. This re-uses the factors model purely for its return predictions, not for any long/short selection rule.

Choice of method

The factors model has the strongest theoretical grounding, since it predicts cross-sectional return from observable signals. The rolling mean rests on an unrealistic i.i.d. returns assumption and serves as a baseline; the EW mean is a variant inspired from the rolling mean, which down-weights distant returns. **We have no prior on which performs best; they can be compared empirically via out-of-sample Spearman rank correlation between predicted and realized returns, computed at each rebalance date across any chosen set of rebalance dates.**

A further alternative to selecting one estimator is to *enrich* whichever one we pick: the Black–Litterman construction (§16.2) treats a chosen estimate as noisy evidence updating a neutral prior, returning a posterior mean. See §16.2.

9.7.2 Covariance matrix

At each rebalance date t_k, $k \in \{0, \dots, P\}$, we require an estimate of the covariance matrix of simple returns over the next period $[t_k, t_{k+1}]$ for all tickers in Ω_{t_k}. With $n_{t_k} := |\Omega_{t_k}|$,

$$\Sigma_{t_k} := \big(\mathrm{Cov}(r_{i,k}, r_{j,k})\big)_{(i,j)\in\Omega_{t_k}^2} \in \mathbb{R}^{n_{t_k}\times n_{t_k}}$$

Empirical estimator: The entries $\mathrm{Cov}(r_{i,k}, r_{j,k})$ are unknown and must be estimated from past data. Fix a window length $W_\Sigma \geq 2$, and let the past W_Σ realised returns be $r_{i,k-1}, \dots, r_{i,k-W_\Sigma}$. With empirical mean $\bar{r}_i := \frac{1}{W_\Sigma}\sum_{w=1}^{W_\Sigma} r_{i,k-w}$, the sample covariance estimator is

$$\widehat{\mathrm{Cov}}(r_{i,k}, r_{j,k}) = \frac{1}{W_\Sigma - 1}\sum_{w=1}^{W_\Sigma}\big(r_{i,k-w} - \bar{r}_i\big)\big(r_{j,k-w} - \bar{r}_j\big)$$

Under the assumption that, for each pair (i, j), the pairs $\big((r_{i,k-w}, r_{j,k-w})\big)_{w\geq 1}$ are i.i.d. with finite second moments, this estimator is unbiased and, by the strong law of large numbers, almost surely convergent to $\mathrm{Cov}(r_{i,k}, r_{j,k})$ as $W_\Sigma \to \infty$. The empirical covariance matrix $\hat{\Sigma}_{t_k,W_\Sigma}$ assembled from these entries is therefore an unbiased and a.s. convergent estimator of Σ_{t_k}.

Caveat on the i.i.d. assumption

The i.i.d. premise is an idealisation adopted for tractability, not a description of the data. Equity returns exhibit volatility clustering, time-varying correlations, and regime shifts across a multi-year window, so the pairs $(r_{i,k-w}, r_{j,k-w})$ are neither independent nor identically distributed in practice. Two consequences follow. First, the unbiasedness and convergence results above should be read as properties of an idealised model, not guarantees on realised estimates. Second, a longer W_Σ reduces estimation variance but increases staleness bias when the true covariance drifts; this bias–variance trade-off, and the absence of any regime adaptation in the fixed window, is revisited in §16.1 (“No regime-switching or adaptive parameters”). The Ledoit–Wolf shrinkage applied below mitigates the estimation-noise side of this trade-off but does not address non-stationarity.

Matrix formulation: For each ticker i, define the centred return vector

$$R_i^{(c)} := \left(r_{i,k-1} - \bar{r}_i,\ \ldots,\ r_{i,k-W_\Sigma} - \bar{r}_i\right)^\top \in \mathbb{R}^{W_\Sigma}$$

and stack these as columns of the data matrix $R_k := \left(R_1^{(c)} \mid \cdots \mid R_{n_{t_k}}^{(c)}\right) \in \mathbb{R}^{W_\Sigma \times n_{t_k}}$. Then

$$\boxed{\hat{\Sigma}_{t_k,W_\Sigma} = \frac{1}{W_\Sigma - 1}\, R_k^\top R_k}$$

Rank deficiency in high dimensions: Each column of R_k is mean-centred, so every column lies in the hyperplane $\{v \in \mathbb{R}^{W_\Sigma} : \mathbf{1}^\top v = 0\}$, of dimension $W_\Sigma - 1$. Hence the column space of R_k has dimension at most $W_\Sigma - 1$, and

$$\operatorname{rank}(\hat{\Sigma}_{t_k,W_\Sigma}) = \operatorname{rank}(R_k) \le \min(W_\Sigma - 1,\ n_{t_k})$$

In our backtest $n_{t_k} \approx 500$ (number of members of the S&P 500) and each inter-rebalance period is roughly one calendar month. Avoiding rank deficiency would require $W_\Sigma \ge n_{t_k} + 1$ (more than four decades of consecutive monthly returns for *every* ticker) which is infeasible given data availability and the fact that many tickers in Ω_{t_k} did not trade publicly that far back. In practice, with for example $W_\Sigma = 36$ (about three years):

$$\operatorname{rank}(\hat{\Sigma}_{t_k,W_\Sigma}) \le 35, \qquad \dim\ker(\hat{\Sigma}_{t_k,W_\Sigma}) \ge 465$$

and shorter windows (e.g. $W_\Sigma = 18$) make the deficiency more severe still.

Consequence for the optimiser: Portfolio variance enters the optimiser as $w^\top \hat{\Sigma}_{t_k,W_\Sigma}\, w$. With $\dim\ker(\hat{\Sigma}_{t_k,W_\Sigma}) \ge 465$, there exist at least 465 linearly independent directions for w along which the *estimated* variance is exactly zero. This is spurious: no non-null portfolio truly has zero return variance, as that would make its return almost surely constant. The optimiser of §9.2 would exploit such directions, reporting portfolios with zero estimated risk and non-zero expected return.

Regularisation (Ledoit–Wolf shrinkage): To eliminate the spurious zero-variance directions and, equally important, to reduce estimation noise in the off-diagonal entries, we apply Ledoit–Wolf shrinkage [15], replacing the sample covariance with a convex combination of $\hat{\Sigma}_{t_k,W_\Sigma}$ and a structured target:

$$\hat{\Sigma}_{t_k}(\rho) := (1-\rho)\,\hat{\Sigma}_{t_k,W_\Sigma} + \rho\, m\, I, \qquad m := \frac{1}{n_{t_k}}\operatorname{tr}\!\left(\hat{\Sigma}_{t_k,W_\Sigma}\right)$$

where $\rho \in (0,1)$ is the shrinkage intensity and mI is a scaled identity whose scale is the average sample variance across tickers. Since $\hat{\Sigma}_{t_k,W_\Sigma} \succeq 0$, its trace is the sum of its non-negative eigenvalues $\lambda_i \ge 0$, and because at least one ticker has strictly positive variance, $m > 0$. The eigenvalues of $\hat{\Sigma}_{t_k}(\rho)$ are $(1-\rho)\lambda_i + \rho m$, and

$$(1-\rho)\lambda_i + \rho m \ \ge\ \rho m \ >\ 0$$

so $\hat{\Sigma}_{t_k}(\rho)$ is **positive definite** with $\ker(\hat{\Sigma}_{t_k}(\rho)) = \{0\}$ for every $\rho \in (0,1)$. All spurious zero-variance directions are therefore removed.

Choice of ρ

Ledoit and Wolf (2004) select ρ to minimise the expected squared Frobenius distance to the true covariance matrix,

$$\rho^* = \arg\min_{\rho\in[0,1]} \mathbb{E}\Big[\big\|\hat{\Sigma}_{t_k}(\rho) - \Sigma_{t_k}\big\|_F^2\Big], \qquad \|A\|_F^2 := \sum_{i,j} A_{i,j}^2$$

and, in the asymptotic regime $n_{t_k}, W_\Sigma \to \infty$ with $n_{t_k}/W_\Sigma \to c \in (0,\infty)$, construct a consistent, closed-form estimator $\hat{\rho}$ of ρ^* from the sample fourth moments of returns. We use $\hat{\rho}$ as the shrinkage intensity.

Per-date universe restriction: Tickers lacking a complete W_Σ-period return history at t_k are excluded from that date's covariance estimate. The estimator returns both the per-date covariance matrices and the per-date shrinkage intensities; an intensity close to 0 indicates a well-conditioned sample.

9.7.3 Betas

At each rebalance date t_k, the beta of each ticker i is estimated by an OLS regression of past inter-rebalance **excess** returns on **market excess** returns, over a rolling window of the previous W_β periods (In the code implementation, we use the default $W_\beta = 12$; values $P_\beta \in \{12, 18, 24\}$ can be compared against each other by comparing the fit of the regression, through the in-sample MSE and R^2, (see §8.11)):

$$r_{i,q} - r_q^f = \alpha_{i,t_k} + \beta_{i,t_k}\big(r_q^m - r_q^f\big) + \varepsilon_{i,q}, \qquad q = k - P_\beta, \dots, k-1$$

where:

- r_q^m is the market return proxy over $[t_q, t_{q+1}]$: the total return of the SPY ETF (SPDR S&P 500 ETF Trust, ticker **SPY**, sourced from `yfinance`);
- r_q^f is the risk-free return proxy over $[t_q, t_{q+1}]$ (see §11);
- $\varepsilon_{i,q}$ is the regression residual.

The OLS slope has the closed form

$$\hat{\beta}_{i,t_k} = \frac{\widehat{\mathrm{Cov}}_{W_\beta}\big(r_{i,\cdot} - r_\cdot^f,\ r_\cdot^m - r_\cdot^f\big)}{\widehat{\mathrm{Var}}_{W_\beta}\big(r_\cdot^m - r_\cdot^f\big)}$$

which coincides with the OLS estimator of β_{i,t_k} in the regression above when an intercept α_{i,t_k} is included.

In the code implementation, tickers with insufficient non-`NaN` history yield `NaN` betas and are dropped before the optimiser call (see §9.7.4).

Portfolio beta: Given the expression for the equity return derived in §9.1, and by bilinearity of covariance and the $\mathcal{F}_{t_k}$-measurability of w_{t_k}, the portfolio beta is linear in the weights:

$$\beta^{\text{Equity}}_{t_k} = \beta_{t_k}^{\top} w_{t_k}, \qquad \beta_{t_k} = \left(\beta_{i,t_k}\right)_{i \in \Omega_{t_k}}$$

which is exactly the quantity constrained by C2 in §9.3.2.

9.7.4 Universe alignment

The optimiser requires μ_{t_k}, Σ_{t_k}, and β_{t_k} to be indexed by a **common** set of tickers. The three estimators above can each drop different tickers (insufficient return history, insufficient beta-regression history, etc.). We therefore form the **aligned universe** as the intersection of the three estimators' ticker indices with the candidate universe of §6, and restrict μ, Σ, β to that set before solving. Tickers dropped during alignment are logged at `WARNING` level.

9.7.5 Computation modes (static vs dynamic)

The framework estimates the optimiser inputs in one of two modes:

- **Static (fixed-calendar) mode:** When the dynamic-rebalance rule is *not* active, all rebalance dates are known in advance, and every estimator at every rebalance date is precomputed in a single batch before the backtest runs.

- **Dynamic mode:** When dynamic rebalancing is active, the rebalance dates are determined during the run, so the estimators are computed on the fly at each rebalance date as it is reached.

The two modes produce identical estimates for any given rebalance date; they differ only in *when* the computation is performed.

10. Caveats of the three strategies

Momentum strategy

As stated in §7.1, momentum is the empirical tendency of stocks that have performed well or badly over a recent window to keep doing so in the near future. We define the momentum of a ticker as the log return of its close price, computed with a one-month buffer. Going long the top k tickers by this signal and short the bottom k is a wager that the long positions will rise and the short positions will fall, in line with that tendency. Nothing forces the tendency to persist over the holding period. The strategy just bets that it will.

Factor models strategy

This strategy (§8) fits a linear model to past data. Suppose it shows skill out of sample, measured primarily by a sign hit rate above the no-information baseline of 0.5 on the tickers actually traded, and, in second place, by a positive Spearman Information Coefficient (IC) between predicted and realized return rankings (§8.11). Going long the top k ranked tickers and short the bottom k wagers two things at once: first, that the predicted *signs* are correct, so that the long positions rise and the short positions fall; second, that the *ordering* is correct, so that the largest predicted returns line up with the largest realized moves and allows exposure to concentrate on the strongest ones. The first is what determines whether each position wins or loses; the second only amplifies the outcome of correctly-signed bets, and amplifies losses just as well when the signs are wrong. Out-of-sample skill on historical data is evidence that the skill might carry forward, but it is not a promise that it will.

Constrained mean-variance portfolio optimization (MVO)

As in §9, the optimizer requires us to model the return of a ticker over a future period as a random variable to try to estimate its expected value, its standard deviation, and its covariance with other tickers' returns over the same period. Each estimate brings its own caveat.

Take the expected return first, estimated by the empirical mean of past returns over a rolling window (§9.7.1). One justification leans on the law of large numbers, under the assumption that each past-period return is a draw from a marginal random variable and that these variables are independent and identically distributed. That i.i.d. property is inconsistent with what is empirically observed (volatility clustering and return autocorrelation being two examples; the covariance caveat of §9.7.2 makes the same point). Once the assumption fails, the LLN no longer backs the estimator, and its accuracy is no longer assured.

A second justification for this estimator splits the historical period into sub-periods, each governed by a single distribution. This accommodates *volatility clustering*, where the standard deviation shifts from one window to the next but is shared by all returns within a sub-period. It can justify a rolling historical average as an estimate of next-period return, provided the sample stays inside one sub-period yet stays large enough for the LLN to deliver accuracy. Those two requirements pull against each other: short enough to remain in one regime, long enough to estimate accurately.

A third approach uses the factor model of §8, with its assumptions (§8.4, §8.6) and out-of-sample metrics (§8.11), to estimate next-period expected returns. The metric hierarchy of §8.11 was established for the top-k/bottom-k selection rule of §8.9, whose profitability crucially relies on the signs and the ordering of the predictions; here the optimizer consumes the predicted *levels*, since $\hat{\mu}$ enters the objective (or the return lower bound) as a magnitude. Therefore the level-fit metrics (R^2 and MSE), of only tertiary importance for the selection rule, become first-order for this use. Here again, even with demonstrable out-of-sample predictive power, applying the model to the coming period bets that it retains that power.

The same questions apply to estimating the covariance matrix and to the assumptions invoked to justify whichever estimator we pick.

Given these inputs, the "max-return" (resp. "min-variance") variant of the MVO optimizer finds the combination of positions, equivalently the equity weights per ticker (§9.1), that maximizes expected portfolio return (resp. minimizes portfolio return variance) subject to the variance cap (resp. return lower bound) and the other constraints (net, gross, beta-neutral). The optimization itself is exact, but it runs on inputs built from the simplifying assumptions above about the random variable representing our next-period equity return (§9.1). Therefore the portfolio that solves the MVO problem carries no guarantee about its future return, and trading on it is a risky bet.

Of these inputs, the expected-return vector μ is both the hardest to estimate and the most consequential to get wrong: Chopra and Ziemba [6] find errors in means far costlier than errors in (co)variances (quantified in §16.2). Improving $\hat{\mu}$ is therefore the highest-value refinement of this strategy. We develop one concrete proposal, the Black–Litterman enrichment of a chosen $\hat{\mu}$ estimator, in §16.2.

Conclusion

These missing guarantees are exactly why we backtest. We simulate deploying each strategy at past dates under specific choices: initial capital, termination rules such as stop loss and return target, and configuration such as dynamic rebalancing. We can then compare the strategies across metrics, as in §14, and reach an assessment that favors, more or less strongly, deploying a given strategy now (§15).

11 Risk-Free Rate

11.1 Choice of rate

We use the **Effective Federal Funds Rate (EFFR)** as a proxy for the overnight U.S. dollar funding rate throughout the backtest. EFFR is the volume-weighted median rate on overnight, unsecured loans in the federal funds market, computed and published each business day by the Federal Reserve Bank of New York. Because it is unsecured it embeds a small amount of bank credit risk, so we treat it as a proxy for the risk-free rate rather than a literal risk-free instrument.

EFFR is used in three places in the backtest:

1. **Cash and debit interest accruals** (§2.6): the broker's cash and debit rates are quoted as EFFR plus or minus a markup.
2. **Excess-return computation**: period returns (§12.1) and per-ticker betas (§9.7.3) are computed against the period's risk-free return (using the compounding from §11.3 below).
3. **Sharpe-ratio computation** (§14.1): the classical Sharpe ratio is defined against a Treasury-bill yield; we compute it using EFFR, so our figures are not directly comparable to published equity Sharpe ratios constructed against T-bills.

Why EFFR rather than a Treasury-bill rate: First, it is a genuine overnight rate: it rolls overnight balances rather than holding a bill to maturity. Second, it is the base off which the broker quotes the cash and debit rates we apply (§2.6), so using it is consistent with the financing rates for our brokerage account. By contrast, the shortest Treasury bill is 4-week, so obtaining a per-calendar-day rate from a bill requires tracking its remaining time-to-maturity at each date and computing an implied daily yield.

Why EFFR rather than SOFR: A secured rate such as SOFR is closer to a true risk-free benchmark and is now the standard reference rate for USD transactions. We nevertheless use EFFR because SOFR was first published on April 3, 2018, so backtests extending earlier cannot use it. (The FRED EFFR series itself begins in July 2000.)

Annualization convention: EFFR is quoted as an annualized percentage on an ACT/360 (money-market) day-count basis. We therefore convert each quote to its decimal value (dividing by 100) and apply it over the relevant day count when accruing interest and computing excess returns (see §11.3).

Data source: The EFFR series is obtained from the St. Louis Federal Reserve's FRED database, series ID `EFFR` [17].

11.2 Daily rate from annualized rate

Given the EFFR ACT/360 convention, the per-day accrual rate is

$$r^f_{d,t} \;=\; \frac{r^{f,\text{annual}}_t}{360}$$

and the return over a holding period of Δ calendar days is $r^f_{d,t} \cdot \Delta$. For a standard overnight loan $\Delta = 1$; for a loan struck on a Friday and maturing Monday (the *weekend block*) $\Delta = 3$, the rate set on Friday being held constant across the weekend in line with standard money-market practice. Interest accrues by simple multiplication within a single accrual period; the compounding *across* periods is handled separately in §11.3.

11.3 Risk-free return between trading dates

The EFFR interest rate from one NYSE trading date $t-1$ to the next trading date t is determined by

$$r^f_t \;=\; r^f_{d,t-1} \cdot (\text{calendar days between } t-1 \text{ and } t)$$

where $r^f_{d,t-1}$ is the EFFR daily rate *for* date $t-1$ (the rate applying to the overnight loan agreed at the close of $t-1$), and the calendar-day count absorbs the weekend block described in §11.2.

Period risk-free return: Our "risk-free return" over an inter-rebalance period $(t_{k-1}, t_k]$ is computed by compounding the daily EFFR returns within the period, assuming earnings are reinvested in subsequent overnight loans at the prevailing EFFR:

$$R^{f,\text{period}}_k \;=\; \prod_{t\in(t_{k-1},t_k]} (1 + r^f_t) \;-\; 1$$

This is the same compounding rule used for the portfolio's period returns (§4.3), and it matches the actual dynamic of rolling overnight loans.

12 Key Performance Indicators

12.1 Overview

Each backtest produces a KPI report in twelve sections, covering the configuration, the run itself, and what it cost in fees, risk, and leverage:

1. **Strategy**: name, parameters, return targets (overall target, and optionally, that used for the dynamic rebalancing rule).
2. **Backtest duration**: date range, number of trading days, cause of termination.

3. **Backtest P&L**: initial and final equity, and the corresponding unrealized return. The return is unrealized because positions are not liquidated on the final backtest date.
4. **Performance between backtest dates**: daily-level return statistics.
5. **Performance per inter-rebalance period**: period-level return statistics.
6. **Drawdowns**: min, max, mean, and standard deviation of the drawdown series.
7. **Trading**: total fees, turnover across rebalances, and the largest single-trade share of daily market volume (with the ticker and date at which it occurred).
8. **Margin requirements**: frequency of maintenance and rebalance MR violations, and a summary of cure events.
9. **Accruals**: per-trading-date distributions of dividends and financing-and-carry.
10. **Leverage**: gross, long, and short leverage statistics.
11. **Daily P&L decomposition**: each P&L stream accumulated over the run, in dollars and as a percentage of the capital deployed, together with the per-date check that the streams reconcile to the realised change in equity (§12.8).
12. **Market exposure**: an ex-post regression of the realised daily equity excess returns on the market's, reporting the estimated beta and intercept with their t-statistics, the R^2, and the statistics of a rolling-window beta.

Additionally, we also report relevant plots, a per-period DataFrame indexed by rebalance date, and a per-trading-date DataFrame indexed by trading day, both carrying three columns (equity return, risk-free return, equity excess return).

12.2 Equity convention for KPIs

All KPI computations use **equity excluding posted margin collateral** ($\widetilde{\text{Equity}}_t$ from §2.2). The posted collateral (when MR violations are handled under Variant A) is external capital that should not inflate the strategy's measured return.

12.3 Drawdowns

The drawdown at t is

$$\text{DD}_t \;=\; \frac{\widetilde{\text{Equity}}_t - \max_{s\le t}\widetilde{\text{Equity}}_s}{\max_{s\le t}\widetilde{\text{Equity}}_s}$$

i.e. the shortfall from the running peak of equity-excluding-collateral. $\text{DD}_t \le 0$ always; $\text{DD}_t = 0$ at every new equity peak.

We report the min (most negative, *i.e*, the maximum drawdown), max (always 0 if the strategy ever reached a new peak after the start of the backtest), avg, and std of the drawdown series.

Conventionally, the *maximum drawdown* is the minimum value of the drawdown series, or equivalently the largest drawdown in absolute value.

12.4 Trading

Five trading metrics are reported:

Total trading fees paid: the sum of all per-share trading commissions paid across the backtest, in dollars.

Total execution cost paid: the difference between the execution price and the close, summed over every trade in the backtest (§2.6.4). Unlike the trading-fee total this is not a booked cash flow, so it appears nowhere in the ledger as an entry of its own; it is reported next to the trading-fee total because the two are of very different magnitudes.

Total gross notional traded: $\sum_t \sum_i p_{t,i} \, |\Delta q_{t,i}|$.

Turnover per trade date: Defined as the gross dollar value traded at t divided by the gross book exposure pre-trade:

$$\text{Turnover}_t \;=\; \frac{\sum_i p_{i,t} \cdot |\Delta q_{i,t}|}{\text{LMV}_t^{\text{pre}} + \text{SMV}_t^{\text{pre}}}$$

We report the min, max, avg, and standard deviation across the dates on which trades were executed, which are the rebalance dates together with any date carrying a deleveraging cure (§3.5.2). A value greater than 1 can be interpreted as if the entire book was "rotated"; values close to 0 signals the strategy held positions across rebalances.

Largest single-trade share of market volume: the maximum of

$$\frac{|\Delta q_{t,i}|}{\text{daily\_volume}_{t,i}}$$

across all trades, with the (date, ticker) at which it occurred. This is a quick check on whether any individual trade would have moved the market: for example, if a trade exceeds 1% of daily volume, the assumed slippage we chose (see §2.5.2) may be too low.

12.5 Margin requirements

Five MR-related metrics are reported, distinguishing maintenance-MR violations (which can occur on any backtest date) from rebalance-MR cures (which occur only on rebalance dates):

- **Maintenance-MR violation count and frequency**: number of dates on which a maintenance-MR cure was executed, and the frequency (count divided by total number of **backtest** dates).

- **Rebalance-MR cure count and frequency**: number of rebalance dates that required a cure (either Variant A or Variant B), and the frequency (count divided by total number of **rebalance** dates).

- **Cure method**: the user-specified `"posting_collateral"` (variant A) or `"shrinking_exposures"` (variant B).

If the cure method was Variant B (shrinking), we report the **shrinking-factor distribution** (min, max, avg, std of α) across cures. Values close to 1 indicate near-feasible portfolios; values close to 0 indicate the amount of intended shares to trade was reduced ("shrunk") proportionally by close to 100%, to meet the rebalance margin requirement.

If the cure method was Variant A (posting collateral), we report **collateral per cure** (min, max, avg, std) and **total collateral posted across the backtest**. The total is useful to reveal the strategy's external capital requirement.

12.6 Accruals

Two accrual streams are reported:

Dividends per trading date: the net cash flow on each trading date from dividend bookings (positive on a long-heavy ex-date, negative on a short-heavy ex-date). We report the min, max, avg, and standard deviation.

Financing costs per trading date: the sum of overnight cash interest, debit interest, borrow fees, rebates, and margin-collateral interest. For long/short portfolios partially financed using debit (margin loans), this is typically negative on average (debit interest dominates).

12.7 Leverage

Three leverage ratios are computed at every backtest date's close and summarised statistically (avg, min, max, std) across the backtest:

- **Long leverage**: $\mathrm{LMV}_t/\mathrm{Equity}_t$
- **Short leverage**: $\mathrm{SMV}_t/\mathrm{Equity}_t$
- **Gross leverage**: $(\mathrm{LMV}_t + \mathrm{SMV}_t)/\mathrm{Equity}_t$

Note that the denominator is **total** equity (including posted margin collateral), because margin requirements are checked against total equity.

12.8 Daily P&L decomposition

This section reports each of the P&L streams of §2.6.6 accumulated over the run, in dollars and as a percentage of initial equity, together with the per-date reconciliation described there.

The first backtest date has no price P&L, since its positions are opened at that close and marked at that same close, and no dividend or financing accrual, since both are booked overnight and there is no prior day. Its equity change is therefore the day's trading costs, together with any collateral posted at that first rebalance under Variant A. For the run of §13.1 this is the \$62.00 discussed in §13.7.

The tolerance, and what it separates: The ledger stores every quantity at full precision, so the only irreducible error in the comparison is the arithmetic itself: on a book of this size, double-precision floating-point cannot resolve differences below roughly 1×10^{-10}. Over the documented run the largest residual on any date is 1.5×10^{-10}. The smallest defect the check must catch is one share of commission, \$0.001. We set the tolerance to a value between these two scales, at 1×10^{-6}: four orders of magnitude above the arithmetic noise and three below the smallest real defect. The computation raises if any date exceeds it, rather than reporting a figure that does not reconcile. An absolute tolerance is appropriate at this book size and would need to be book-relative on a much larger book.

The section reports the number of dates tested, the largest absolute daily residual, the tolerance, and the number of dates exceeding it. An example is shown in Figure 11 of §13.7.

Takeaway

Reporting this decomposition is what makes the cost of running the strategy legible. On the documented run of §13.1, a gross price P&L of +12.08% of initial equity becomes a net +9.39% after three cost streams, the largest of which by an order of magnitude is execution costs at 1.66%.

13 Diagnostic Outputs and Backtest Run Example

This section presents the full set of diagnostic outputs the backtest produces, walked through on an example. The example is the **factors-model strategy** described in §8, configured with the parameters listed below. In §13.11, we also run the other strategies as additional examples.

13.1 Configuration of the backtest example (strategy: factors model)

Strategy : Factors model (see §8) with $k = 10$ tickers per side, fixed share count $Q = 100$, $W_{\text{factor}} = 6$ rebalance periods, $W_{vol} = 60$ trading-day volatility window, momentum window $W_{mom} = 4 \times 23 = 92$ trading days with buffer $b_{mom} = 23$ trading days. Also, configured to performing OLS regressions (not Ridge).

Calendar (see §4): Start `2023-01-05`, rebalance frequency type `ndays` with $N = 23$ trading days, $P = 12$ inter-rebalance periods. The size of the set of training rebalance dates prepended to the rebalance calendar is set by the maximum estimator window, here $W_{\text{factor}} = 6$ periods.

Costs and rules (see §2.6, §2.8, §3, §5):

- Trading: half-spread 1 bps, slippage 3 bps (conservative), commission $0.001/share.
- Margin requirements: Reg-T initial (50% long, 50% short) and FINRA maintenance (25% long, 30% short).
- Cure method of margin requirement violations: Variant B (shrinking exposures).
- Stop-loss: triggered at 50% of initial equity.
- Strategy return target: 9%.
- Dynamic rebalancing rule: enabled, with the inter-rebalance return target set to 3%/12 = 0.25% per period.
- Initial cash (which coincides with our initial equity): $100,000.

Universes of investable tickers (see §6) : For each rebalance, the pool of tickers from which we may select our new positions are built from the S&P 500 historical members with point-in-time forward-data filtering, and we apply no further ADV capping .

Note that in this example, we **enable** the dynamic-rebalance rule (§5.4).

13.2 Equity and drawdowns

Figure 1 shows the equity time series, both including and excluding margin collateral, together with the drawdown of equity excluding margin collateral. In this case, the two equity series coincide, with the green line overlaying the black line. The equity curves are shown in the upper panel, with the drawdown displayed below.

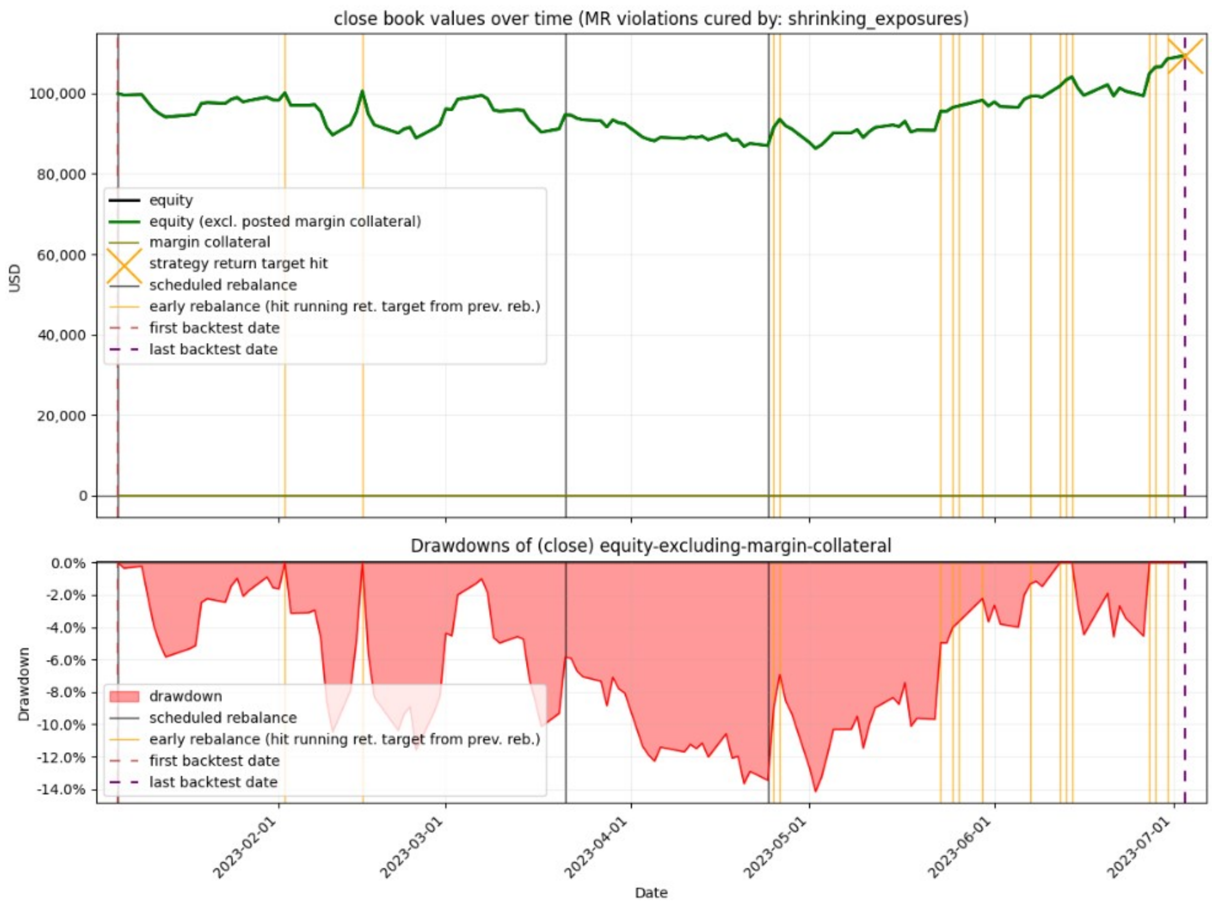

Figure 1: Equity and drawdowns.

The strategy reached a 9.39% total equity return over 123 trading days (backtest dates), as precisely reported by the KPI report in §13.7. The backtest terminated early at the first close when the 9% strategy return target was surpassed. As noted previously, at the last backtest date we do not simulate the liquidation (closing) of our positions, so the return is an ***unrealized*** return. Figure 1 plots the equity at close as an interpolating line through the close markers; the markers themselves are omitted here but shown in Figure 2 for illustration. This figure therefore displays only close values. At any given backtest date, the snapshot of equity at the open corresponds to the value of equity after accounting for the overnight P&L from financing costs on the previous day's book values (§2.6.1) and for dividends (§2.6.2), and after marking to market the positions in our book at the day's open prices. The snapshot at the *close* is derived from marking to market the book's long/short positions at the day's close prices, after having possibly updated the positions from the open as a result of a rebalance (simulation of the execution of the corresponding trades, §2.8), which can reduce equity both through the simulated slippage / market-impact model (§2.5.2) and through trading fees (§2.6.4). Trades can be carried out either to rebalance our portfolio according to the new portfolio composition dictated by our strategy, or to cure a maintenance-MR violation by shrinking our long and short exposures (under Variant B, §3.5.2).

In this backtest, posted margin collateral is always \$0, as we run Variant B, so total equity equals equity excluding posted collateral throughout the run. We can see that during the backtest the dynamic rebalancing rule was triggered on multiple dates (shown by the vertical orange lines), each time the running equity return since the last rebalance surpassed the target of 0.25%; this triggered an early rebalance and rescheduled the subsequent rebalance dates from then on. The vertical grey line indicates a date where the portfolio was rebalanced as scheduled since the previous rebalance. The strategy experienced varying drawdowns, reaching a maximum drawdown (in absolute value) of 14.15%.

13.3 Equity and margin requirement

Figure 2 shows the addition of the maintenance margin requirement as an orange line and overlay markers on dates where MR cures were executed.

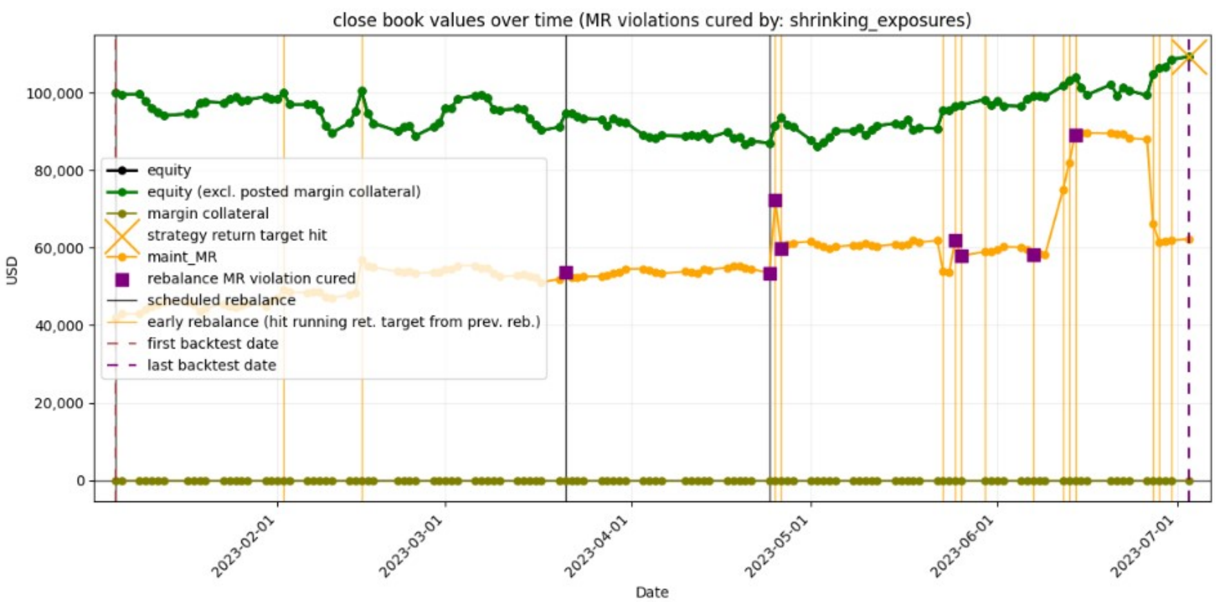

Figure 2: Close values of equity and maintenance margin requirement.

The orange line is the value of the **maintenance margin requirement (MR)** at the close of each date, after any MR cure executed on that date (and thus after shrinking our exposures (Variant B) or posting collateral (Variant A)). Two distinct cures can occur: a *rebalance-time MR cure*, applied when a rebalance requests more gross exposure than equity can support, and a *maintenance-MR cure*, applied if equity falls below the requirement at the close. In this run the maintenance-MR cure never fires (there is **no maintenance-MR violation** anywhere (these would show as orange-triangle markers), consistent with the KPI report's `maint_MR_violation_frequency = 0.00%`) so the close-of-day MR line reflects only the rebalance-time cures. Per our backtest framework design, maintenance MR is checked at the close of every backtest date, and in this backtest it remained covered by equity throughout. This line lies below the equity line at every backtest date, representing compliance with the maintenance margin requirement throughout the run. Rises in the maintenance

MR are explained by increases in our gross long/short exposure at rebalances, as revealed by the formula (3) in §3.2.

The purple square markers indicate dates where a **rebalance-time MR cure was executed**. In this run (under Variant B), some of the rebalances triggered such a cure: the strategy was requesting more gross exposure than equity could support at default Reg-T haircuts, and the bisection on the shrinking factor α (§3.4.4) reduced the requested portfolio to a feasible one. The KPI report (§13.7) confirms this: rebalance-MR cure frequency is 44.44%, with shrink factors in the range 22.00%–98.00%, meaning the post-shrink portfolio was sized at 22.00%–98.00% of the requested portfolio (i.e., scaling every requested long and short position by that same factor).

13.4 LMV / SMV / total short proceeds and leverage ratios

Figure 3 displays the evolution of the aggregate composition of the book:

Figure 3: Plot of the values of LMV, SMV, total short proceeds, and leverage ratios at close.

Upper panel: Because of the time scale involved, Figure 3 plots the values of the respective accounts **only at the close** (omitting their values at the open), interpolating the markers at each date (the markers themselves are not shown). The sudden jumps in the lines, visible at the vertical orange or grey lines (rebalance dates), arise because when we rebalance the portfolio on a given date the exposures may change significantly, as dictated by the strategy, so the line connecting the close markers of two consecutive dates can show an acute ascent or descent. To show this

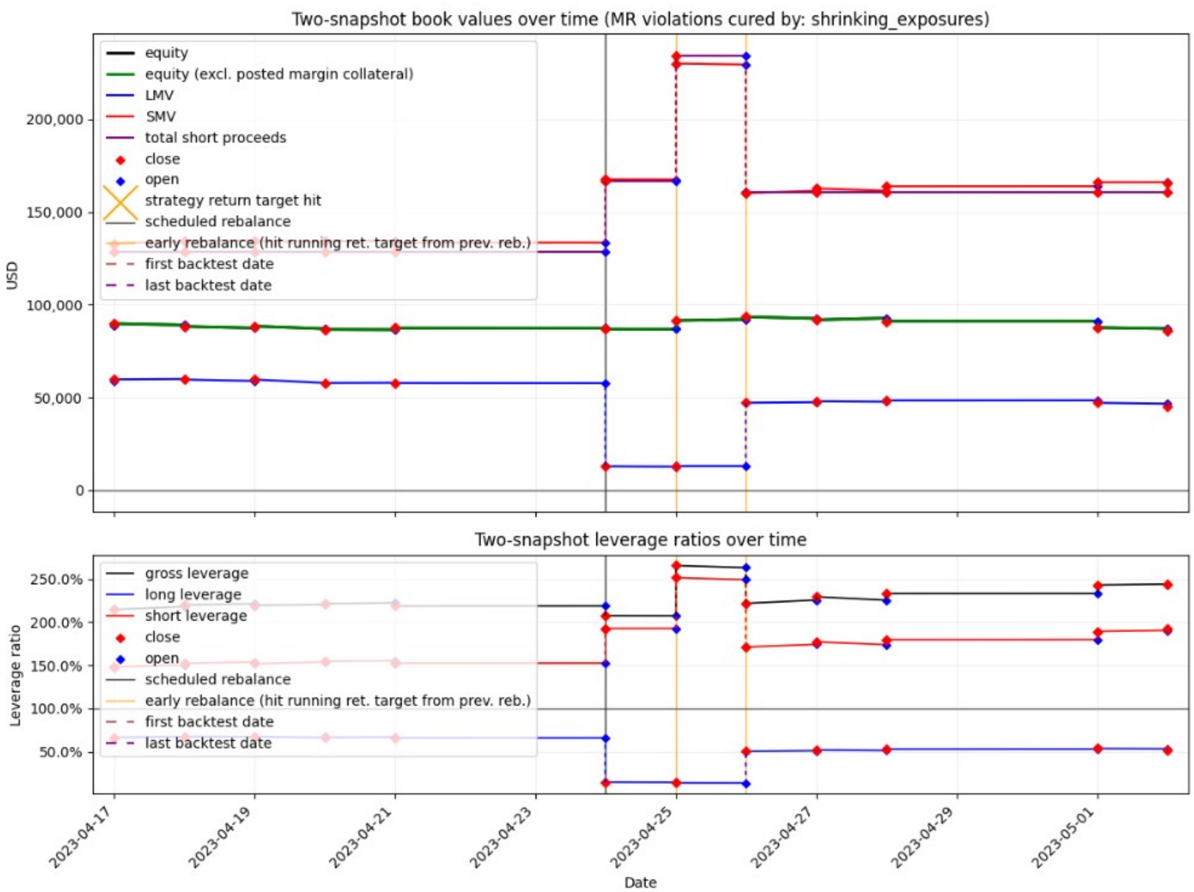

Figure 4: Section of the plot of the two-snapshot book values of LMV, SMV,total short proceeds, and leverage ratios.

precisely, Figure 4 localizes the plot from Figure 3 to the period of the three rebalances surrounding 2023-05-01, where both the open and close values of every account are plotted. The discontinuous jumps reflect the changes in positions (between the open and the close of that date) as a result of rebalancing.

Remark

The purple "total short proceeds" series sits near the SMV series. We recall from §2.2 that ShortProceeds_t is the total short proceeds, and the gap

$$\text{ShortProceeds}_t - \text{SMV}_t$$

equals the unrealized P&L that *would* be realized by closing all short positions at t (excluding trading costs and price slippage, which would reduce the realized gain in the adverse direction). When the "total short proceeds" line lies **above** the SMV line, the short book is net profitable in aggregate; when it lies **below**, it is at an aggregate loss. Inspecting the first rows of the book-values table (Figure 7) shows that for the first backtest dates (2023-01-05 to 2023-01-26) the gap at the close varies in the range \$0–\$10,000 in absolute value.

Lower panel: Leverage ratios are taken relative to equity (including any margin collateral), as defined in §2.4), which for this backtest (Variant B) coincides with equity excluding margin collateral, since posted collateral is \$0. The KPI report (Figure 10) indicates that the strategy had, on average, an 86.90% long leverage ratio at close (×0.86) and a 127.20% short leverage ratio at close (×1.27), for an average gross leverage ratio of 214.20% at close (×2.14).

13.5 Cash, debit, and equity

Figure 5 shows the cash and debit balances alongside equity and equity excluding margin collateral (here both are equal, as explained previously).

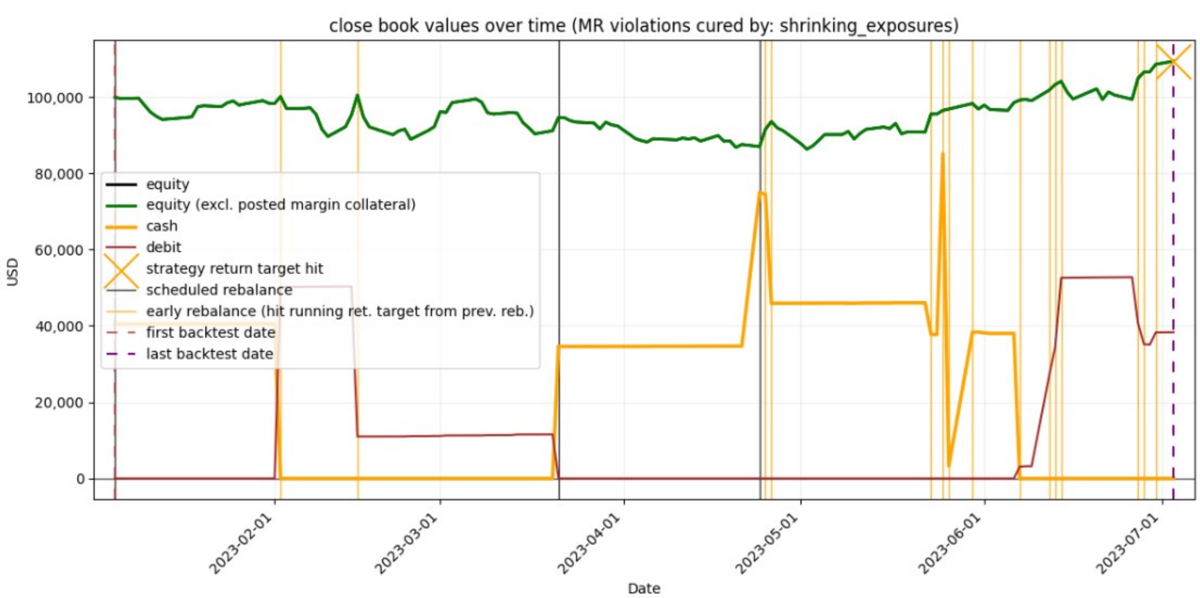

Figure 5: Plot of cash, debit, and equity at close.

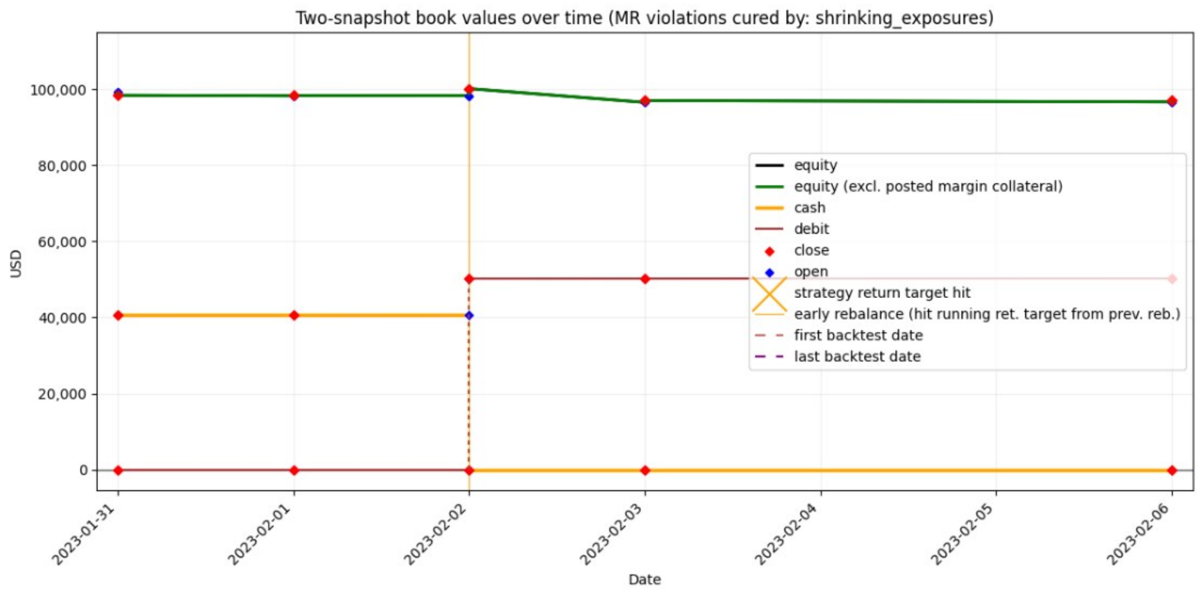

Figure 6: Section of the plot of two-snapshot book values of cash, debit, and equity at close.

The orange **cash** series and the brown **debit** series are never both positive at a date's close (at most one is nonzero) by construction of the cash-sweep policy (§2.3), which is invoked after every cash-moving operation (each financing accrual, each dividend accrual, and each trade simulation). Between rebalances, debit and cash therefore still move from day to day: financing costs and dividend flows are swept against the prevailing balance, drawing debit down (or, once debit is cleared, building positive cash) when the net flow is a credit, and adding to debit when it is a charge. At a rebalance, debit may additionally decrease when cash raised from liquidating positions is swept against it (§2.3), or increase when we increase our margin loan to finance new long exposures.

Figure 6 localizes the plot from Figure 5 to the time window surrounding the rebalance near `2023-02-01`, plotting both the open and close daily values of cash and debit, illustrating the intraday discontinuous fluctuations at a rebalance date. In this example, increasing our long exposure at that rebalance consumed all available cash; the remaining cost was financed by drawing on the margin loan, an instance of the cash-to-debit reclassification of §2.3 (free cash hits zero, and the shortfall is reclassified as debit).

13.6 Book-values table

Figure 7 presents the first 15 rows of the daily ledger, giving the full state of every account in our book, with both daily snapshots (at open and close) shown for each.

13.7 KPI summary report

The structured KPI report described in §12.1 is shown for this backtest run in Figures 8, 9, 10 and 11. It can be saved to a file `kpis_report.txt` and can also be displayed inline in the Jupyter notebook cell.

[25]:
```python
book_table = kpis.build_book_ledger_across_dates(backtest_results)
n_rows = 15 # set to None to display all rows
kpis.show_book(book_table.iloc[:n_rows])
```

[25]:

account	equity		equity_excluding_margin_collateral		margin_collateral		cash		debit		LMV		SMV		total_short_proceeds	
moment	open	close	open	close	open	close	open	close	open	close	open	close	open	close	open	close
date																
2023-01-05	100,000.00	99,938.00	100,000.00	99,938.00	0.00	0.00	100,000.00	40,500.21	0.00	0.00	0.00	59,474.00	0.00	90,521.00	0.00	90,484.79
2023-01-06	99,583.07	99,585.27	99,583.07	99,585.27	0.00	0.00	40,503.98	40,503.98	0.00	0.00	59,960.00	60,941.00	91,365.70	92,344.50	90,484.79	90,484.79
2023-01-09	98,837.75	99,708.35	98,837.75	99,708.35	0.00	0.00	40,515.25	40,515.25	0.00	0.00	61,217.00	61,121.00	93,379.30	92,412.70	90,484.79	90,484.79
2023-01-10	99,737.10	97,833.40	99,737.10	97,833.40	0.00	0.00	40,519.01	40,519.01	0.00	0.00	61,080.00	62,160.00	92,346.70	95,330.40	90,484.79	90,484.79
2023-01-11	97,756.54	96,041.94	97,756.54	96,041.94	0.00	0.00	40,522.75	40,522.75	0.00	0.00	62,209.00	62,776.00	95,460.00	97,741.60	90,484.79	90,484.79
2023-01-12	96,125.27	94,928.96	96,125.27	94,928.96	0.00	0.00	40,511.47	40,511.47	0.00	0.00	63,223.00	63,014.00	98,094.00	99,081.30	90,484.79	90,484.79
2023-01-13	94,985.28	94,111.78	94,985.28	94,111.78	0.00	0.00	40,515.19	40,515.19	0.00	0.00	62,251.00	63,392.00	98,265.70	100,280.20	90,484.79	90,484.79
2023-01-17	93,653.99	94,615.59	93,653.99	94,615.59	0.00	0.00	40,530.00	40,530.00	0.00	0.00	63,168.00	63,527.00	100,528.80	99,926.20	90,484.79	90,484.79
2023-01-18	93,995.00	94,802.20	93,995.00	94,802.20	0.00	0.00	40,533.71	40,533.71	0.00	0.00	63,986.00	62,719.00	101,009.50	98,935.30	90,484.79	90,484.79
2023-01-19	95,441.52	97,480.42	95,441.52	97,480.42	0.00	0.00	40,537.43	40,537.43	0.00	0.00	62,169.00	60,874.00	97,749.70	94,415.80	90,484.79	90,484.79
2023-01-20	97,175.06	97,704.97	97,175.06	97,704.97	0.00	0.00	40,516.17	40,516.17	0.00	0.00	61,170.00	62,538.00	94,995.90	95,834.00	90,484.79	90,484.79
2023-01-23	97,885.97	97,487.97	97,885.97	97,487.97	0.00	0.00	40,527.38	40,527.38	0.00	0.00	63,012.00	64,565.00	96,138.20	98,089.20	90,484.79	90,484.79
2023-01-24	98,740.59	98,492.59	98,740.59	98,492.59	0.00	0.00	40,531.10	40,531.10	0.00	0.00	64,349.00	63,883.00	96,624.30	96,406.30	90,484.79	90,484.79
2023-01-25	99,992.92	98,961.02	99,992.92	98,961.02	0.00	0.00	40,534.83	40,534.83	0.00	0.00	63,051.00	63,532.00	94,077.70	95,590.60	90,484.79	90,484.79
2023-01-26	98,796.16	97,859.66	98,796.16	97,859.66	0.00	0.00	40,538.57	40,538.57	0.00	0.00	64,660.00	64,237.00	96,887.20	97,400.70	90,484.79	90,484.79

Figure 7: First 15 rows of the table of book values.

Remark

From the *Daily P&L decomposition* section of the KPI report (Figure 11), the first backtest date contributes no price P&L and an equity change of exactly −\$62.00: −\$2.00 of trading fees and −\$60.00 of execution costs. The reason is that we trade at the execution price, which is worse than the pre-trade price (proxied by the close price of the day): we buy above it and short-sell below it, as per our execution price model (2) from §2.5.2. Our positions at the close are then marked at the close price, therefore producing a loss on equity, because for a buy, we paid above the close for shares then marked at the close; for a short sale, we receive the short sale proceeds from the execution price, and the short market value is then marked at the higher close price (therefore reducing equity as well). This simulated procedure therefore results in a loss of our equity by the amount

$$\sum_i \left[\lambda_{i,t} + p_{i,t} \cdot \frac{\text{halfspread\_bps}_{i,t} + \text{slip\_bps}_{i,t}}{10\,000} \right] |q_{i,t}|,$$

where $\lambda_{i,t}$ is the trading commission per share of ticker i traded at t.

Remark

From the *Trading* section of the KPI report (Figure 9), the **maximum percentage of market volume traded** for any ticker during the entire backtest was 0.12%, reached by ticker NVR on `2023-05-26`. This metric is useful for assessing whether the slippage and market-impact assumptions of the execution price model (§2.5.2) are unrealistically low. In general, a larger fraction of daily market volume traded implies greater market impact, which can lead to higher execution slippage and adverse short-term price movements, not captured by our current backtest framework.

13.8 Time series of returns

The two return DataFrames are displayed with a red/white/green colormap for intuitive visual scanning, and both show three columns: the return of equity (excluding margin collateral), the risk-free return (§11.1), and the excess return (equity return minus the risk-free return).

Inter-rebalance period returns (Figure 12):

The first row (`2023-01-05`) corresponds to the first rebalance at the start of the backtest and has no previous period, so its return is `N/A`. A *period* here designates the time window between the dates of two consecutive rows (corresponding to two consecutive rebalance dates), with the exception of the last row, which is indexed by the **last backtest day** (the backtest terminated early once the return target was reached, per §13.2). For example, between the **close** of `2023-01-05` and the **close** of the date in the next row (`2023-02-05`), equity (excluding posted margin collateral) increased by 0.20%.

Returns between consecutive backtest dates (Figure 13):

Here each row gives the return from the previous backtest date's close to the current date's close.

13.9 Audit logs

Three audit logs can be saved for any given backtest run (default output directory: `outputs/`):

```
--------------------------------------------------------------------------------
                                   STRATEGY
--------------------------------------------------------------------------------
  strategy_name                                   top_bottom_k_from_factors_model
  return_target_for_strategy                                                9.00%
  inter_rebalance_periods_of_same_duration                                     no
  return_target_for_inter_rebalance_period                                  0.25%
  strategy_parameters:
    top_k_bottom_k                                                             10
    shares_per_ticker                                                         100
    rolling_window_trading_days_for_momentum                                   92
    buffer_trading_days_for_momentum                                           23
    rolling_window_trading_days_for_volatility                                 60
    rolling_periods_for_factors_model_regression                                6

--------------------------------------------------------------------------------
                               BACKTEST DURATION
--------------------------------------------------------------------------------
  number_of_backtest_days                                                     123
  first_backtest_date                                                  2023-01-05
  last_backtest_date                                                   2023-07-03
  cause_of_backtest_termination                    hit_return_target_for_strategy

--------------------------------------------------------------------------------
                                 BACKTEST P&L
--------------------------------------------------------------------------------
  initial_equity                                                       100,000.00
  final_equity                                                         109,394.48
  unrealized_return_of_equity                                               9.39%

--------------------------------------------------------------------------------
                        PERFORMANCE BETWEEN BACKTEST DATES
--------------------------------------------------------------------------------
  number_of_backtest_dates                                                    123
  avg_excess_return_between_backtest_dates                                  0.08%
  std_of_excess_returns_between_backtest_dates                              1.96%
  min_excess_return_between_backtest_dates                                 -5.68%
  max_excess_return_between_backtest_dates                                  5.54%
```

Figure 8: KPI report (part 1).

```
--------------------------------------------------------------------------------
                       PERFORMANCE PER INTER-REBALANCE PERIOD
--------------------------------------------------------------------------------
 number_of_periods                                                            18
 periods_of_same_duration                                                     no
 avg_period_excess_return                                                  0.42%
 std_of_period_excess_return                                               3.11%
 min_period_excess_return                                                 -8.53%
 max_period_excess_return                                                  5.21%

--------------------------------------------------------------------------------
                                    DRAWDOWNS
--------------------------------------------------------------------------------
 drawdown of equity-excluding-margin-collateral:
   min                                                                   -14.15%
   max                                                                     0.00%
   avg                                                                    -6.02%
   std                                                                     4.11%

--------------------------------------------------------------------------------
                                     TRADING
--------------------------------------------------------------------------------
 total_trading_fees_paid_in_backtest                                       30.35
 total_execution_cost_paid_in_backtest                                  1,657.90
 total_gross_notional_traded_in_backtest                            4,144,758.19
 turnover:
   min                                                                    23.67%
   max                                                                   211.88%
   avg                                                                   112.36%
   std                                                                    62.45%
 max_pct_of_daily_market_volume_traded_for_a_ticker_during_backtest:
   max_pct_of_daily_market_volume                                          0.12%
   for_ticker                                                                NVR
   at_backtest_date                                                   2023-05-26
```

Figure 9: KPI report (part 2).

```
-------------------------------------------------------------------------------
                              MARGIN REQUIREMENTS
-------------------------------------------------------------------------------
 maint_MR_violation_number_of_backtest_days                                   0
 maint_MR_violation_frequency                                             0.00%
 rebalance_MR_cure_number_of_backtest_days                                    8
 rebalance_MR_cure_frequency                                             44.44%
 cure_method                                                shrinking_exposures
 shrink_factor_to_cure_an_MR_violation:
   min                                                                   22.00%
   max                                                                   98.00%
   avg                                                                   61.40%
   std                                                                   27.70%
 collateral_per_cure                                                        N/A
 total_collateral_posted                                                    N/A

-------------------------------------------------------------------------------
                                    ACCRUALS
-------------------------------------------------------------------------------
 dividends_per_trading_date:
   min                                                                  -196.50
   max                                                                   100.00
   avg                                                                    -7.45
   std                                                                    34.35
 financing_costs_per_trading_date:
   min                                                                   -51.98
   max                                                                    14.82
   avg                                                                    -0.67
   std                                                                    10.32

-------------------------------------------------------------------------------
                                    LEVERAGE
-------------------------------------------------------------------------------
 gross_leverage:
   min                                                                  150.10%
   max                                                                  326.40%
   avg                                                                  214.20%
   std                                                                   36.50%
 long_leverage:
   min                                                                   11.70%
   max                                                                  157.00%
   avg                                                                   86.90%
   std                                                                   37.00%
 short_leverage:
   min                                                                   39.00%
   max                                                                  251.40%
   avg                                                                  127.20%
   std                                                                   42.00%
```

Figure 10: KPI report (part 3).

```
--------------------------------------------------------------------------------
                         DAILY P&L DECOMPOSITION
--------------------------------------------------------------------------------
  cumulative_pnl_in_dollars:
    price_pnl                                                        12,081.99
    dividend_pnl                                                       -916.69
    financing_pnl                                                       -82.57
    trading_fees                                                        -30.35
    execution_costs                                                  -1,657.90
    posted_collateral                                                     0.00
    total_equity_change                                               9,394.48
  cumulative_pnl_as_pct_of_initial_equity:
    price_pnl                                                           12.08%
    dividend_pnl                                                        -0.92%
    financing_pnl                                                       -0.08%
    trading_fees                                                        -0.03%
    execution_costs                                                     -1.66%
    posted_collateral                                                    0.00%
    total_equity_change                                                  9.39%
  identity_check:
    number_of_dates_tested                                                 123
    max_absolute_daily_residual                                       1.50e-10
    tolerance_per_day                                                    1e-06
    number_of_dates_above_tolerance                                          0
```

Figure 11: KPI report (part 4): the daily P&L decomposition and its identity check (§12.8).

- `position_changes.txt` (see Figure 14);
- `trades.txt` (see Figure 15);
- `short_proceeds_changes.txt` (explained in §2.2) (see Figure 16)

Each log shows the chronological records.

[29]:
```
n_rows = None  # set to None to display all rows
kpis.style_returns_df(backtest_kpis.period_returns.iloc[:n_rows])
```

[29]:

rebalance dates (except last date)	previous_period_equity_returns	previous_period_rf_returns	previous_period_equity_excess_returns
2023-01-05	N/A	N/A	N/A
2023-02-02	0.20%	0.337%	-0.14%
2023-02-15	0.40%	0.165%	0.23%
2023-03-21	-5.85%	0.433%	-6.28%
2023-04-24	-8.08%	0.456%	-8.53%
2023-04-25	5.23%	0.013%	5.21%
2023-04-26	2.21%	0.013%	2.19%
2023-05-23	2.12%	0.376%	1.74%
2023-05-25	0.98%	0.028%	0.96%
2023-05-26	0.37%	0.014%	0.35%
2023-05-30	1.50%	0.056%	1.44%
2023-06-07	0.92%	0.113%	0.81%
2023-06-12	2.63%	0.071%	2.56%
2023-06-13	1.45%	0.014%	1.43%
2023-06-14	0.81%	0.014%	0.79%
2023-06-27	0.76%	0.183%	0.58%
2023-06-28	1.54%	0.014%	1.53%
2023-06-30	1.91%	0.028%	1.89%
2023-07-03	0.76%	0.042%	0.72%

Figure 12: Table of all the inter-rebalance period returns.

```
[30]:
1 n_rows = 15 # set to None to display all rows
2 kpis.style_returns_df(backtest_kpis.returns_between_backtest_dates.iloc[:n_rows])
```

[30]:

backtest dates	equity_return_since_previous_backtest_date	rf_return_since_previous_backtest_date	equity_excess_return_since_previous_backtest_date
2023-01-05	N/A	0.012%	N/A
2023-01-06	-0.35%	0.012%	-0.36%
2023-01-09	0.12%	0.036%	0.09%
2023-01-10	-1.88%	0.012%	-1.89%
2023-01-11	-1.83%	0.012%	-1.84%
2023-01-12	-1.16%	0.012%	-1.17%
2023-01-13	-0.86%	0.012%	-0.87%
2023-01-17	0.54%	0.012%	0.52%
2023-01-18	0.20%	0.012%	0.19%
2023-01-19	2.83%	0.012%	2.81%
2023-01-20	0.23%	0.012%	0.22%
2023-01-23	-0.22%	0.036%	-0.26%
2023-01-24	1.03%	0.012%	1.02%
2023-01-25	0.48%	0.012%	0.46%
2023-01-26	-1.11%	0.012%	-1.12%

Figure 13: First 15 rows of the table of daily returns.

```
Strategy: top_bottom_k_from_factors_model
MR cure method: shrinking_exposures
Backtest period: 2023-01-05 to 2023-07-03
Number of backtest dates: 123


---------------      CHRONOLOGICAL ORDER OF CHANGES IN POSITIONS DURING BACKTEST      ---------------


shares in book on backtest date #1 (2023-01-05), at open:

ticker
A       0
AAL     0
AAP     0
AAPL    0
ABBV    0
       ..
YUM     0
ZBH     0
ZBRA    0
ZION    0
ZTS     0
Length: 460, dtype: int64
--------------------------------------------------

shares in book on backtest date #1 (2023-01-05), at close:

ticker
AIZ      100
APA     -100
CEG     -100
D        100
ENPH    -100
ETSY    -100
FCX     -100
FSLR    -100
GNRC     100
KMX      100
LUMN     100
NFLX    -100
NWL      100
RCL     -100
SLB     -100
STLD    -100
STX      100
SWK      100
TSN      100
VFC      100
dtype: int64
--------------------------------------------------

shares in book on backtest date #20 (2023-02-02), at close:

ticker
AIZ     -100
ETSY     100
FSLR     100
ALGN     100
BBWI     100
CCL      100
CME     -100
```

Figure 14: Beginning of the log of positions changes.

```
Strategy: top_bottom_k_from_factors_model
MR cure method: shrinking_exposures
Backtest period: 2023-01-05 to 2023-07-03
Number of backtest dates: 123


---------------      BACKTEST TRADES IN CHRONOLOGICAL ORDER      ---------------

(first trade date: 2023-01-05; last trade date: 2023-06-30)


shares traded on backtest date #1 (2023-01-05):

AIZ      100
APA     -100
CEG     -100
D        100
ENPH    -100
ETSY    -100
FCX     -100
FSLR    -100
GNRC     100
KMX      100
LUMN     100
NFLX    -100
NWL      100
RCL     -100
SLB     -100
STLD    -100
STX      100
SWK      100
TSN      100
VFC      100
Name: shares_traded, dtype: int64
--------------------------------------------------


shares traded on backtest date #20 (2023-02-02):

AIZ     -200
ALGN     100
APA      100
BBWI     100
CCL      100
CEG      100
CME     -100
CZR      100
D       -100
ENPH     100
ETSY     200
FCX      100
FSLR     200
GNRC    -100
HRL     -100
KDP     -100
KMX     -100
LUMN    -100
MRNA     100
NFLX     100
NVDA     100
NWL     -100
RCL      100
SEDG     100
SLB      100
STLD     100
STX     -100
SWK     -100
TSLA     100
TSN     -100
VFC     -100
Name: shares_traded, dtype: int64
--------------------------------------------------
```

Figure 15: Beginning of the log of trades.

```
Strategy: top_bottom_k_from_factors_model
MR cure method: shrinking_exposures
Backtest period: 2023-01-05 to 2023-07-03
Number of backtest dates: 123


--------------     CHRONOLOGICAL ORDER OF CHANGES IN SHORT-PROCEEDS LOTS DURING BACKTEST     --------------

      (on any given date, changes occur when opening, increasing, reducing, or closing short positions)


short-proceeds lots on backtest date #1 (2023-01-05), at open:

{}
------------------------------------------------

short-proceeds lots on backtest date #1 (2023-01-05), at close:

{'APA': deque([(100, 41.74)]), 'CEG': deque([(100, 82.38)]), 'ENPH': deque([(100, 240.27)]), 'ETSY': deque([(100, 119.51)]), 'FCX': deque([(100, 39.82)]), 'FSLR': deque([(100, 145.4)]), 'NFLX': deque([(100, 30.96)]), 'RCL': deque([(100, 53.36)]), 'SLB': deque([(100, 52.65)]), 'STLD': deque([(100, 98.75)])}
------------------------------------------------

short-proceeds lots on backtest date #20 (2023-02-02), at close:

{'AIZ': deque([(100, 133.53)]), 'CME': deque([(100, 175.93)]), 'HRL': deque([(100, 45.32)]), 'KDP': deque([(100, 35.2)])}
------------------------------------------------

short-proceeds lots on backtest date #29 (2023-02-15), at close:

{'AIZ': deque([(100, 133.53)]), 'HRL': deque([(100, 45.32)]), 'KDP': deque([(100, 35.2)]), 'CHRW': deque([(100, 105.14)]), 'CVS': deque([(100, 88.31)]), 'D': deque([(100, 58.23)]), 'EXR': deque([(100, 162.12)]), 'RSG': deque([(100, 125.76)]), 'TSN': deque([(100, 61.23)]), 'WM': deque([(100, 152.08)])}
------------------------------------------------

short-proceeds lots on backtest date #52 (2023-03-21), at close:

{'BSX': deque([(67, 48.82)]), 'LMT': deque([(67, 474.57)]), 'NKE': deque([(67, 125.56)]), 'PM': deque([(67, 95.66)]), 'ROST': deque([(67, 104.1)]), 'TJX': deque([(67, 76.33)]), 'ULTA': deque([(67, 515.7)]), 'V': deque([(67, 221.86)]), 'YUM': deque([(67, 128.5)]), 'ZBH': deque([(67, 127.74)])}
------------------------------------------------

short-proceeds lots on backtest date #75 (2023-04-24), at close:

{'ULTA': deque([(21, 515.7)]), 'ZBH': deque([(21, 127.74)]), 'AVGO': deque([(21, 63.43)]), 'BKNG': deque([(21, 107.14)]), 'CDNS': deque([(21, 212.89)]), 'CPRT': deque([(21, 39.26)]), 'FICO': deque([(21, 713.27)]), 'GE': deque([(21, 79.9)]), 'NVR': deque([(21, 5964.83)]), 'WYNN': deque([(21, 114.43)])}
------------------------------------------------

short-proceeds lots on backtest date #76 (2023-04-25), at close:

{'ULTA': deque([(21, 515.7), (9, 541.64)]), 'AVGO': deque([(21, 63.43), (9, 62.18)]), 'BKNG': deque([(21, 107.14), (9, 105.76)]), 'CDNS': deque([(21, 212.89), (9, 203.82)]), 'CPRT': deque([(21, 39.26), (9, 38.75)]), 'FICO': deque([(21, 713.27), (9, 709.24)]), 'GE': deque([(21, 79.9), (9, 78.53)]), 'NVR': deque([(21, 5964.83), (9, 5752.41)]), 'WYNN': deque([(21, 114.43), (9, 111.59)]), 'PHM': deque([(30, 64.92)])}
------------------------------------------------
```

Figure 16: Beginning of the log of changes of short-proceeds lots.

13.10 What this set of outputs is meant to show

The set of output diagnostics shown across section §13 serve to answer these questions:

1. **"Did the strategy make money?"**: visible from the equity plot, P&L section (KPI report), returns DataFrames.

2. **"Did it use too much risk?"**: visible from the leverage plot, leverage section (KPI report), and complementarily, the drawdown plot.

3. **"Where did the P&L come from?"**: visible in the book table, the LMV/SMV/short-proceeds plot, accruals section (financing, and dividends) and trading fees (KPI report).

4. **"What trades were carried out?"**: visible in the audit logs (trades and position changes).

5. **"How many margin requirement violations and subsequent cures were needed?"**: visible from MR-cure markers on the MR plot, and MR statistics (KPI report).

13.11 Additional backtest run examples

As further illustrations, we present diagnostics for backtest runs using different strategies (momentum, MVO (max return), and MVO (min variance)). We use use the same backtest calendar, costs and rules, and construction method for the investable tickers at each rebalance date as those specified in §13.1. We equally set the strategy return target to 9%. We also run these backtests under variant B: curing MR violations by shrinking our long and short exposures uniformely across all positions.

13.11.1 MVO strategies

For the next two backtest run examples using the MVO strategies (max-return and min-variance), we disable the dynamic rebalancing rule for these two runs, because MVO is optimizing for the entire upcoming period ending at the next scheduled rebalance date. This does not prevent us testing these strategies with the dynamic rebalancing rule enabled (which we do, as a comparative, when computing the Sharpe ratio of the strategies in section §14.1.6). Disabling the dynamic rebalancing rule implies that even if the running return since the last rebalance spikes before the next scheduled rebalance, we maintain our positions until period-end and rebalance only on the scheduled dates.

The constraints for the two MVO variants are set as presented in table 3.

Parameter	Max return	Min variance
Lower return bound	3%	—
Variance cap	—	4%
Beta cap	0.001	0.001
Gross exposure cap	200%	200%
Net exposure cap	0.001	0.001
Max shares per ticker	10,000	10,000

Table 3: Optimizer configurations for the two MVO variants. The beta cap and net exposure cap are set to 0.001 (i.e. effectively zero) to enforce an approximately beta-neutral and dollar-neutral book.

Furthermore, for each rebalance, to run the optimizer we need to provide the following inputs:

1. The **estimated expected return** for the upcoming period, for all tickers from the investable universe. We compute these estimations using the **factors models**, as explained in §9.7.1.3, configured with the following estimation parameters: $W_{\text{factor}} = 6$ rolling periods preceding the rebalance date, $W_{mom} = 92$ (trading days), $b_{mom} = 23$ (trading days), and $W_{vol} = 60$ (trading days).

2. The **estimated covariance matrix of the returns** for the upcoming period, for all tickers from the investable universe. As explained in §9.7.2, we compute the empirical covariance matrix choosing $W_{\Sigma} = 18$ rolling periods preceding the rebalance date, and then apply the Ledoit-Wolf regularization procedure.

3. The **estimated beta** for all tickers from the investable universe, as of the rebalance date. We follow the procedure from §9.7.3, choosing $W_{\beta} = 12$ rolling periods preceding the rebalance date.

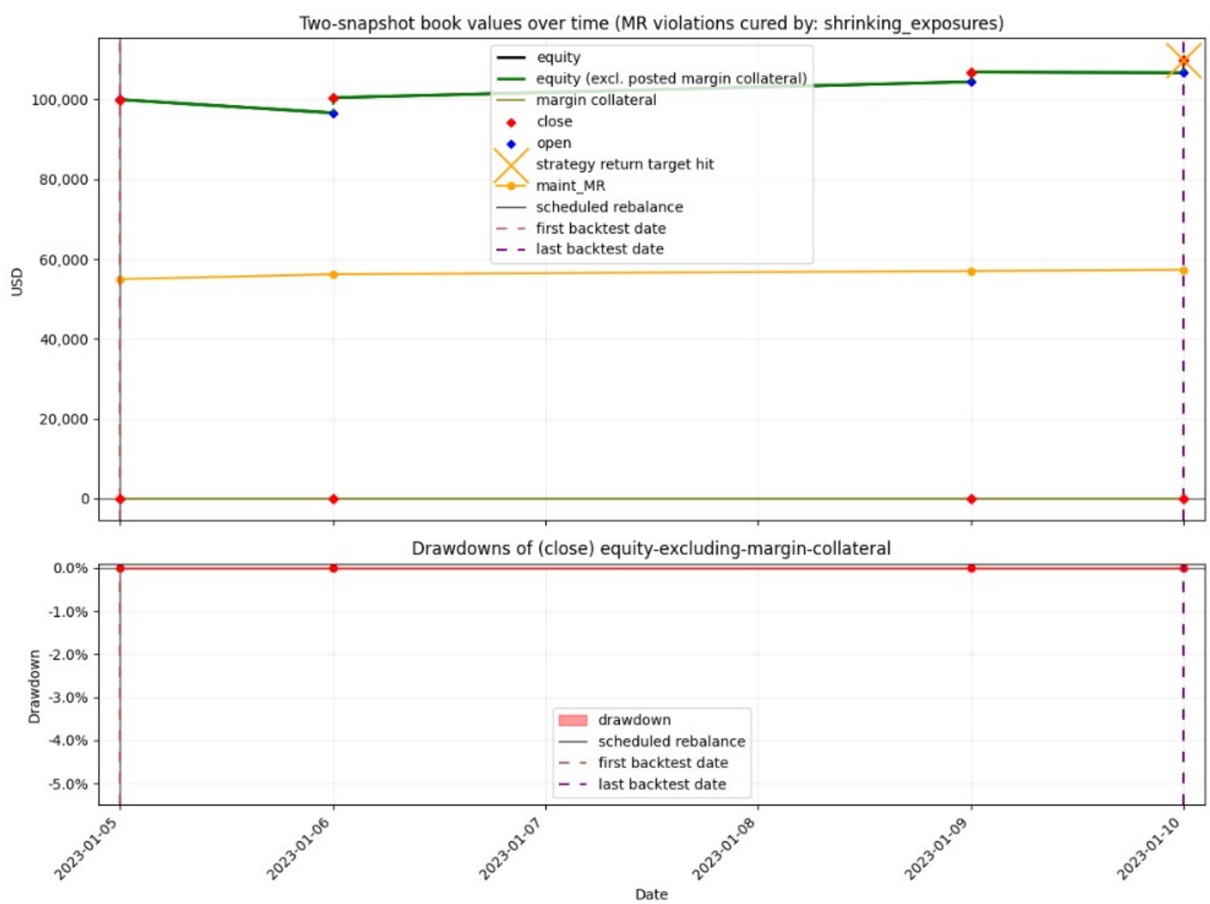

Figure 17: Equity, drawdown, and MR (strategy: MVO (max return)).

MVO (max return): In this backtest, the strategy reached an unrealized return on equity of 9.93% in just 4 backtest days (trading days), starting on `2023-01-05` and ending at `2023-01-10` (this precise information for the unrealized return is available from the KPI report for this run, which we do not show). As revealed by the log of position changes in our book shown in Figure 19,

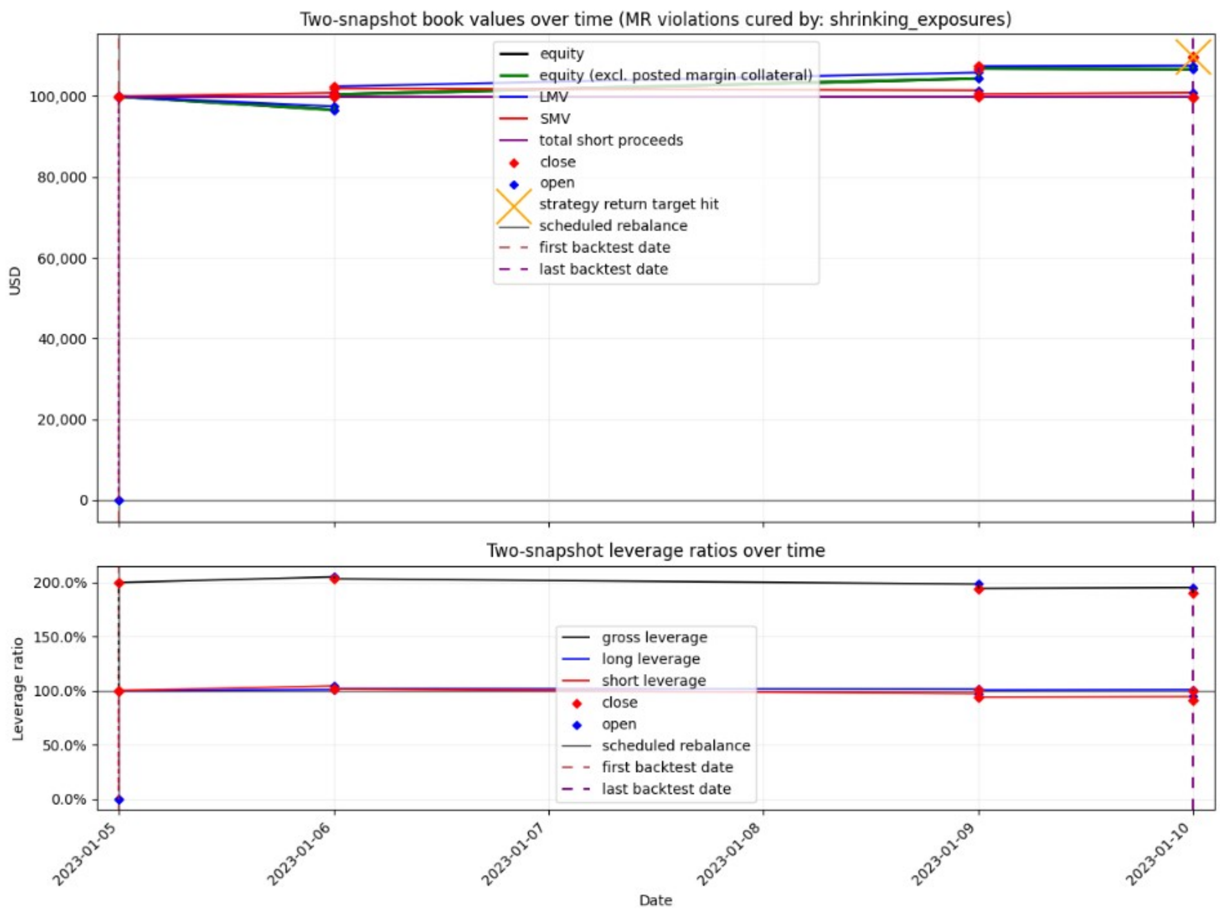

Figure 18: LMV, SMV, total short proceeds, Equity, and leverage ratios (strategy: MVO (max return)).

```
Strategy: return_mean_variance_optimization
MR cure method: shrinking_exposures
Backtest period: 2023-01-05 to 2023-01-10
Number of backtest dates: 4


---------------     CHRONOLOGICAL ORDER OF CHANGES IN POSITIONS DURING BACKTEST     ---------------


shares in book on backtest date #1 (2023-01-05), at open:

ticker
A       0
AAL     0
AAP     0
AAPL    0
ABBV    0
       ..
YUM     0
ZBH     0
ZBRA    0
ZION    0
ZTS     0
Length: 460, dtype: int64
--------------------------------------------------

shares in book on backtest date #1 (2023-01-05), at close:

ticker
GNRC    536
SJM    -627
TSLA    428
dtype: int64
--------------------------------------------------

shares in book on the last backtest date (2023-01-10), at the close:

(unchanged from the last logged change above)
--------------------------------------------------
```

Figure 19: Log of position changes (strategy: MVO (max return)).

[26]:

```
book_table = kpis.build_book_ledger_across_dates(backtest_results)
n_rows = None # set to None to display all rows
kpis.show_book(book_table.iloc[:n_rows])
```

[26]:

account	equity		equity_excluding_margin_collateral		margin_collateral		cash		debit		LMV		SMV		total_short_proceeds	
moment	open	close	open	close	open	close	open	close	open	close	open	close	open	close	open	close
date																
2023-01-05	100,000.00	99,918.47	100,000.00	99,918.47	0.00	0.00	100,000.00	60.21	0.00	0.00	0.00	99,898.24	0.00	99,937.53	0.00	99,897.55
2023-01-06	96,611.35	100,404.83	96,611.35	100,404.83	0.00	0.00	59.53	59.53	0.00	0.00	97,469.60	102,391.68	100,815.33	101,943.93	99,897.55	99,897.55
2023-01-09	104,385.42	106,830.26	104,385.42	106,830.26	0.00	0.00	57.42	57.42	0.00	0.00	105,860.24	107,402.20	101,429.79	100,526.91	99,897.55	99,897.55
2023-01-10	106,663.29	109,840.70	106,663.29	109,840.70	0.00	0.00	56.73	56.73	0.00	0.00	107,561.96	109,591.96	100,852.95	99,705.54	99,897.55	99,897.55

Figure 20: Table of book values (strategy: MVO (max return)).

backtest dates	equity_return_since_previous_backtest_date	rf_return_since_previous_backtest_date	equity_excess_return_since_previous_backtest_date
2023-01-05	N/A	0.012%	N/A
2023-01-06	0.49%	0.012%	0.47%
2023-01-09	6.40%	0.036%	6.36%
2023-01-10	2.82%	0.012%	2.81%

Figure 21: Table of daily returns (strategy: MVO (max return)).

out of all S&P 500 members at the initial rebalance date `2023-01-05`, the optimizer resolved the following optimal positions (after the two-step procedure to convert optimal weights to integer share counts, explained in section §9.6): long 536 shares of ticker GNRC, long 428 shares of ticker TSLA, and short 627 shares of ticker SJM. The table of daily book values shown in Figure 20 reveals that it was the long positions (LMV) which were overwhelmingly responsible for increasing the equity. Figure 21 shows the daily returns up to the end of the backtest, occurring at the first date the backtest return target of 9% was exceeded, which happened only 4 trading days after the start on `2023-01-05`; the 9.93% figure quoted above is simply the (realized) unrealized-return at the close of that first exceeding date. (Note that this 9% figure is the backtest's exit target and is distinct from the optimizer's 3% lower return bound listed in table 3.)

Figure 17 shows that there were no maintenance MR violations, nor did we need to invoke Variant B's cure mechanism (shrinking exposures) to address a rebalance MR violation. This is because the optimizer yielded optimal positions with small enough exposure, thereby preserving our existing equity above the rebalance margin requirement and above the maintenance margin requirement throught the backtest.

Figure 18 shows the close values of LMV and SMV time series (upper plot), and the leverage ratios (bottom plot), defined in §2.4. We can see how at the close of `2023-01-05` (thus after rebalancing on that date), the gross constrained cap of 200% fed to the optimizer was met.

MVO (min variance): This backtest achieved a 9.52% unrealized return on initial equity after 11 backtest days (trading days), and as shown in Figure 22, equity experienced two negligible drawdowns on only two dates. The main comparison remark we can make with respect to the max-return variant of the MVO backtest above is that, in contrast to the concentrated positions yielded by the max-return optimizer for the first rebalance, shown in Figure 19 (hundreds of shares across only 3 tickers), the optimizer for this MVO min-variance strategy opened positions across 39 tickers for the first rebalance, as shown in Figure 23. This fact resonates with the idea of "diversification", which is intuitively related to minimizing the variance of the return of the portfolio, as done by the optimizer of the min-variance variant of the MVO problem.

13.11.2 Momentum strategy

We test the momentum strategy (§7) with $k = 10$, $Q = 100$.

We use the same backtest calendar, costs and rules, and construction method for the investable tickers at each rebalance date as those specified in §13.1, with one exception: for the two runs below we **disable** the dynamic rebalancing rule (§5.4), so the portfolio is rebalanced only on the scheduled dates. We equally set the strategy return target to 9%.

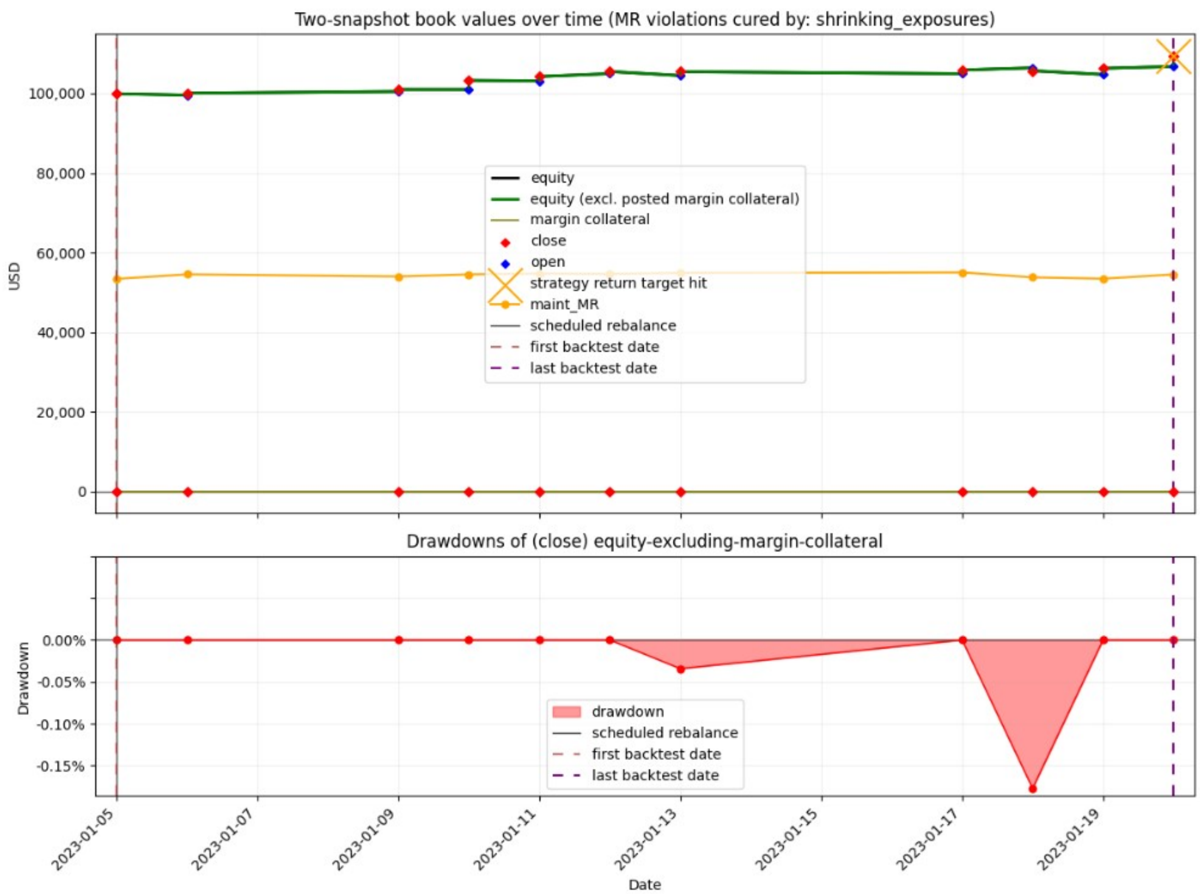

Figure 22: Equity, drawdown, and MR (strategy: MVO (min variance)).

```
Strategy: return_mean_variance_optimization
MR cure method: shrinking_exposures
Backtest period: 2023-01-05 to 2023-01-20
Number of backtest dates: 11


---------------     CHRONOLOGICAL ORDER OF CHANGES IN POSITIONS DURING BACKTEST     ---------------


shares in book on backtest date #1 (2023-01-05), at open:

ticker
A       0
AAL     0
AAP     0
AAPL    0
ABBV    0
       ..
YUM     0
ZBH     0
ZBRA    0
ZION    0
ZTS     0
Length: 460, dtype: int64
---------------------------------------------------

shares in book on backtest date #1 (2023-01-05), at close:

ticker
ABBV      -7
AFL      -73
AIG      -58
AON       -5
AWK       -5
AZO       -1
BBWI     185
BSX     -115
CAG      -81
CCL      339
CL       -77
CVS       -3
DRI      -27
DVA      164
FIS      328
GNRC      43
HRL      -37
IT        -5
JNJ       -4
KO       -52
LLY       -3
LNC      646
MCD      -33
MDLZ     -69
MET      -28
META     112
MTCH      50
ORLY    -129
PEP      -32
PG       -68
REGN      -1
RSG      -28
SEDG       2
SJM      -26
TJX       -1
WBD      967
WDC       32
WM       -20
YUM      -27
dtype: int64
---------------------------------------------------
```

Figure 23: Log of position changes (strategy: MVO (min variance)).

[26]:
```
book_table = kpis.build_book_ledger_across_dates(backtest_results)
n_rows = None # set to None to display all rows
kpis.show_book(book_table.iloc[:n_rows])
```

[26]:

account	equity		equity_excluding_margin_collateral		margin_collateral		cash		debit		LMV		SMV		total_short_proceeds	
moment	open	close	open	close	open	close	open	close	open	close	open	close	open	close	open	close
date																
2023-01-05	100,000.00	99,918.26	100,000.00	99,918.26	0.00	0.00	100,000.00	421.29	0.00	0.00	0.00	99,535.01	0.00	95,113.25	0.00	95,075.21
2023-01-06	99,582.16	100,054.48	99,582.16	100,054.48	0.00	0.00	420.67	420.67	0.00	0.00	100,073.72	101,708.56	95,987.44	97,149.96	95,075.21	95,075.21
2023-01-09	100,482.91	100,993.77	100,482.91	100,993.77	0.00	0.00	676.82	676.82	0.00	0.00	101,732.60	101,105.99	97,001.72	95,864.25	95,075.21	95,075.21
2023-01-10	100,988.58	103,262.67	100,988.58	103,262.67	0.00	0.00	676.22	676.22	0.00	0.00	101,027.20	103,287.78	95,790.06	95,776.54	95,075.21	95,075.21
2023-01-11	103,152.22	104,248.10	103,152.22	104,248.10	0.00	0.00	675.63	675.63	0.00	0.00	103,517.84	104,225.10	96,116.45	95,727.85	95,075.21	95,075.21
2023-01-12	104,952.88	105,500.38	104,952.88	105,500.38	0.00	0.00	664.68	664.68	0.00	0.00	105,098.36	104,691.42	95,885.37	94,930.93	95,075.21	95,075.21
2023-01-13	104,486.83	105,464.28	104,486.83	105,464.28	0.00	0.00	653.92	653.92	0.00	0.00	103,340.55	105,174.61	94,582.85	95,439.45	95,075.21	95,075.21
2023-01-17	104,949.08	105,879.67	104,949.08	105,879.67	0.00	0.00	651.55	651.55	0.00	0.00	104,781.16	105,622.31	95,558.83	95,469.39	95,075.21	95,075.21
2023-01-18	106,418.91	105,692.17	106,418.91	105,692.17	0.00	0.00	650.96	650.96	0.00	0.00	105,989.91	103,276.52	95,297.17	93,310.52	95,075.21	95,075.21
2023-01-19	104,761.61	106,298.93	104,761.61	106,298.93	0.00	0.00	586.48	586.48	0.00	0.00	102,052.90	103,002.20	92,952.98	92,364.96	95,075.21	95,075.21
2023-01-20	106,760.42	109,433.11	106,760.42	109,433.11	0.00	0.00	549.72	549.72	0.00	0.00	103,542.72	106,710.48	92,407.22	92,902.30	95,075.21	95,075.21

Figure 24: Table of book values (strategy: MVO (min variance)).

We test both Variant A (curing MR violations by posting margin collateral) and Variant B (curing MR violations by shrinking our long and short exposures uniformly across all tickers).

As reported by the *Backtest P&L* section of the KPI report for each run (not shown here), Variant A reached an unrealized return on equity of 10.38% after 169 backtest days (trading days), and Variant B reached an unrealized return on equity of 11.07% after 171 backtest days (trading days). The backtest stopped at the first date for which the value of equity **excluding margin collateral** at the close represented an unrealized return of 9% or higher relative to initial equity.

The plots in Figure 25 show the values at close for the equity including margin collateral, the equity excluding margin collateral, the margin collateral posted, the maintenance MR, as well as the MR violation cure events. Figure 27 shows the MR violation events for both runs. We see that no maintenance MR violations occurred during either run (our equity was above the maintenance margin requirement throughout). We also see that running this backtest under Variant A required posting \$49,139.59 as margin collateral to cure all the rebalance margin requirement violations during the backtest (only 2). Under Variant B, the number of rebalance MR violations was higher (6) than with Variant A.

Figure 26 shows that the evolution of our long and short exposures (and the leverage ratios) was similar, although not identical, across the two variants.

Need for a systematic comparison of strategies across backtest runs

These examples show the need for a methodology to compare backtest performance systematically across configurations: different historical periods, return targets, the dynamic rebalancing rule enabled or disabled, curing MR violations by shrinking exposures (Variant B) or posting margin collateral (Variant A), different strategy settings and estimation parameters of the inputs, etc. This is addressed to a certain extent in §14.

14 Systematic Performance Comparison of Strategies

14.1 One comparison metric for risk-adjusted performance between strategies: the Sharpe ratio

We wish to compare the risk-adjusted performance of the strategies over a given historical period: between dates d_1 and d_2 (e.g. $d_1 =$ `2010-03-05` and $d_2 =$ `2020-01-16`).

Let $\{1, \ldots, N\}$ index all consecutive backtest dates (which coincide with all trading days) in the window $]d_1, d_2]$.

Consider the discrete-time stochastic process of daily excess returns $(r_k^{d,e})_{k\in\{1,\ldots,N\}}$, where

$$r_k^{d,e} = r_k^d - r_k^f,$$

with

$$r_k^d = \ln\left(\frac{\widetilde{\text{Equity}}_k}{\widetilde{\text{Equity}}_{k-1}}\right)$$

the daily log-return (of equity excluding our posted margin collateral) between consecutive days $k-1$ and k, and r_k^f the daily log-return of the risk-free asset.

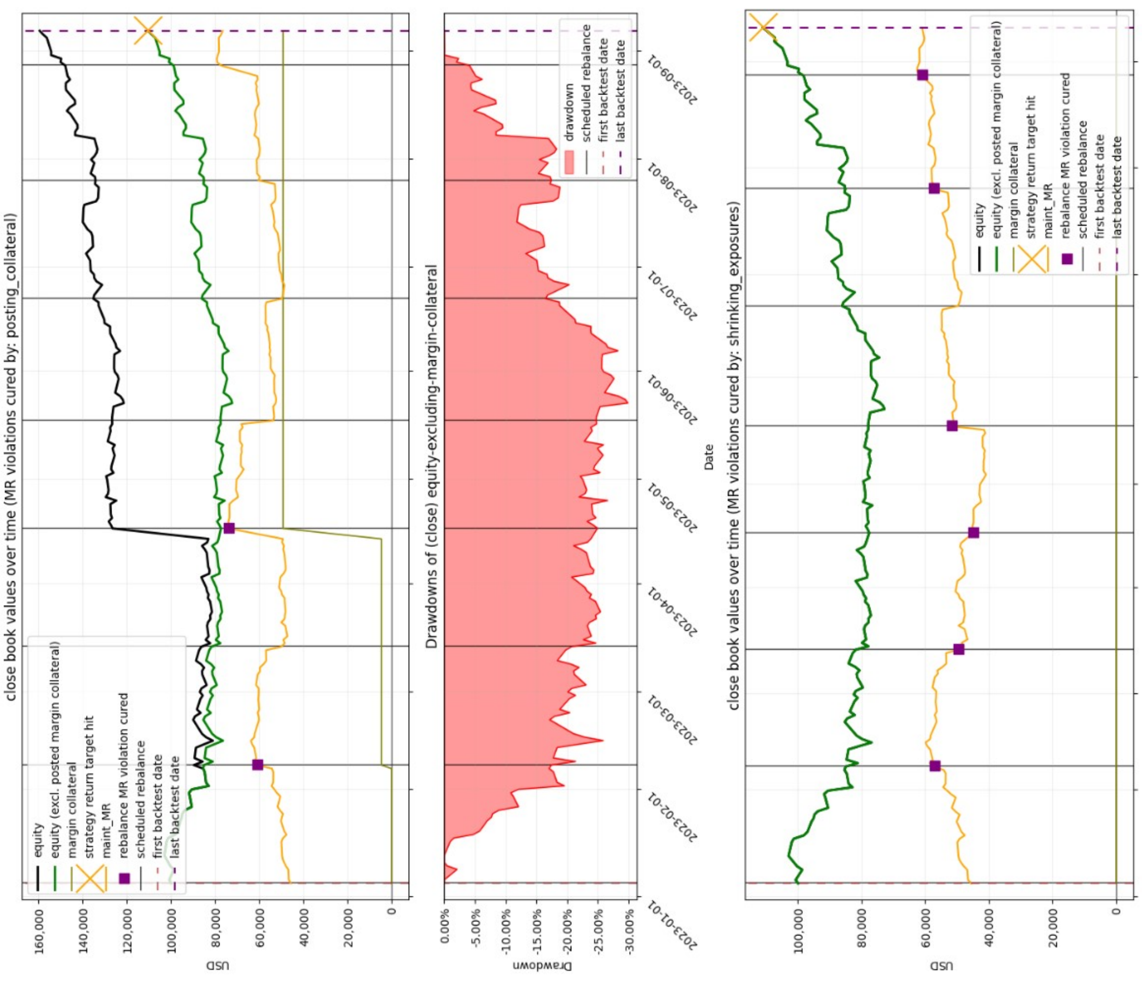

Figure 25: Plot of equity, drawdown, and MR at close (momentum strategy, Variant A (top), Variant B (bottom)).

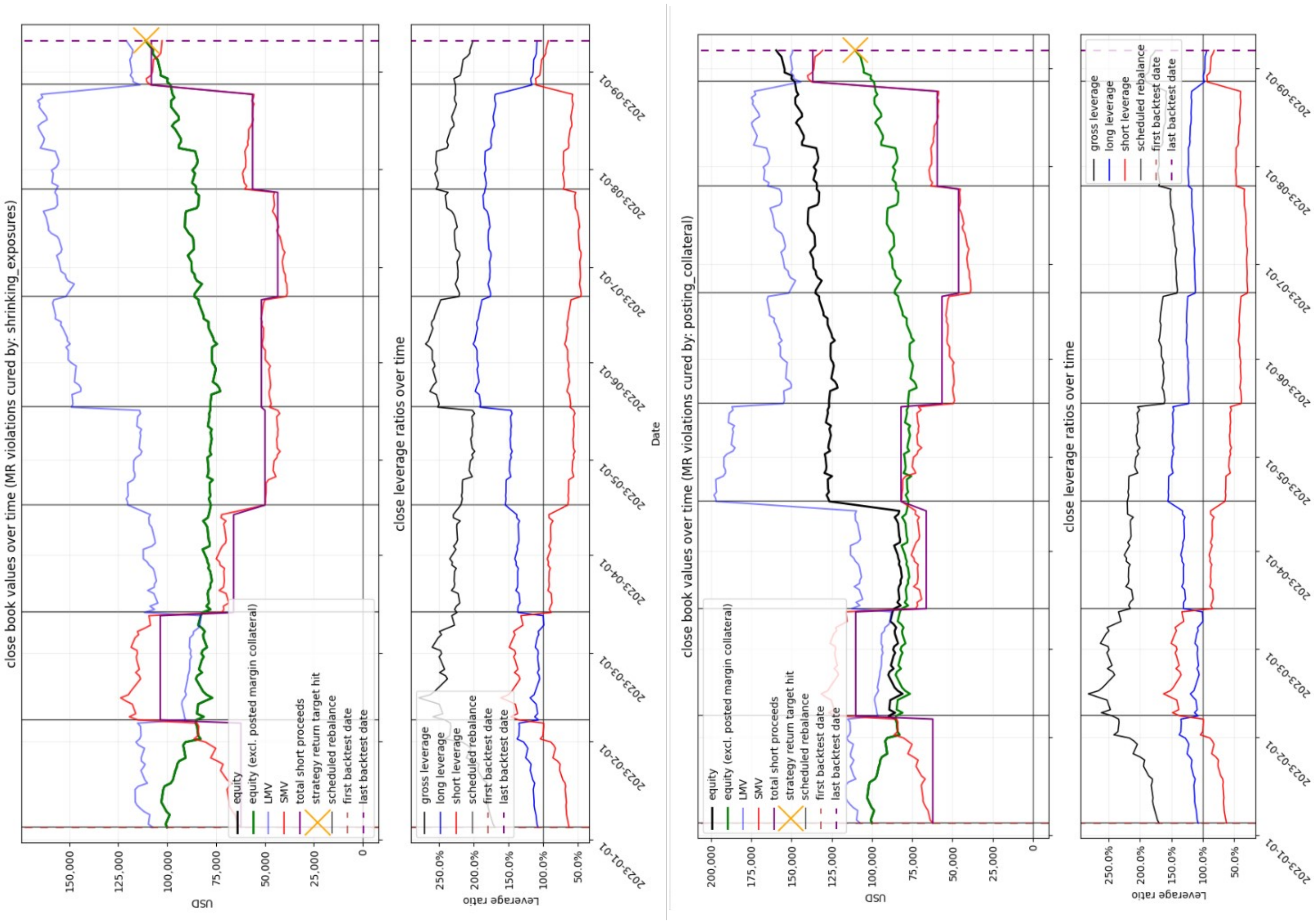

Figure 26: Plot of the values of LMV, SMV, total short proceeds, and leverage ratios at close (momentum strategy, Variant B (top), Variant A (bottom)).

```
--------------------------------------------------------------------------------
                              MARGIN REQUIREMENTS
--------------------------------------------------------------------------------
 maint_MR_violation_number_of_backtest_days                                    0
 maint_MR_violation_frequency                                              0.00%
 rebalance_MR_cure_number_of_backtest_days                                     2
 rebalance_MR_cure_frequency                                              25.00%
 cure_method                                                  posting_collateral
 shrink_factor_to_cure_an_MR_violation                                       N/A
 collateral_per_cure:
   min                                                                  4,583.66
   max                                                                 44,555.93
   avg                                                                 24,569.80
   std                                                                 19,986.13
 total_collateral_posted                                               49,139.59
```

```
--------------------------------------------------------------------------------
                              MARGIN REQUIREMENTS
--------------------------------------------------------------------------------
 maint_MR_violation_number_of_backtest_days                                    0
 maint_MR_violation_frequency                                              0.00%
 rebalance_MR_cure_number_of_backtest_days                                     6
 rebalance_MR_cure_frequency                                              75.00%
 cure_method                                                 shrinking_exposures
 shrink_factor_to_cure_an_MR_violation:
   min                                                                    62.00%
   max                                                                   100.00%
   avg                                                                    88.30%
   std                                                                    13.40%
 collateral_per_cure                                                         N/A
 total_collateral_posted                                                     N/A
```

Figure 27: *Margin Requirements* section from the KPI report for both runs: Variant A (top), Variant B (bottom).

Remark

To observe a realization of the stochastic process of daily (excess) log-returns for window $[d_1, d_2]$, we must run a backtest for the configured strategy, and disable any termination rule (stop-loss, or target-hit), in order to let the backtest run through the entire window $[d_1, d_2]$.

If we assume the stochastic process of daily excess log-returns $(r_k^{d,e})_{k\in\{1,\dots,N\}}$ satisfies

$$\begin{cases} \forall k, & \mathbb{E}[r_k^{d,e}] = \mu^e \\ \forall k, & \mathrm{Var}(r_k^{d,e}) = \sigma^2 \end{cases}$$

i.e. that daily excess returns have *constant mean* and *constant variance* (in particular, are *homoskedastic*) across all dates, then we can define the **daily Sharpe ratio** (here for the log-return of equity excluding posted margin collateral) as

$$\mathrm{SR}^{\mathrm{daily}} = \frac{\mu^e}{\sigma}.$$

These assumptions on the mean and variance of daily returns are **not realistic**. The variance assumption is the more clearly violated one: equity returns (including those of the S&P 500 constituents forming the equity) exhibit well-documented volatility clustering (time-varying conditional variance), so the homoskedasticity $\mathrm{Var}(r_k^{d,e}) = \sigma^2$ fails. The constant-mean assumption is also questionable, given documented time variation in expected return (e.g. momentum).

Nevertheless, without assuming any property of the underlying process, we can compute the empirical mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$ of the sample of **observed** daily excess-log returns, $(\widehat{r}_k^{d,e})_{\{1,\dots,N\}}$, and form the empirical ratio $\hat{\mu}/\hat{\sigma}$, where

$$\hat{\mu} := \frac{1}{N}\sum_{k=1}^{N} \hat{r}_k^{d,e}, \qquad \hat{\sigma} := \sqrt{\frac{1}{N-1}\sum_{k=1}^{N}\left(\hat{r}_k^{d,e} - \hat{\mu}\right)^2}.$$

This empirical daily Sharpe ratio is therefore equal to the **realized** (excess) daily return (of equity excluding margin collateral) **per unit of realized (excess) daily volatility**, over the chosen historical period $[d_1, d_2]$.

It is a descriptive statistic of the realized return path: it requires no assumption to be well-defined, only the **interpretation** that it summarizes what was realized over **this specific window**, **not** an estimate of a forward-looking population parameter.

This is the daily statistic we compute across all strategies, in their chosen configuration (including the initial capital deployed, which we set the same for all strategies), and use as a comparison metric for risk-adjusted daily performance between strategies across the selected historical periods.

Remark

Because our backtesting framework already reflects trading costs, financing costs, dividends, and margin-requirement enforcement in equity, the daily excess returns used to compute the Sharpe ratio already **account for these simulated real-world frictions**.

14.1.1 Interpretation of the Sharpe Ratio

The Sharpe ratio is the *expected excess return per unit of risk*. It admits two complementary interpretations.

The first is geometric, and arises from the two-fund separation result of Markowitz portfolio optimization theory [16]. In a universe of finitely many investable assets, the portfolio of risky assets with the maximum Sharpe ratio is the *tangency portfolio*. It has the following optimality property: if an investor allocates wealth between the tangency portfolio and the risk-free asset, then, for any chosen level of risk (standard deviation), the expected return of the blended portfolio is greater than or equal to that of any portfolio composed only of risky assets, with equality attained only when all wealth is invested entirely to the tangency portfolio itself. (Risk levels above that of the tangency portfolio are reached by leveraging it, i.e. allocating more than the invested wealth to it and borrowing at the risk-free rate.)

The second interpretation is probabilistic: under specific assumptions, the Sharpe ratio controls the probability that a single-period excess return is positive. The remainder of this subsection develops this second interpretation.

In this subsection, let $(r_k)_k$ denote the time series of excess returns over a fixed horizon (daily, annual, or some other window.). For what follows, we assume **weak stationarity**:

$$\forall k, \quad \mathbb{E}[r_k] = \mu, \qquad \mathrm{Var}(r_k) = \sigma^2,$$

with $\sigma > 0$, and the Sharpe ratio is defined as:

$$S = \frac{\mu}{\sigma}.$$

A Gaussian interpretation: If we further assume each period excess return is normally distributed ($r_k \sim \mathcal{N}(\mu, \sigma^2)$), then we have

$$\begin{aligned}
\mathbb{P}(r_{k+1} > 0) &= \mathbb{P}\left(\frac{r_{k+1} - \mu}{\sigma} > -\frac{\mu}{\sigma}\right) \\
&= \mathbb{P}\big(Z > -S\big), \qquad Z \sim \mathcal{N}(0,1) \\
&= \Phi(S),
\end{aligned}$$

where Φ is the standard normal CDF. Note that this holds for *any* sign of the Sharpe ratio S.

A distribution-free interpretation Without a distributional assumption we cannot compute $\mathbb{P}(r_{k+1} > 0)$ exactly. Nevertheless, a one-sided bound holds for *every* distribution with finite mean and variance. We can invoke *Cantelli's inequality*: for the random variable r_{k+1} with mean μ and variance σ^2, and any $\lambda > 0$,

$$\mathbb{P}\big(r_{k+1} - \mu \le -\lambda\sigma\big) \le \frac{1}{1+\lambda^2}.$$

The event $\{r_{k+1} \le 0\}$ can be written $\{r_{k+1} - \mu \le -\mu\} = \{r_{k+1} - \mu \le -S\sigma\}$, which matches Cantelli with $\lambda = S$. This requires $\lambda = S > 0$, therefore the following bound applies only for a **positive** Sharpe ratio:

$$\boxed{\mathbb{P}(r_{k+1} \le 0) \le \frac{1}{1+S^2} \quad \Longleftrightarrow \quad \mathbb{P}(r_{k+1} > 0) \ge \frac{S^2}{1+S^2}} \qquad (S > 0).$$

In words: under the weak stationarity assumption above, a Sharpe ratio $S > 0$ guarantees that the probability of a positive excess return in any single period is *at least* $\dfrac{S^2}{1+S^2}$, regardless of the shape of the distribution.

Comparison

S	Distribution-free: $\mathbb{P}(r_{k+1} > 0) \geq \dfrac{S^2}{1+S^2}$	Gaussian: $\mathbb{P}(r_{k+1} > 0) = \Phi(S)$
0.5	20.0%	69.1%
1	50.0%	84.1%
2	80.0%	97.7%
3	90.0%	99.87%

The distribution-free bound is much weaker than the Gaussian result: at $S = 1$ it only guarantees a coin flip (50%), versus 84% under normality. The gap is the price of distributional agnosticism.

Caveat: The result above describes only the **marginal distribution** of a single period return r_{k+1}, using just its mean and variance, under the assumption of weak stationarity of the stochastic process $(r_k)_k$. The marginal result says nothing about temporal dependence. In particular,

$$\mathbb{P}(r_{k+1} > 0 \mid r_k < 0) \neq \mathbb{P}(r_{k+1} > 0)$$

in general, and the value of $\mathbb{P}(r_{k+1} > 0 \mid r_k < 0)$, which we have not estimated, is what answers the question "after a loss from the previous period, will we have a gain at the next period?".

14.1.2 Model 1: Annualization under constant mean, constant variance, and zero autocovariance at all lags

If we additionally assume the underlying process for daily log-returns has constant mean and variance across all dates and zero autocovariance at every nonzero lag (i.e. assume ***weak stationarity of daily log-returns***), then $\mathrm{SR}^{\text{daily}}$ is well-defined, and we can ask: what is the **annualized** risk-adjusted performance of each strategy? Can we define an annualized Sharpe ratio independent of the date? Under these assumptions, the answer is **yes**.

Throughout this subsection we write $\mu := \mathbb{E}[r_t^d]$ for the (constant) *raw* daily mean and $r^{f,d}$ for the (a.s. constant) daily risk-free rate, so that the constant excess daily mean is $\mu^e := \mu - r^{f,d}$ (consistent with the daily Sharpe ratio $\mathrm{SR}^{\text{daily}} = \mu^e/\sigma$ defined in §14.1).

Fix any date t and define $r^{\text{annual}}(t, t+252)$, the annual log-return between t and $t + 252$ trading days:

$$r^{\text{annual}}(t, t+252) = \ln\left(\frac{\widetilde{\text{Equity}}_{t+252}}{\widetilde{\text{Equity}}_t}\right) = \sum_{k=0}^{251} \ln\left(\frac{\widetilde{\text{Equity}}_{t+k+1}}{\widetilde{\text{Equity}}_{t+k}}\right) = \sum_{k=0}^{251} r_{t+k}^d$$

The log-return telescopes into a **sum** of daily log-returns (this additive structure is the reason the derivation works for log-returns specifically, and therefore why we chose log-returns instead of simple returns).

Taking expectations and using linearity:

$$\mathbb{E}\Big[r^{\text{annual}}(t,t+252)\Big] = \sum_{k=0}^{251} \mathbb{E}\Big[r^d_{t+k}\Big] = 252\,\mu \qquad \text{(independent of } t\text{)},$$

and hence the excess annual mean is

$$\mathbb{E}\Big[r^{\text{annual},e}\Big] = 252\,(\mu - r^{f,d}) = 252\,\mu^e.$$

Note

The risk-free rate is treated as an almost-surely (a.s.) constant random variable; subtracting a constant shifts the mean but leaves all variances and covariances unchanged.

For the variance, since $\mathrm{Var}(X - c) = \mathrm{Var}(X)$ for a constant c, the excess and raw annual returns have the same variance:

$$\begin{aligned}
\mathrm{Var}\Big(r^{\text{annual},e}(t,t+252)\Big) &= \mathrm{Var}\Big(r^{\text{annual}}(t,t+252) - r^{f,\text{annual}}\Big) \\
&= \mathrm{Var}\Big(r^{\text{annual}}(t,t+252)\Big) \\
&= \mathrm{Var}\left(\sum_{k=0}^{251} r^d_{t+k}\right) \\
&= \sum_{k=0}^{251} \mathrm{Var}\Big(r^d_{t+k}\Big) + \sum_{\substack{i,j=0\\ i\neq j}}^{251} \mathrm{Cov}\Big(r^d_{t+i}, r^d_{t+j}\Big)
\end{aligned} \tag{5}$$

If we assume the process $(r^d_t)_t$ is homoskedastic (i.e., $\mathrm{Var}(r^d_t) = \sigma^2$ for all t), **and** has zero autocovariance at all nonzero lags, i.e.

$$\forall t,\ \forall k \in \mathbb{Z}^*, \quad \mathrm{Cov}\Big(r^d_t, r^d_{t+k}\Big) = 0,$$

then the off-diagonal terms in (5) vanish and $\mathrm{Var}(r^{\text{annual},e}) = 252\,\sigma^2$.

The annual Sharpe ratio of the excess return (equity excl. posted margin collateral) is then

$$\mathrm{SR}^{\text{annual}} = \frac{252\,\mu^e}{\sqrt{252\,\sigma^2}} = \sqrt{252}\,\frac{\mu^e}{\sigma} = \sqrt{252}\,\mathrm{SR}^{\text{daily}}.$$

14.1.3 Model 2: Annualization under constant mean, constant variance, and time-invariant autocovariance (Lo's correction)

We now relax the zero-autocovariance assumption while retaining weak stationarity. Concretely, we assume the autocovariance of $(r^d_t)_t$ is time-invariant: a function of the lag only, not of calendar time t (the autocovariances at different lags may take different values from one another):

$$\forall t,\ \forall k \in \mathbb{Z}, \quad \mathrm{Cov}\Big(r^d_t, r^d_{t+k}\Big) = \gamma(k) \qquad \text{(independent of } t\text{)}.$$

We can then define the **autocorrelation function** (ACF):

$$\rho(k) = \frac{\mathrm{Cov}\left(r_t^d, r_{t+k}^d\right)}{\sqrt{\mathrm{Var}\left(r_t^d\right)}\sqrt{\mathrm{Var}\left(r_{t+k}^d\right)}} \quad \text{(independent of } t\text{)} \tag{6}$$

Given the constant-variance assumption, $\mathrm{Var}(r_t^d) = \mathrm{Var}(r_{t+k}^d) = \gamma(0) = \sigma^2 > 0$, this collapses to

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} \tag{7}$$

(Without assuming constant variance, the expression (6) is the *Pearson correlation*, always in $[-1, 1]$, and the expression (7) is the *ACF*; they coincide precisely because of the stationarity assumption.)

Remark 1 (boundedness and the equality case)

By the Cauchy–Schwarz inequality applied to the inner product $\langle X, Y\rangle = \mathbb{E}[XY]$ on the space $L^2(\Omega, \mathcal{F}, \mathbb{P})$ of finite-second-moment random variables, together with the definition of covariance,

$$\forall k \in \mathbb{Z}, \quad |\rho(k)| \leq 1.$$

Equality $|\rho(k)| = 1$ is exactly the case of **affine dependence**: there exists $\alpha \neq 0$ with

$$r_{t+k}^d - \mu = \alpha\,(r_t^d - \mu) \quad \text{a.s.},$$

where $\mu = \mathbb{E}[r_t^d]$ (independent of t), and $\mathrm{sign}(\alpha) = \rho(k)$.

Given the constant-variance assumption, taking variances of both sides gives $\sigma^2 = \alpha^2\,\sigma^2$; since $\sigma^2 > 0$, this yields $|\alpha| = 1 = |\rho(k)|$, so $\alpha = \rho(k) \in \{-1, +1\}$. Therefore:

$$\begin{aligned}
\rho(k) = +1 &\iff \left(\forall t,\ r_{t+k}^d - \mu = +(r_t^d - \mu) \text{ a.s.}\right) &&\iff \left(\forall t,\ r_{t+k}^d = r_t^d \text{ a.s.}\right)\\
\rho(k) = -1 &\iff \left(\forall t,\ r_{t+k}^d - \mu = -(r_t^d - \mu) \text{ a.s.}\right) &&\iff \left(\forall t,\ r_{t+k}^d = \mu + (\mu - r_t^d) \text{ a.s.}\right)
\end{aligned}$$

These equality cases are boundary curiosities: each forces a deterministic a.s. relationship between returns k steps apart that, imposed *for all* t under constant variance, cannot be realized by a genuine (non-degenerate) stationary return series. They serve only to characterize when the Cauchy–Schwarz bound is attained.

Remark 2 (parity)

$$\forall k \in \mathbb{Z}, \quad \gamma(k) = \gamma(-k) = \gamma(|k|), \qquad \rho(k) = \rho(-k) = \rho(|k|).$$

Under these assumptions,

$$\mathrm{Var}\left(\sum_{k=0}^{251} r_{t+k}^d\right) = 252\,\sigma^2 + \sum_{\substack{i,j=0\\ i\neq j}}^{251} \mathrm{Cov}\left(r_{t+i}^d, r_{t+j}^d\right),$$

with, using the parity of $\gamma(\cdot)$,

$$\mathrm{Cov}\Big(r^d_{t+i}, r^d_{t+j}\Big) = \gamma\big((t+j)-(t+i)\big) = \gamma(|i-j|).$$

Let $n := 251$ (so that the annual horizon spans $n+1 = 252$ daily returns). Then

$$\sum_{\substack{i,j=0\\ i\neq j}}^{n} \mathrm{Cov}\Big(r^d_{t+i}, r^d_{t+j}\Big) = \sum_{k=1}^{n} \gamma(k)\, n_k, \qquad n_k := \#\big\{(i,j)\in\{0,\dots,n\}^2 \mid i\neq j,\ |i-j| = k\big\}.$$

For $(i,j)\in\{0,\dots,n\}^2$ with $i \neq j$, we have $|i-j| \in \{1,\dots,n\}$. Fix $k \in \{1,\dots,n\}$. Then

$$|i-j| = k \iff (i = j+k) \text{ or } (i = j-k).$$

Counting the valid pairs in each case (j ranges so that both indices stay in $\{0,\dots,n\}$) gives $n-k+1$ each, hence

$$n_k = 2(n-k+1),$$

so that

$$\sum_{\substack{i,j=0\\ i\neq j}}^{n} \mathrm{Cov}\Big(r^d_{t+i}, r^d_{t+j}\Big) = \sum_{k=1}^{n} 2(n-k+1)\,\gamma(k).$$

Recalling $\gamma(k) = \rho(k)\,\gamma(0) = \sigma^2\rho(k)$ (where we have used $\gamma(0) = \mathrm{Var}(r^d_t) = \sigma^2$),

$$\mathrm{Var}\left(\sum_{k=0}^{251} r^d_{t+k}\right) = 252\,\sigma^2 + \sigma^2 \sum_{k=1}^{n} 2(n-k+1)\,\rho(k), \qquad n = 251.$$

Therefore, writing $m := n+1 = 252$ for the horizon length (so that $n = m-1$ and $n-k+1 = m-k$),

$$\begin{aligned}
\mathrm{SR}^{\mathrm{annual}} &= \frac{\mathbb{E}[r^{\mathrm{annual},e}]}{\sqrt{\mathrm{Var}(r^{\mathrm{annual},e})}} \\
&= \frac{252\,\mu^e}{\sqrt{\sigma^2\Big(252 + \sum_{k=1}^{m-1} 2(m-k)\,\rho(k)\Big)}} \\
&= \sqrt{252}\,\frac{\mu^e}{\sigma}\,\frac{1}{\sqrt{1 + \frac{2}{m}\sum_{k=1}^{m-1}(m-k)\,\rho(k)}},
\end{aligned}$$

i.e.

$$\mathrm{SR}^{\mathrm{annual}} = \sqrt{252}\cdot \mathrm{SR}^{\mathrm{daily}}\cdot \eta(m), \qquad m = 252,$$

where

$$\eta(m) = \left[1 + \frac{2}{m} \sum_{k=1}^{m-1} (m-k)\,\rho(k) \right]^{-1/2}$$

is **Lo's correction term** (Lo, 2002, *The Statistics of Sharpe Ratios*, Financial Analysts Journal).

Remark 1

- If $\rho(k) > 0$ for all lags $k \in \{1, \ldots, m-1\}$ (positive autocorrelation, i.e. *"persistence"* of daily returns at all lags k), then $\eta(m) < 1$, so annualizing the daily Sharpe by the factor $\sqrt{252}$ **overstates** the annual Sharpe;
- If $\rho(k) < 0$ for all lags $k \in \{1, \ldots, m-1\}$ (negative autocorrelation, i.e. *"mean reversion"* of daily returns at all lags k), then $\eta(m) > 1$, so annualizing the daily Sharpe by the factor $\sqrt{252}$ **understates** the annual Sharpe;
- If $\rho(k) = 0$ for all lags $k \in \{1, \ldots, m-1\}$ (uncorrelated daily returns at all lags k), then $\eta(m) = 1$, recovering the same annual Sharpe as Model 1.

More generally, only the sign of the weighted sum $\sum_{k=1}^{m-1}(m-k)\rho(k)$ matters; uniform-sign autocorrelations are a sufficient but not necessary condition for over- or under-statement.

Remark 2

We see, from the sum $\sum_{k=1}^{m-1}(m-k)\,\rho(k)$, that the dominant weights are on the **low** lags (weight 251 on $\rho(1)$, decaying linearly to weight 1 on $\rho(251)$). So η is driven mainly by short-lag autocorrelation.

14.1.4 What neither annualization repairs (model 1 or 2)

To annualize the daily Sharpe ratio, both the $\sqrt{252}$-annualization (Model 1) and the Lo-corrected annualization (Model 2) still require the **mean and variance to be constant over the window**. Against a series with drift in μ^e or σ^2 (which daily strategy returns exhibit empirically, as documented in the literature) neither annualized Sharpe ratio is a well-defined population quantity. Lo's correction repairs the *autocorrelation* term only; it does not rescue a non-stationary mean and variance. We therefore read the annualized Sharpe ratio more like a **realized, descriptive, rescaled comparative diagnostics across strategies over an identical window**, and additionally **rely on the max-drawdown** value over the backtest run for the considered historical period to provide an enriched description of how risk-adjusted performance varies across said historical periods (because max-drawdown is a *path* statistic, directly sensitive to the realized regime drift in μ^e and σ^2 that the Sharpe ratio does not capture).

14.1.5 Conclusion: what we report in practice

For each strategy and historical period we report two figures:

1. **the daily statistic** $\hat{\mu}/\hat{\sigma}$, as it is a descriptive statistic of the realized return path, well-defined with no assumptions on the underlying process, and computed identically for every strategy over an identical period.

2. **the annualized Sharpe under Model 1,** $\sqrt{252}\,\hat{\mu}/\hat{\sigma}$, reported only to place (1) on the familiar annual scale. We read it as a rescaling of (1), not as an estimate of a forward-looking population parameter (*i.e.* a forward-looking risk-adjusted figure assumed to persist into the future).

We do not report the Lo-corrected figure (Model 2), as Lo's correction repairs the autocorrelation term of an annualization whose constant-mean / constant-variance premise is already violated on our data (see §14.1.4). Correcting the smaller, autocorrelation-induced distortion while leaving the larger non-stationarity untouched would lend the annual figure a false air of precision. The dominant threat to any annualized Sharpe here is **regime drift** in μ^e and σ^2, which η does not address; we therefore prefer the assumption-free daily statistic, the plain $\sqrt{252}$ rescaling, and (for the risk dimension that no single Sharpe captures) the max-drawdown (see §12.5).

Remark (how serial correlation would be assessed)

Pursuing the Lo correction, given a sample of M consecutive daily excess returns, one would test the null hypothesis $H_0 : \rho(1) = \cdots = \rho(h) = 0$ at a small lag h (a typical rule of thumb is $h \approx \ln M$: h must grow with the sample size M (more data supports probing more lags) but only slowly, since genuine serial dependence in returns concentrates at short lags, so each additional high lag adds mostly sampling noise to the test) via a portmanteau test (Ljung–Box), and, only upon rejection, estimate $\rho(1), \ldots, \rho(h)$ by the sample ACF

$$\hat{\rho}(k) = \frac{\sum_{t=1}^{M-k}\left(\hat{r}^d_{t+k} - \bar{r}_M\right)\left(\hat{r}^d_t - \bar{r}_M\right)}{\sum_{t=1}^{M}\left(\hat{r}^d_t - \bar{r}_M\right)^2}, \qquad \bar{r}_M := \frac{1}{M}\sum_{t=1}^{M}\hat{r}^d_t,$$

setting $\rho(k) = 0$ for $k > h$ (as an approximation, justified since the weight $(m-k)$ in $\eta(m)$ is largest at low lags, where the estimates $\hat{\rho}(k)$ are also most reliable).

But the null already assumes what §14.1.4 denies. The scalar $\rho(k)$ exists only once time-invariance collapses the family $\left(\mathrm{Cov}(r^d_t, r^d_{t+k})\right)_t$ into a single $\gamma(k)$. So $H_0 : \rho(k) = 0$ is a question *inside* the stationary model, not a verdict *on* it: the portmanteau test assumes the autocorrelation, and cannot audit the stationarity that grounds this assumption. We would be testing within a model whose premise we have already rejected.

14.1.6 Results

We evaluated each of the four strategies (momentum, factors-model, MVO max-return, MVO min-variance) over a panel of windows chosen to approximate single market regimes (each of at least one year of data, to avoid extrapolating an annualized Sharpe ratio from a short window of a few days or months), in addition to computing a full-sample Sharpe ratio (daily and annualized) from `2010-03-05` to `2026-02-02`. As explained in the previous subsections, the resulting Sharpe ratios serve as cross-strategy comparators, not as forward-looking estimates.

We tested the strategies' risk-adjusted performance over the following **historical periods**:

- `2010-03-05` to `2026-02-02` (full sample of daily returns)
- `2010-03-05` to `2020-01-16` (post-GFC bull)
- `2022-10-05` to `2026-01-26` (bull)

Note that the full sample includes both sub-periods.

Configuration of each strategy tested: We used the following configuration for each strategy.

- **Momentum** (§7): Same configuration as that specified in §13.11.2
- **Factors model** (§8): Same configuration as that specified in §13.1
- **MVO max-return** (§9.2.1) and **min-variance** (§9.2.2): Same configuration as that specified in §13.11.1

Calendar (see §4): rebalance frequency type `ndays` with $N = 23$ trading days.

Costs and rules (see §2.6, §2.8, §3, §5):

- Trading: half-spread 1 bps, slippage 3 (conservative) bps, commission \$0.001/share.
- Margin requirements: Reg-T initial (50% long, 50% short) and FINRA maintenance (25% long, 30% short).
- Cure method of margin requirement violations: Variant B (shrinking exposures). We choose this variant to keep the strategies running without requiring additional capital injections to comply with the maintenance or initial margin requirements.
- Stop-loss: triggered at 0% of initial equity, so that each strategy runs through the entire historical period without stopping unless all initial capital is lost (**equivalently**, unless the max drawdown reaches 100%).
- Strategy return target hit rule: **disabled**, so that each strategy runs through the entire historical period.
- Dynamic rebalancing rule: We compute the Sharpe ratios both for the dynamic rebalancing rule enabled and with it disabled, to observe the effect it has. In the case where we enable it, we set the inter-rebalance return target to $3\%/12 = 0.25\%$ per period.
- Initial cash deployed (which coincides with the initial equity): \$100,000.

Universes of investable tickers (see §6) : For each rebalance, the pool of tickers from which we may select our new positions are built from the S&P 500 historical members with point-in-time forward-data filtering, and we apply no further ADV capping.

The results are presented in the tables from Figure 28.

Discussion of the results: For these particular strategies and backtest configuration, the effect of enabling the dynamic rebalancing rule is **mixed and window-dependent**: it improves neither metric uniformly, and degrades neither uniformly. On the daily (and hence annualized) Sharpe ratio, only the momentum strategy improves in all three windows; the factors model improves in two of three (including `2022-10-05` – `2026-01-26`, where it moves from -0.54 to $+0.22$), while both MVO strategies degrade in two of three. On the maximum drawdown (in absolute terms, i.e. made less negative), only MVO min-variance improves in all three windows; momentum improves in two of three, and the factors model and MVO max-return in one of three each. No strategy improves on both metrics across all three windows.

[22]:
```
1 sc.style_sharpe_table(sharpe_table_with_dynamic_reb_disabled)
```

[22]:

	n_daily_returns			daily_sharpe			annualized_sharpe			max_drawdown		
historical window	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)
strategy												
factors_model	2484	4002	828	-0.02	-0.03	-0.03	-0.27	-0.46	-0.54	-69.80%	-99.05%	-81.65%
momentum	2484	4002	828	0.01	0.01	-0.02	0.10	0.12	-0.26	-33.53%	-61.65%	-65.45%
mvo_max_return	2484	4002	828	-0.02	-0.01	-0.03	-0.35	-0.23	-0.51	-98.44%	-99.48%	-86.08%
mvo_min_variance	2484	4002	828	-0.01	-0.01	-0.03	-0.22	-0.23	-0.51	-58.00%	-77.15%	-50.43%

[26]:
```
1 sc.style_sharpe_table(sharpe_table_with_dynamic_reb_enabled)
```

[26]:

	n_daily_returns			daily_sharpe			annualized_sharpe			max_drawdown		
historical window	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)	(2010-03-05) - (2020-01-16)	(2010-03-05) - (2026-02-02)	(2022-10-05) - (2026-01-26)
strategy												
factors_model	2484	4002	828	-0.02	-0.03	0.01	-0.32	-0.40	0.22	-82.37%	-99.56%	-55.65%
momentum	2484	4002	828	0.01	0.01	0.01	0.14	0.16	0.17	-27.14%	-72.49%	-52.16%
mvo_max_return	2484	4002	828	-0.01	-0.03	-0.04	-0.23	-0.43	-0.69	-94.41%	-99.87%	-92.99%
mvo_min_variance	2484	4002	828	-0.01	-0.02	-0.05	-0.21	-0.34	-0.86	-45.34%	-69.22%	-47.34%

Figure 28: Sharpe ratios of the four strategies across the three historical windows, with the dynamic rebalancing rule disabled (top) and enabled (bottom). All other settings identical.

We see that across all the historical periods, strategies, and settings for the dynamic rebalancing rule (enabled or disabled), the maximum annualized Sharpe achieved is 0.22 (achieved for the factors model over the historical window `2022-10-05` – `2026-01-26`, with the dynamic rebalancing rule enabled). Of the 24 cells across the two tables, 18 are negative.

14.2 Comparison of the performance of strategies when deployed at specific dates, with specific return targets

14.2.1 Purpose

A single backtest answers the question: How did a particular strategy, with a specific set of strategy parameters, perform when deployed on a given start date under a particular backtest configuration (including termination rules, initial cash allocation, the choice of MR-violation cure method (variant A or B), and the dynamic rebalancing rule (disabled or enabled with a specified inter-rebalancing return target))? Strategy performance can be affected by changes to any of these parameters. In this section we illustrate how strategies compare across multiple start dates and return targets, holding the other parameters fixed. This is one of many parameter combinations we might sweep over. Specifically, we build the following two tables:

- **Table A** stores the relevant KPIs (specified in §14.2.4) as we sweep over strategies and strategy return targets, with the **dynamic rebalancing rule disabled** and under **Variant B** (curing any margin-requirement violations by uniformly shrinking our long and short exposures).

- **Table B** reports the same KPIs while sweeping over the strategy return target for a single strategy (the momentum strategy, used here for illustration). It compares runs with the dynamic rebalancing rule **enabled** (using a specified, **constant** inter-rebalancing return target across runs) against runs with the rule **disabled**. This isolates the effect of the dynamic rebalancing rule on performance. (Recall that under the dynamic rebalancing rule, a rebalance is triggered whenever the cumulative return since the previous rebalance reaches the specified target return before the next scheduled rebalance date. In such cases, the portfolio is rebalanced immediately to the positions deemed optimal by the strategy at that time.)

Setting a strategy return target that, once hit, systematically closes our positions is an important risk-control measure; Tables A and B sweep over this target precisely to show how it shapes realized performance. As an illustration, we sweep each strategy across 12 different start dates and 6 increasing return targets. We sweep start dates to assess each strategy's performance over different periods. A larger number of start dates would make the comparison more statistically meaningful; we limit ourselves to 12, just enough to make the methodology clear, to keep the total computation runtime manageable.

Estimating a strategy's capacity

A strategy's ***capacity*** is the largest amount of capital it can deploy before net performance **degrades** below an acceptable threshold. We can define capacity as the initial deployed cash at which the net Sharpe ratio (§13.1) of a given historical period falls to a chosen fraction of its baseline value (the Sharpe at negligible deployed capital). The main driving mechanism is market impact: as capital grows, orders become larger relative to available liquidity, so filling them walks further up (or down) the order book (temporary impact) while, when execution leaves a lasting footprint, it shifts the price persistently (permanent impact), thereby driving down the value of our booked positions.

To assess how initial cash affects the strategy's Sharpe ratio, we can sweep this initial-cash parameter across a set of increasing values, running the backtest with all other configurations held fixed.

14.2.2 Configuration held fixed across all comparison runs

The parameters we keep the same across all runs used to build Table A and Table B are those whose choice is specified in §14.1.6, with the exception of the stop-loss threshold, which we here set at 50%, and the dynamic rebalancing rule (which we enable to build Table A and disable to build Table B).

14.2.3 Parameters that vary across runs

For both Tables A and B:

- **Start dates**: 12 start dates spaced quarterly from `2021-01` to `2023-10`.
- **Strategy return target**: 1%, 3%, 5%, 7%, 10%, 12%.

For Table A, we swept across all four strategies:

- **Strategy**: momentum, factor model, MVO (max return), MVO (min variance).

For Table B (illustrated on the momentum strategy only):

- **Dynamic rebalancing rule**: two settings: (i) *enabled*, with a constant inter-rebalance return target of $3\%/12 = 0.25\%$ per period; and (ii) *disabled*.

The total number of backtests for Table A is therefore

$$12 \text{ start dates} \times 6 \text{ return targets} \times 4 \text{ strategies} = \mathbf{288} \text{ backtests}$$

and that for Table B is

$$12 \text{ start dates} \times 6 \text{ return targets} \times 2 \text{ rebalance-rule settings} = \mathbf{144} \text{ backtests (momentum only).}$$

14.2.4 Reported per-cell aggregates

Each (strategy, start date, return target) triple produces one backtest. Each backtest terminates in one of three ways:

1. **Strategy return target hit** before the one-year scheduled backtest end.
2. **Stop-loss triggered**: equity (excluding margin collateral) fell below the stop-loss threshold (here set to 50% of initial equity).
3. **Reached the last scheduled date** without either of the above firing.

For every (strategy, return-target) cell of the table we report four aggregate metrics **over the backtest start dates** tested:

- **Number of start dates tested**: here, 12.
- **Frequency of return-target hit**: $\dfrac{\text{number of runs that hit the return target}}{\text{number of start dates tested}}$.

- **Frequency of stop-out**: $\dfrac{\text{number of runs that triggered the stop-loss}}{\text{number of start dates tested}}$.
- **Average time to target** (*only over runs that hit the target*): the average number of trading days from the start of the backtest (deployment) until the target-hit date. A lower value means the target was reached faster on average. If no run hit the target, the value is reported as "N/A".

14.2.5 Table A: strategy × return-target sweep

The table is reported in Figure 29.

A given row tells us how each strategy performed when given the same return target. A given column tells us how a single strategy performed across the spectrum of return targets.

What we look for:

- *Frequency of target hits across the different targets:* Because the stop-loss threshold is the same across all targets, any run that hits a higher target necessarily hit every lower target first; the frequency is therefore a non-increasing function of the return target.
- *Time-to-target conditional on a hit:* Between two strategies that hit the same return target, we prefer a shorter time-to-target.

When the two criteria conflict (one strategy hits more often, the other hits faster) we may weight hit-frequency first.

For example, at the 7% strategy annual return target:

- **Momentum**: 83.3% frequency of target hits, average time-to-target 71.8 backtest dates (trading days).
- **Factor model**: 50.0% frequency of target hits, average time-to-target 19.3 backtest dates.

At this target, one could argue that the momentum strategy is the stronger of the two across the 12 start dates tested: 33.3 percentage points higher in frequency of target hits, at an average time-to-target before reaching the 7% return of 71.8 trading days.

14.2.6 Termination causes by strategy

Figure 30 reveals the termination causes by strategy across all backtest runs used to build Table A.

We see that `mvo_min_variance` is the only strategy with 0 stop-loss terminations across all 288 backtest runs, which we are tempted to attribute to the fact that this strategy minimizes the estimated variance of the equity return while keeping the estimated expected return above a lower bound (§9.2.2).

14.2.7 Table B: effect of the dynamic rebalancing rule on one strategy

Table B (see Figure 31) isolates the effect of the dynamic rebalancing rule, holding the strategy fixed (we illustrate this with the momentum strategy).

```
[9]: 1 sc.style_comparison_table(table_a)
```

[9]:

metric	n_start_dates_tested				frequency_target_hit				avg_time_to_target_if_target_hit			
strategy_name	momentum	factors_model	mvo_max_return	mvo_min_variance	momentum	factors_model	mvo_max_return	mvo_min_variance	momentum	factors_model	mvo_max_return	mvo_min_variance
return_target												
1.0%	12	12	12	12	100.0%	83.3%	91.7%	91.7%	16.2	18.5	23.3	27.3
3.0%	12	12	12	12	83.3%	66.7%	83.3%	91.7%	40.8	21.8	18.8	37.5
5.0%	12	12	12	12	83.3%	58.3%	83.3%	66.7%	67.6	33.1	20.8	37.6
7.0%	12	12	12	12	83.3%	50.0%	83.3%	66.7%	71.8	19.3	30.5	66.5
10.0%	12	12	12	12	83.3%	50.0%	75.0%	66.7%	80.8	37.8	36.7	87.6
12.0%	12	12	12	12	75.0%	50.0%	58.3%	66.7%	89.9	48.7	48.6	120.9

Figure 29: Table A: strategy × return-target sweep.

cause_of_termination	hit_return_target_for_strategy	reached_last_scheduled_backtest_date	stop_loss_termination	All
strategy_name				
factors_model	43	8	21	72
momentum	61	1	10	72
mvo_max_return	57	0	15	72
mvo_min_variance	54	18	0	72
All	215	27	46	288

Figure 30: Termination causes by strategy across all backtest runs.

14.2.8 Termination causes with and without the dynamic rebalancing rule enabled

Figure 32 reveals the termination causes across all backtest runs used to build Table B.

14.3 Ex-post market exposure of the realised equity curve

The Sharpe ratios reported in §14.1 measure excess return per unit of the strategy's own volatility, but they do not indicate what drives that return. A long/short strategy can generate a positive excess return simply by maintaining net exposure to the market and benefiting from its performance; its Sharpe ratio alone cannot distinguish such market-driven returns from returns generated by skill in stock selection or by other strategy-specific behaviour.

This subsection therefore measures the strategy's realised market exposure ex post, from the realised equity curve. The methodology is set out and illustrated on the documented backtest of §13.1, whose results are read off Figure 33 in §14.3.4 and Figure 35 in §14.3.5. The same analysis is then reported for every strategy and configuration tested in §14.2, in §14.3.7.

14.3.1 The model and its estimator

Let $\{1, \ldots, N\}$ index the backtest dates, with $N = 122$. Let r_k^d be the *simple* daily return of equity excluding posted margin collateral, r_k^f the risk-free return of §11.3, and r_k^m the simple total return of the SPY ETF, the same market proxy used by the beta estimator of §9.7.3. Define the daily excess returns

$$r_k^{d,e} = r_k^d - r_k^f, \qquad r_k^{m,e} = r_k^m - r_k^f.$$

We then fit the single-factor market model

$$r_k^{d,e} = \alpha + \beta r_k^{m,e} + \varepsilon_k, \qquad k = 1, \ldots, N. \tag{8}$$

Here, β measures the realised sensitivity of the strategy's excess return to the market's excess return over the window, while α is the average daily excess return not accounted for by this market exposure. The interpretation of α is deliberately limited to this one-factor model: whether it can be regarded as evidence of skill, rather than exposure to other systematic factors, is considered later in §14.3.6.

In matrix form, with $y := \left(r_k^{d,e}\right)_k \in \mathbb{R}^N$ and $r^{m,e} := \left(r_k^{m,e}\right)_k \in \mathbb{R}^N$,

$$y = X\theta + \varepsilon, \qquad X = \left[\,\mathbf{1} \mid r^{m,e}\,\right] \in \mathbb{R}^{N \times p}, \qquad \theta = (\theta_0, \theta_1)^\top = (\alpha, \beta)^\top \in \mathbb{R}^2, \qquad p = 2.$$

[15]:
```
1 sc.style_comparison_table(table_b)
```

[15]:

metric	n_start_dates_tested		frequency_target_hit		avg_time_to_target_if_target_hit	
rule_setting	**with_inter_rebalance_rule**	**without_inter_rebalance_rule**	**with_inter_rebalance_rule**	**without_inter_rebalance_rule**	**with_inter_rebalance_rule**	**without_inter_rebalance_rule**
return_target						
1.0%	12	12	100.0%	100.0%	10.9	16.2
3.0%	12	12	83.3%	83.3%	11.0	40.8
5.0%	12	12	83.3%	83.3%	40.3	67.6
7.0%	12	12	75.0%	83.3%	48.8	71.8
10.0%	12	12	75.0%	83.3%	57.7	80.8
12.0%	12	12	75.0%	75.0%	58.7	89.9

Figure 31: Table B: effect of the dynamic rebalancing rule (momentum strategy).

cause_of_termination	hit_return_target_for_strategy	reached_last_scheduled_backtest_date	stop_loss_termination	All
use_inter_rebalance_rule				
with_rule	59	0	13	72
without_rule	61	1	10	72
All	120	1	23	144

Figure 32: Termination causes for the momentum strategy across all backtest runs.

```
--------------------------------------------------------------------------------
                              MARKET EXPOSURE
--------------------------------------------------------------------------------
  n_observations                                                          122
  null_hypothesis_for_t_and_p                              alpha = 0 and beta = 0
  p_value_reference_distribution                       standard normal (asymptotic)
  beta                                                                  0.331
  hac_standard_error_beta                                              0.2659
  classical_standard_error_beta                                        0.1948
  t_stat_beta_hac                                                        1.24
  p_value_beta_two_sided                                                0.213
  beta_95pct_confidence_interval_hac:
    lower                                                              -0.190
    upper                                                               0.852
  alpha_daily                                                        0.000352
  hac_standard_error_alpha_daily                                     0.001625
  classical_standard_error_alpha_daily                               0.001780
  t_stat_alpha_hac                                                       0.22
  p_value_alpha_two_sided                                               0.829
  alpha_annualization_convention           compounded: (1+alpha_daily)^252 - 1
  alpha_annualized                                                      9.27%
  alpha_annualized_95pct_confidence_interval_hac:
    lower                                                             -51.09%
    upper                                                             143.48%
  r_squared                                                             0.023
  hac_lags_for_standard_errors                                              5
  rolling_beta_window_in_trading_days                                      60
  rolling_beta:
    min                                                                -0.432
    max                                                                 0.913
    avg                                                                 0.254
    std                                                                 0.421
  avg_net_exposure_pct_of_equity                                      -40.31%

--------------------------------------------------------------------------------
```

Figure 33: KPI report (part 5): ex-post market exposure of the realised equity curve, for the documented run of §13.1.

The OLS estimator is

$$\hat{\theta} = (X^\top X)^{-1} X^\top y,$$

provided that $\operatorname{rank}(X) = 2$, which fails only if the market excess return is constant over the window (which it is not). Under the conditional exogeneity assumption

$$\mathbb{E}[\varepsilon \mid X] = 0,$$

OLS is unbiased:

$$\mathbb{E}[\hat{\theta} \mid X] = \theta, \qquad \text{and hence} \qquad \mathbb{E}[\hat{\theta}] = \mathbb{E}\Big[\mathbb{E}[\hat{\theta} \mid X]\Big] = \theta.$$

Nothing is assumed yet about the covariance structure or distribution of ε; those assumptions enter only when the coefficients are tested.

The market proxy is a total-return series, adjusted for distributions (dividends) of the underlying stocks, unlike the prices against which our own positions are marked (§2.5.1). Using a total-return proxy ensures that the market factor represents the full realised return to holding the market, including distributions. A price-only proxy would omit the dividend component and therefore make the market factor inconsistent with the total-return quantity being used to explain the strategy's excess return.

Relation to the regression of §8.4

Both are linear models fitted by OLS, but in §8.4 the index runs over *tickers* at one date and the intercept belongs to a return-forecasting model, whereas here it runs over *time*, over the daily returns of one realised equity path. The two intercepts are therefore unrelated quantities. A second difference is that the second column of X is here a realised market return, so the design matrix is random, which is why exogeneity is stated conditionally as $\mathbb{E}[\varepsilon \mid X] = 0$.

14.3.2 What is tested

For each coefficient α and β, we test the hypotheses

$$H_0^\alpha : \ \alpha = \alpha_0 \qquad \text{and} \qquad H_0^\beta : \ \beta = \beta_0,$$

where $\alpha_0, \beta_0 \in \mathbb{R}$ are the null values specified for the two coefficients. In this analysis, we test whether the realised equity curve earned anything beyond what its market exposure delivered,

$$H_0^\alpha : \ \alpha = 0,$$

and whether it carried any market exposure over the window,

$$H_0^\beta : \ \beta = 0.$$

These are the *null hypotheses*. The testing procedure asks whether the observed data provide sufficient evidence to reject either one. Rejecting H_0^β would provide evidence that the strategy had non-zero market exposure over the window ($\beta \neq 0$); rejecting H_0^α would provide evidence that its average excess return was non-zero after accounting for the market exposure represented in the model ($\alpha \neq 0$).

The essence of the procedure is to choose a statistic T, computable from the observed data and the hypothesised value alone, whose distribution under H_0 is known, and then to ask how probable a value at least as extreme as the observed one would be if H_0 held. That probability is the *p-value*. A threshold λ is fixed in advance, conventionally $\lambda = 5\%$, and H_0 is rejected when $p < \lambda$. We write λ rather than the customary α, which is already the intercept of (8).

The classical case is set out first, because it fixes the shape of the statistic; its reference distribution is then replaced in §14.3.3, where the assumption behind it is examined. Under the classical assumption $\varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 I_N)$, the statistic

$$T_i \;=\; \frac{\hat{\theta}_i - \theta_i^0}{s\sqrt{c_{ii}}}, \qquad s^2 = \frac{\|\hat{\varepsilon}\|^2}{N-p}, \qquad c_{ii} = \big[(X^\top X)^{-1}\big]_{ii}, \qquad i \in \{0,1\},$$

follows an exact t_{N-p} distribution under H_0. Since both nulls stated above set $\theta_i^0 = 0$, the statistic reduces in this analysis to $T_i = \hat{\theta}_i/(s\sqrt{c_{ii}})$.

Write $T_i = Z/\sqrt{W/(N-p)}$ with $Z = (\hat{\theta}_i - \theta_i^0)/(\sigma\sqrt{c_{ii}})$ and $W = (N-p)s^2/\sigma^2$. Under H_0, $Z \sim \mathcal{N}(0,1)$ and $W \sim \chi^2_{N-p}$. The pair $(\hat{\theta}, \hat{\varepsilon})$ is jointly Gaussian with $\mathrm{Cov}(\hat{\theta}, \hat{\varepsilon}) = 0$, hence independent; as s^2 is a function of $\hat{\varepsilon}$ alone, Z and W are independent. The ratio is therefore t_{N-p} by definition.

The test is two-sided: the alternative admits departures on either side of θ_i^0. Writing T_i^{obs} for the realised value of T_i, the p-value is therefore

$$p_i \;=\; \mathbb{P}_{H_0}\big(|T_i| \geq |T_i^{\mathrm{obs}}|\big) \;=\; 2\Big(1 - F_{N-p}\big(|T_i^{\mathrm{obs}}|\big)\Big),$$

the second equality following from the symmetry of the student distribution about zero, where F_{N-p} denotes its c.d.f.

The assumption $\varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 I_N)$ is not realistic for our equity curve, for the reasons set out in §14.3.3. The statistic and its reference distribution are replaced accordingly there.

Reading a p-value

The p-value is $\mathbb{P}_{H_0}$(a statistic at least this extreme), not $\mathbb{P}(H_0 \mid \text{the observation})$. A small p-value means the observation would be surprising were H_0 true, and so counts against it: the smaller the value, the stronger the evidence against the null.
A *large* value is not evidence *for* H_0. It says only that the data are not incompatible with it. Failing to reject $\beta = 0$ therefore does not establish that $\beta = 0$, that is, that the book was market-neutral.

14.3.3 Why the classical assumptions fail here

The classical assumption about the distribution of the vector of daily noises conditional on the design matrix, $\varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 I_N)$, combines three premises: normality, homoskedasticity, and zero serial autocovariance.

The first two are contradicted by what is known of the data. Daily asset returns display volatility clustering, so that the conditional variance changes from date to date, and have heavier tails than the Gaussian; both are among the stylised facts surveyed in [7].

Consequently, we replace the classical assumption with the weaker pair

$$\text{(A1)}\quad \mathbb{E}[\varepsilon \mid X] = 0, \qquad\qquad \text{(A2)}\quad \operatorname{Var}(\varepsilon \mid X) = \Omega,$$

where Ω is an unknown symmetric positive semi-definite matrix whose diagonal entries may differ from one date to the next and whose off-diagonal entries need not vanish. Figure 34 draws the shape the estimator below is built to approximate: unequal variances on the diagonal, autocovariances decaying with the lag off it.

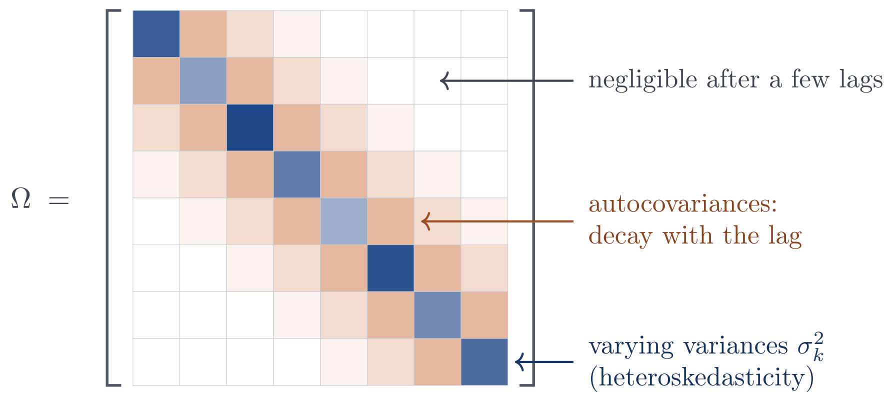

Figure 34: The shape of Ω that the Newey–West estimator below is built for. The diagonal holds the individual variances σ_k^2, which differ from one observation to the next (heteroskedasticity); the off-diagonal bands hold the autocovariances, which decay as the lag grows.

Under (A1)–(A2), OLS remains unbiased ($\mathbb{E}[\hat{\theta} \mid X] = \theta$, so that $\mathbb{E}[\hat{\theta}] = \theta$ by the tower property) and the conditional covariance of $\hat{\theta}$ is the "sandwich"

$$\operatorname{Var}(\hat{\theta} \mid X) \;=\; (X^\top X)^{-1} \big(X^\top \Omega X\big) (X^\top X)^{-1}.$$

The symmetric matrix Ω holds $N(N+1)/2 = 7{,}503$ unknowns, considerably more than the $N = 122$ observations. But Ω itself is not needed: only the 2×2 matrix $X^\top \Omega X$, and

$$X^\top \Omega X \;=\; \sum_{k=1}^{N} \sum_{l=1}^{N} \mathbb{E}[\varepsilon_k \varepsilon_l \mid X]\, x_k x_l^\top,$$

where $x_k^\top$ is the k-th row of X. Grouping the double sum by the lag $j = k - l$ gives

$$X^\top \Omega X \;=\; N\, S_N, \qquad S_N \;=\; \Gamma_0 + \sum_{j=1}^{N-1} \big(\Gamma_j + \Gamma_j^\top\big), \qquad \Gamma_j = \tfrac{1}{N} \sum_{k=j+1}^{N} \mathbb{E}[\varepsilon_k \varepsilon_{k-j} \mid X]\, x_k x_{k-j}^\top,\; j \ge 0,$$

where Γ_0 carries the heteroskedasticity and Γ_j ($j \in \{1, \ldots, N-1\}$) the autocovariances. Each Γ_j is estimated by

$$\widehat{\Gamma}_j = \tfrac{1}{N} \sum_{k=j+1}^{N} \hat{\varepsilon}_k \hat{\varepsilon}_{k-j}\, x_k x_{k-j}^\top,$$

where the $\hat{\varepsilon}_k$ are the OLS residuals. Truncating at a lag cap $L \in \{0, 1, \ldots, N-1\}$ and applying Bartlett weights yields the Newey–West estimator [18]:

$$\widehat{S}_L \;=\; \widehat{\Gamma}_0 + \sum_{j=1}^{L} \underbrace{\Big(1 - \tfrac{j}{L+1}\Big)}_{w_j} \big(\widehat{\Gamma}_j + \widehat{\Gamma}_j^\top\big), \qquad \widehat{S}_0 \;=\; \widehat{\Gamma}_0,$$

with $L = 0$ recovering the heteroskedasticity-only estimator of [23]. The weights w_j are what make $\widehat{S}_L$ positive semi-definite, and so guarantee a non-negative estimated variance; the truncated but unweighted sum carries no such guarantee.

Having chosen L (see the box below), we define our estimators of $\mathrm{Var}(\hat\theta \mid X)$ and of the standard errors as

$$\widehat{\mathrm{Var}}^{\mathrm{HAC}}(\hat\theta \mid X) = N\,(X^\top X)^{-1}\,\widehat{S}_L\,(X^\top X)^{-1}, \qquad \widehat{\mathrm{SE}}^{\mathrm{HAC}}(\hat\theta_i) = \sqrt{\Big[\widehat{\mathrm{Var}}^{\mathrm{HAC}}(\hat\theta \mid X)\Big]_{ii}}, \quad i \in \{0,1\}.$$

HAC stands for *heteroskedasticity- and autocorrelation-consistent*. Under the regularity conditions of [18, 2], and along any bandwidth sequence with $L_N \to \infty$ and $L_N/N \to 0$, we have $\widehat{S}_{L_N} \xrightarrow{p} S$, and so, under $H_0 : \theta_i = \theta_i^0$,

$$\boxed{T_i = \frac{\hat\theta_i - \theta_i^0}{\widehat{\mathrm{SE}}^{\mathrm{HAC}}(\hat\theta_i)} \xrightarrow[N\to\infty]{d} \mathcal{N}(0,1),}$$

and we may report the asymptotic two-sided p-value

$$\boxed{p_i \;=\; \mathbb{P}\big(|\mathcal{N}(0,1)| \ge |T_i^{\mathrm{obs}}|\big) \;=\; 2\Big(1 - \Phi\big(|T_i^{\mathrm{obs}}|\big)\Big),}$$

which approximates $\mathbb{P}_{H_0}\big(|T_i| \ge |T_i^{\mathrm{obs}}|\big)$ with an error that vanishes as $N \to \infty$.

Choice of the lag truncation

The run uses $L = 5$, which admits autocovariances out to one trading week, with Bartlett weights $5/6, 4/6, 3/6, 2/6, 1/6$ at lags 1 to 5. The fixed rule of thumb $\lfloor 4(N/100)^{2/9} \rfloor$, widely used as a software default, gives 4 at $N = 122$; it is often attributed to [19]. Either way the choice is a trade-off: too small an L discards genuine autocovariance, too large adds noisily estimated terms.

What a fixed L costs

The result above holds along a bandwidth sequence $L_N \to \infty$ with $L_N/N \to 0$; in the run L is fixed at 5. With L fixed, $\widehat{S}_L$ converges not to S but to the truncated, Bartlett-weighted sum $\Gamma_0 + \sum_{j=1}^{L} w_j(\Gamma_j + \Gamma_j^\top)$: the standard error targets the autocovariances out to one trading week, and misses whatever lies beyond. If those omitted terms are positive, the standard error is too small and the test over-rejects.
Pointing the same way, HAC t-tests are known to over-reject in samples of this size [2, 13, 14]. The true p-values are therefore, if anything, larger than those reported in §14.3.4, so the failures to reject there are conservative.

14.3.4 Results

	estimate	HAC SE	T^{HAC}	two-sided p	reject at 5%?
$\hat\beta$	0.331	0.266	1.24	0.213	no
$\hat\alpha$	0.000352/day	0.001625/day	0.22	0.829	no

with $R^2 = 0.023$.

Neither null hypothesis is rejected at 5%, and neither would be at any conventional level.

What this does and does not say

The 95% interval for β is $[-0.19,\ 0.85]$: neither a book that was market-neutral nor one that ran a beta of 0.85 can be ruled out at 5%. The interval for α runs from -51% to $+143\%$ annualised. At $N = 122$ the test is underpowered: the data do not resolve either coefficient, and the annualised α of 9.27% carries no evidence of skill.

Annualisation of α

Equation (8) is fitted to *simple* daily returns, and simple returns compound, so $\hat{\alpha}$ is annualised geometrically: $(1 + 0.000352)^{252} - 1 = 9.27\%$. The interval is annualised by applying the same map to its endpoints, which is legitimate because $x \mapsto (1 + x)^{252} - 1$ is strictly increasing: the daily interval $[-0.00283,\ 0.00354]$ maps to $[-51.1\%,\ +143.4\%]$.

Interpretation of the R^2: Given the OLS fit with an intercept and a single regressor, we have

$$R^2 \;=\; \frac{\hat{\beta}^2\,\widehat{\mathrm{Var}}(r^{m,e})}{\widehat{\mathrm{Var}}(r^{d,e})} \;=\; \widehat{\mathrm{corr}}\big(r^{d,e}, r^{m,e}\big)^2 \qquad (\S 8.11),$$

so $R^2 = 0.023$ gives $\widehat{\mathrm{corr}}(r^{d,e}, r^{m,e}) = +0.152$, the sign following that of $\hat{\beta}$. About 2.3% of the day-to-day variation in the strategy's excess returns is accounted for by the market over this window. This is a description of the sample, not an inference about the population. The remaining 97.7% is not explained *by the market*, which is not the same as idiosyncratic: it may contain exposure to other systematic factors, as noted in §14.3.6.

14.3.5 The rolling beta

Figure 35 plots the 60-day rolling estimate of β over the backtest; its summary statistics (minimum, maximum, mean, standard deviation) are reported in Figure 33. The estimate swings between -0.432 and 0.913, with a standard deviation of 0.2659 across windows, and its pointwise 95% interval contains the full-sample estimate $\hat{\beta} = 0.331$ at almost every date.

Each window holds 60 observations, so the per-window HAC standard errors rest on less data than the 122-day figures of §14.3.4, and the normal reference distribution is correspondingly less reliable. The intervals are also pointwise rather than simultaneous: they are not a confidence band for the path as a whole, and the fact that most of them cover $\hat{\beta}$ is not evidence that β was constant over the window.

14.3.6 Conclusion, and the natural extension

1. **Are the equity's excess returns explained by the market?**

   The factors-model strategy selects names by predicted return (§8.8, §8.9) and holds a fixed share count per name (§7.3), so whatever beta the book carries is a by-product of selection and never a targeted quantity. In contrast, the MVO strategies of §9 impose a beta cap (§9.3.2) and a net-exposure cap (§9.3.3), both set to 0.001 in §13.11.1 (but those are constraints on *predicted* beta at rebalance, and nothing forces the realised beta of the subsequent path to respect them).

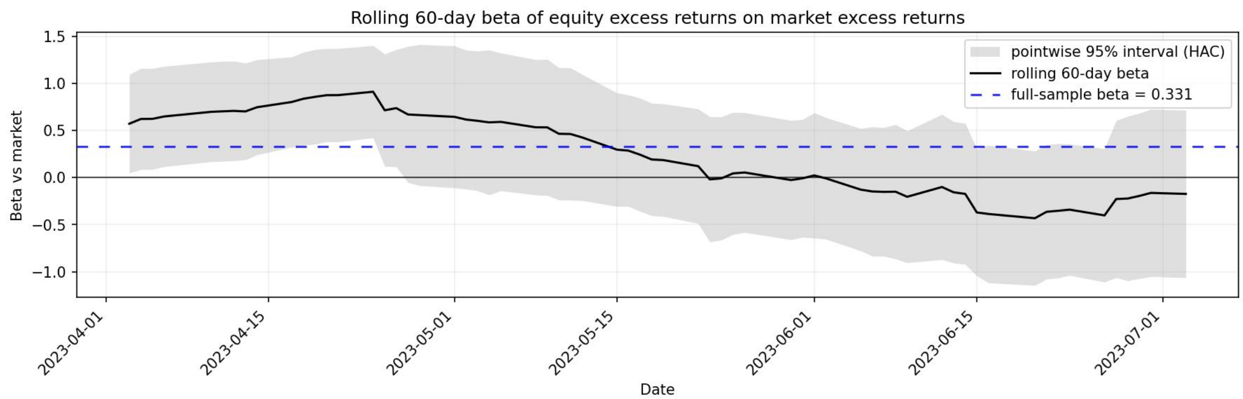

Figure 35: Rolling 60-day beta of the realised equity excess returns on the market's, with a pointwise 95% interval from the per-window HAC standard errors. The dashed line is the full-sample estimate $\hat{\beta} = 0.331$.

Descriptively, the market accounts for little of this run: $R^2 = 0.023$ over the 122 days. From statistical inference, we cannot say more than that we failed to reject $\beta = 0$ ($T^{\mathrm{HAC}} = 1.24$, $p = 0.213$), and that failure is not a demonstration of neutrality. The 95% interval runs from -0.19 to 0.85, so a substantial long market exposure remains entirely compatible with what was observed.

2. **Is the α evidence of skill?**

   No. With $p = 0.829$ and an interval spanning roughly -51% to $+143\%$ annualised, the estimate of 9.27% is uninformative. And independently of the width of that interval, the run terminated on its return target, so its final date is by construction a point at which cumulative return reached a high-water mark; a window whose endpoint is selected that way has a sample mean return biased upward, and $\hat{\alpha}$ with it (selection biais).

   Beyond that, **a well-estimated α would still be the wrong α.** The intercept of (8) is the part of the daily excess return not explained by *the market*, a single factor. It is not the part unexplained by *known risk factors*. The natural and necessary extension is to replace (8) with

$$r_k^{d,e} \;=\; \alpha \;+\; \sum_{j=1}^{J} b_j\, f_{j,k} \;+\; \varepsilon_k,$$

   estimated by OLS with the same HAC inference, where $f_{1,k} = r_k^{m,e}$ is the market excess return retained as the first factor and the remaining $f_{j,k}$ are daily returns of reference long–short factor portfolios: size (SMB) and value (HML) from the Fama–French three-factor model [11], profitability (RMW) and investment (CMA) from its five-factor extension [10], and Carhart's momentum factor (UMD) [5]. Daily series for all six are published in [12].

   An α that survives these known risk factors would be far stronger evidence than one that survives the market alone, whereas an α that vanishes once they are included would say the strategy is harvesting known premia rather than generating new return. What the extension does *not* do is fix the problem diagnosed above: it changes what α means, not how precisely it is measured. With N unchanged at 122 and seven regressors instead of two, the confidence interval around α would still be so wide as to be uninformative.

14.3.7 Results: ex-post market exposure analytics for each strategy

Equation (8) is refitted, with the same HAC inference, for each of the twelve strategy and window combinations of §14.2, once with the dynamic rebalancing rule disabled and once with it enabled. Figure 36 reports the twenty-four fits. Each window carries between 828 and 4,002 daily observations against $N = 122$ for the documented run.

1. **The constrained strategies are more neutral:** the two MVO strategies of §9 realise $|\hat{\beta}| \leq 0.272$ and an average net exposure within $[-11.7\%, +3.2\%]$ across all twelve of their runs, while the two strategies carrying no beta or net-exposure cap reach $\hat{\beta} = +1.590$ (momentum, 2022–2026, rule enabled) and average net exposures of 43.5% (factors) and 81.9% (momentum). §14.3.6 noted that the caps of §9.3.2 and §9.3.3 bind on *predicted* beta at rebalance, and that nothing forces the realised path to respect them; over these windows the realised path largely does. The two MVO strategies share the same caps, yet minimum-variance is the tighter of the two on both measures ($|\hat{\beta}| \leq 0.155$ and $|\text{net}| \leq 3.2\%$, against 0.272 and 11.7%).
2. **The dynamic rebalancing rule adds market sensitivity:** enabling it raises $\hat{\beta}$ in nine of the twelve pairs, with a median change of +0.148. The clearest single case is the factors model over 2010–2020, where $\hat{\beta}$ rises from +0.102 to +0.516.
3. **No run shows evidence of skill:** the annualised $\hat{\alpha}$ is negative in nineteen of the twenty-four runs, with a median of −11.8%. However, only one run rejects $\alpha = 0$ at 5%, the factors model over 2010–2026 with the rule enabled, and its estimate is a *loss* of −23.1% annualised.

15 Assessing the risk of deploying a strategy in the present from the above comparison metrics

Deploying any of the three strategies currently shipped with the framework carries the risks explained in §10. We can, nevertheless, use the comparison methodology of §14.1 and §14.2 to be more or less confident that a strategy will perform well if deployed, with the important caveat that this confidence is conditional on the future resembling the tested past (favorable past performance raises confidence without removing the wager).

For example, following the methodology of §14.2, suppose a strategy hits a return target of 5% in 80% of cases when deployed at each of 300 start dates scattered across the last three years, with an average time-to-target of 100 trading days. Then this would be encouraging. If, in addition, the annualized Sharpe ratio for the same configuration (measured over the historical window that spans all those start dates and ends in the present) exceeds 1, this adds additional encouragement. The first metric measures how often a target is reached across many backtest start dates, while the Sharpe ratio measures risk-adjusted return over a single continuous window. **Strong comparison metrics constitute weak positive evidence: they are a necessary condition one would want any deployable strategy to satisfy, and a strategy that failed them should clearly not be deployed.**

However, as explained in §10, nothing removes the risk inherent in deploying any of these three systematic strategies.

16 Limitations and Extensions

strategy	historical window	n_observations	beta	beta_95pct_ci_lower	beta_95pct_ci_upper	t_stat_beta_hac	alpha_annualized	t_stat_alpha_hac	r_squared	avg_net_exposure
factors_model	(2010-03-05) - (2020-01-16)	2484	0.102	-0.045	0.248	1.36	-7.46%	-1.02	0.004	40.06%
	(2010-03-05) - (2026-02-02)	4002	-0.094	-0.349	0.161	-0.72	-16.57%	-1.80	0.001	43.12%
	(2022-10-05) - (2026-01-26)	828	0.308	-0.015	0.630	1.87	-24.74%	-1.24	0.013	21.02%
momentum	(2010-03-05) - (2020-01-16)	2484	0.077	-0.051	0.205	1.18	0.95%	0.15	0.003	18.54%
	(2010-03-05) - (2026-02-02)	4002	0.205	0.078	0.331	3.18	0.94%	0.13	0.013	27.89%
	(2022-10-05) - (2026-01-26)	828	0.789	0.331	1.246	3.38	-21.78%	-1.11	0.081	60.37%
mvo_max_return	(2010-03-05) - (2020-01-16)	2484	-0.272	-0.449	-0.096	-3.02	-14.81%	-0.86	0.005	-1.94%
	(2010-03-05) - (2026-02-02)	4002	-0.218	-0.357	-0.079	-3.07	-10.12%	-0.70	0.004	-2.62%
	(2022-10-05) - (2026-01-26)	828	0.092	-0.376	0.560	0.39	-27.16%	-0.93	0.001	-1.61%
mvo_min_variance	(2010-03-05) - (2020-01-16)	2484	-0.110	-0.185	-0.035	-2.87	-2.63%	-0.44	0.007	0.36%
	(2010-03-05) - (2026-02-02)	4002	-0.109	-0.166	-0.053	-3.79	-3.37%	-0.65	0.008	1.72%
	(2022-10-05) - (2026-01-26)	828	0.155	-0.003	0.312	1.93	-13.42%	-1.20	0.011	3.19%

strategy	historical window	n_observations	beta	beta_95pct_ci_lower	beta_95pct_ci_upper	t_stat_beta_hac	alpha_annualized	t_stat_alpha_hac	r_squared	avg_net_exposure
factors_model	(2010-03-05) - (2020-01-16)	2484	0.516	0.294	0.738	4.55	-15.40%	-1.69	0.060	36.60%
	(2010-03-05) - (2026-02-02)	4002	0.269	-0.070	0.608	1.56	-23.07%	-2.16	0.009	43.52%
	(2022-10-05) - (2026-01-26)	828	0.233	-0.472	0.937	0.65	6.09%	0.25	0.007	29.29%
momentum	(2010-03-05) - (2020-01-16)	2484	0.105	-0.001	0.211	1.94	1.20%	0.21	0.007	18.14%
	(2010-03-05) - (2026-02-02)	4002	0.360	0.185	0.534	4.05	1.32%	0.15	0.028	33.32%
	(2022-10-05) - (2026-01-26)	828	1.590	1.237	1.943	8.83	-15.30%	-0.60	0.198	81.93%
mvo_max_return	(2010-03-05) - (2020-01-16)	2484	-0.132	-0.362	0.099	-1.12	-9.60%	-0.63	0.001	-7.01%
	(2010-03-05) - (2026-02-02)	4002	-0.106	-0.276	0.065	-1.22	-19.77%	-1.67	0.001	-5.29%
	(2022-10-05) - (2026-01-26)	828	-0.085	-0.616	0.445	-0.31	-37.31%	-1.23	0.000	-11.70%
mvo_min_variance	(2010-03-05) - (2020-01-16)	2484	0.056	-0.003	0.115	1.85	-4.11%	-0.77	0.003	-0.85%
	(2010-03-05) - (2026-02-02)	4002	0.077	0.021	0.132	2.71	-6.79%	-1.58	0.006	0.41%
	(2022-10-05) - (2026-01-26)	828	0.060	-0.145	0.265	0.58	-17.43%	-1.71	0.002	2.53%

Figure 36: Ex-post market exposure of each strategy over each historical window of §14.2, with the dynamic rebalancing rule disabled (top) and enabled (bottom). Coefficients are estimated by (8) with the HAC standard errors of §14.3.3; `alpha_annualized` is compounded as in §14.3.4, and `avg_net_exposure` is the mean over backtest dates of long leverage minus short leverage at close.

16.1 Known limitations of the current framework

- **Daily frequency only:** Trades execute at the close. Intraday bid–ask spread evolution is outside the scope of the simulation. Therefore, strategies that require intraday rebalancing are not supported.

- **Static cost assumptions:** Trading costs (half-spread, slippage, commission) are parameter inputs, applied uniformly across dates and stocks unless overridden by the user. In reality, the half-spread and slippage realized for a market order vary with order size, volatility, and time of day. The framework supports per-(date, ticker) overrides of the size-independent half-spread and slippage (§2.5.2) for stress-testing at any date, but it does not model cost as a function of trade size, as explained in §2.5.2.

- **No risk-management overlay:** The strategies described in §7–§9 produce a target portfolio per rebalance date, without enforcing stop-losses on individual positions, or capping exposure to sectors or industries, or sizing positions by recent volatility (inverse-volatility weighting, whereby the more volatile stock gets a smaller allocation because it can swing much more). These would be important additions.

- **Survivorship-bias mitigation is partial:** Although historical S&P 500 membership is taken from Wikipedia (which may not be fully accurate), we also introduce a small survivorship bias by retaining, at each rebalance date, only tickers with available price data for the next inter-rebalance period (see §6.4). We do this because our framework currently does not support the handling of a delisting of a ticker while we hold positions on that ticker, and because our free price data feed (`yfinance`) may have gaps in the daily prices required for marking to market until the next rebalance. Requiring full price availability over the entire next inter-rebalance window can systematically drop names that were delisted or acquired mid-period, which biases returns upward.

- **Limited corporate-action handling:** Stock dividends are already modelled; however, splits, spin-offs, mergers, ticker changes, and special dividends are handled only insofar as `yfinance`'s close-price series accommodates them.

- **No short-availability model, and recall risk ignored:** We assume stocks are always available to borrow for short-selling and that the short positions are maintainable. In practice, hard-to-borrow names carry meaningful recall risk: a broker can force-close a short at any time if the underlying lender (typically another of the broker's clients) recalls the shares. Because we model neither borrow availability nor recall events, in practice, the strategy's shorts may close prematurely (on the broker's schedule rather than ours) forcing deviations from the strategy's intended positions. This limitation matters most for strategies whose shorts concentrate in small-cap or heavily-shorted names.

- **No order size constraints:** The trade simulator does not check trade size against the rolling average daily volume (ADV). A position representing, say, 5% of a stock's ADV could, depending on the participation rate at which it is executed, incur material price slippage and a permanent market impact. Our framework's current simulator, however, would still fill the order at the execution price from §2.5.2, which does not factor in trade size. To flag such cases, we report the metric `max_pct_of_daily_market_volume_traded_for_a_ticker_during_backtest` in the *Trading* section of the KPI report (see the example in Figure 9 from §13.7).

- **No tax accounting:** Pre-tax performance only. Wash sales, short-term vs long-term gains, and the asymmetric tax treatment of dividends on long versus short positions are outside the scope of our current framework's implementation.

- **No regime-switching adaptive parameters:** The estimator windows for the strategies (μ window, Σ window, β window, factor-regression window) are set once at the start of the backtest and held fixed. In practice, market conditions change over time, and we could adapt these parameters dynamically.

Example

- During a high-volatility crisis regime (e.g. March 2020), one might shorten the covariance-estimation window from 252 trading days to 30 or 60 so the risk model reacts more quickly to rapidly changing correlations.
- During a stable low-volatility regime, longer estimation windows reduce noise in covariance and beta estimates.

16.2 Possible extensions

Our framework's architecture supports several extensions without restructuring.

- **Improving the expected-return input $\hat{\mu}$ of the MVO strategy.** The estimates feeding the MVO optimiser (§9.7) are the weakest point in Strategy 3. Errors in the expected-return vector μ are far more damaging than errors in the covariance matrix Σ: Chopra and Ziemba [6] find that, for an investor of risk tolerance 50, errors in means are about 11 times as costly as errors in variances and about 21 times as costly as errors in covariances, the relative cost of mean errors growing further at higher risk tolerances. Their measure is the *cash-equivalent loss* (CEL): the optimiser is run on a perturbed input, the resulting portfolio is evaluated under the *true* parameters via the mean–variance utility $U(w) = \mu^\top w - \frac{1}{2\lambda} w^\top \Sigma w$ (λ the risk tolerance), and the CEL is the monetary compensation that equates this to the utility of the portfolio built on the true inputs. Improving the estimation of μ is therefore the highest-value target for improving this strategy. We could pursue, for example, this one concrete direction:
  - **Bayesian combination (Black–Litterman).** Rather than selecting between the historical-mean or factor-model estimators of §9.7.1 by out-of-sample performance (see the remark in §9.7.1.3), the Black–Litterman framework [4] "enriches" either of them: a neutral prior on μ (for a market-neutral book, centred on 0) is updated, via Bayes' rule, by our choice (say, the factor-model prediction $\hat{\mu}^{\text{factor}}$), treated as noisy evidence, yielding a posterior mean μ^{BL} that is fed to the optimiser in place of the raw estimate $\hat{\mu}^{\text{factor}}$. The construction is detailed in the remark below.

Black–Litterman posterior for μ

The true vector of expected returns $\mu = \mathbb{E}[R]$ is a fixed (unknown) constant, exactly as in §8. Black–Litterman adopts the Bayesian framework of describing our *knowledge* of that constant by a probability distribution: the distribution represents our degree of certainty about μ, not any physical variability of μ. This is a different probabilistic interpretation from the frequentist one of §8, and the "randomness" of μ below should be read in this sense.

Prior distribution: Before consulting the factor model, we have no reason to expect a directional preference for the return of any of the stocks, and thus centre our **belief** on a zero expected return for all of them:

$$\mu \sim \mathcal{N}\big(0,\ \tau\Sigma\big), \qquad 0 < \tau < 1,$$

where Σ is the return covariance matrix (§9.7.2) and the small scalar τ tightens the prior, reflecting that our uncertainty about a *mean* return is less dispersed than returns themselves.

Evidence (the factor model): The prediction $Q := \hat{\mu}^{\text{factor}}$ of §8.8 is modelled as a noisy reading of μ,

$$Q \mid \mu \ \sim\ \mathcal{N}(\mu,\ \Omega), \qquad \text{i.e.} \qquad Q = \mu + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Omega),$$

where the covariance matrix Ω encodes how much we trust the prediction (larger entries of Ω mean less trust). This model treats $\hat{\mu}^{\text{factor}}$ as an *unbiased* estimator of μ, which the factor model may not always be (for example under the Ridge regression of §8.10). It is, in those cases, an incorrect model; however, this modelling yields a *shrinkage* of the factor prediction toward the prior (see below), which remains useful (as explained below). Choosing Ω is a modelling decision, left to future work.

Posterior distribution: By Bayes' rule the posterior density satisfies $p(\mu \mid Q) \propto p(Q \mid \mu)\, p(\mu)$. Both factors are Gaussian in μ, so their product has Gaussian form: the log-posterior is a quadratic in μ, and completing the square identifies it as a Gaussian $\mu \mid Q \sim \mathcal{N}(\mu^{\text{BL}}, \Sigma_\mu^{\text{BL}})$ with

$$\Sigma_\mu^{\text{BL}} = \Big[(\tau\Sigma)^{-1} + \Omega^{-1}\Big]^{-1}, \qquad \boxed{\mu^{\text{BL}} = \Big[(\tau\Sigma)^{-1} + \Omega^{-1}\Big]^{-1} \Omega^{-1}\, \hat{\mu}^{\text{factor}}}$$

(the posterior precision $(\tau\Sigma)^{-1} + \Omega^{-1}$ is a sum of positive-definite matrices, hence invertible). The posterior mean μ^{BL} is a precision-weighted blend of the prior mean 0 and the prediction $\hat{\mu}^{\text{factor}}$:

$$\Omega \to \infty \ \Rightarrow\ \mu^{\text{BL}} \to 0 \qquad \text{and} \qquad \Omega \to 0 \ \Rightarrow\ \mu^{\text{BL}} \to \hat{\mu}^{\text{factor}},$$

i.e. worthless evidence collapses the estimate to the neutral prior, and perfect evidence recovers the raw factor prediction. In between, it is a shrunk version of the factor prediction, pulled toward 0 by exactly the amount our distrust Ω (relative to the prior $\tau\Sigma$) dictates.

We use this by feeding μ^{BL} to the optimiser of §9.2 in place of the rolling or factor estimate, keeping the §9.7.2 covariance. The two free parameters (the prior tightness τ and the view-confidence Ω) would need to be calibrated against observed strategy performance and validated out-of-sample; we leave their specification as future work.

- **Additional strategies:** Given the modular structure of the framework's codebase, adding a new strategy would be relatively simple.

- **Multi-factor neutrality for the constrained MVO strategy.** All betas are computed against SPY returns as a proxy for the US large-cap equity market; neutrality therefore means zero exposure to this single market factor. A multi-factor model (e.g. Fama–French 3- or 5-factor) would additionally neutralize exposures to factors such as size (SMB) and value (HML), whose returns are constructed from long–short reference portfolios sorted on market capitalization and book-to-market ratios. This provides a richer notion of *factor neutrality*. Replacing the single-β SPY neutrality constraint with Fama–French-style factor neutralization requires: (a) a factor-return panel aligned with the rebalancing grid; (b) per-stock rolling factor loadings estimated by OLS using a common factor design matrix over a specified estimation window; and (c) one linear optimization constraint per factor, driving the corresponding portfolio exposure toward zero.

Remark

For any single factor, the portfolio's factor loading equals the w_i-weighted sum of the individual tickers' loadings on that factor (provided every ticker is regressed on the same factor design matrix over the same window) exactly as for market beta (derivation in §9.7.3).

- **Risk-management overlay:** With regards to the limitation "No risk-management overlay" from §16.1, we could implement sector caps, single-name caps, and inverse-volatility sizing as a post-processing step on the `StrategyOutput` without modifying the strategies themselves.

- **Walk-forward hyperparameter optimisation:** The current framework runs with fixed estimator windows. A walk-forward variant would re-tune those windows at each rebalance using only the preceding history (see the examples under "No regime-switching or adaptive parameters" in §16.1). The `evaluate_rolling_periods_by_hit_rate(...)` helper for the factors model (from `modules/returns_prediction_model.py`) is a single-window proof of concept. We could generalize it to roll forward across the full backtest, refitting at each rebalance, and storing the chosen parameters alongside the trade log.

- **Live trading integration:** The framework currently generates per-rebalance target portfolios that could in principle be passed to a real broker. The required additions are: **(a)** replacing `yfinance` with a real-time market data feed; **(b)** replacing simulated trading execution with broker API calls; **(c)** reconciling the simulated book against the actual broker statement at least daily; **(d)** a kill-switch and reconciliation-alert layer.

Example

If the reader has an account at the online brokerage platform E*TRADE (by Morgan Stanley), its API documentation [20] provides a clear starting point for implementing (a), (b), (c), and (d).

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Appendix: Bad-Ticker Filter

Section 6.5 introduced the bad-ticker pre-filter applied once to the full price panel of all historical S&P 500 members before the investable universe is built for each rebalance date.

A1.1 Symptoms of bad data

`yfinance`, the free Python module fetching the price data from Yahoo Finance API endpoints that we use for this backtest framework, has some failures:

- **Recycled ticker symbols:** A ticker belongs to one company at a time, not permanently. When the company currently using a ticker is acquired, renamed, or delisted, that ticker is freed up and can be assigned later to an unrelated company. `yfinance` may then return the new company's price history under the old ticker, with no flag marking the handover (as strongly suggested by A.2).
- **Instrument swaps:** `yfinance` may silently return foreign listings or other instruments sharing the ticker, producing price data that mixes unrelated securities under a single ticker label (as strongly suggested by A.3).

A1.2 Example 1: `WPI`, July 2010

```python
import yfinance as yf
from IPython.display import display

raw_price_data = yf.download(
    tickers = ["WPI"],
    start = pd.Timestamp("2010-07-10"),
    end = pd.Timestamp("2010-08-05"),
    auto_adjust = False,
    group_by = "ticker",
    progress = False,
    threads = True,
)

display(raw_price_data)
```

Python Script 2: Downloading raw `WPI` price data via `yfinance`

What `WPI` was in 2010: Watson Pharmaceuticals common stock, listed on the NYSE under `WPI`. Watson acquired the Actavis Group in 2012 and, on 24 January 2013, renamed itself Actavis, Inc. and changed its NYSE ticker from `WPI` to `ACT` [22]. The `WPI` symbol was therefore freed for reassignment only *after* January 2013.

In July 2010, Watson was an established large-cap pharmaceutical name: it reported full-year 2010 net revenues of roughly \$3.6 billion and non-GAAP earnings per share of \$3.42 on about 122 million shares outstanding [21]. An equity with that level of earnings trades in the tens of dollars per share, not in cents.

What `yfinance` returns: The data downloaded from Python Script 2, presented in Figure 37, shows prices stuck around \$0.25–\$0.27 throughout the window `2010-07-10`–`2010-08-05`, an order of magnitude below the expected level and inconsistent with any plausible NYSE-listed large-cap

Ticker						WPI
Price	Open	High	Low	Close	Adj Close	Volume
Date						
2010-07-12	0.270	0.270	0.270	0.270	0.270	10000
2010-07-13	0.260	0.270	0.260	0.260	0.260	0
2010-07-14	0.250	0.270	0.250	0.250	0.250	0
2010-07-15	0.260	0.270	0.260	0.260	0.260	0
2010-07-16	0.235	0.270	0.235	0.235	0.235	0
2010-07-19	0.260	0.270	0.260	0.260	0.260	0
2010-07-20	0.260	0.270	0.260	0.260	0.260	0
2010-07-21	0.260	0.260	0.260	0.260	0.260	40000
2010-07-22	0.260	0.270	0.260	0.270	0.270	290000
2010-07-23	0.260	0.270	0.260	0.260	0.260	0
2010-07-26	0.260	0.270	0.260	0.260	0.260	0
2010-07-27	0.260	0.270	0.260	0.260	0.260	0
2010-07-28	0.270	0.270	0.270	0.270	0.270	1320000
2010-07-29	0.260	0.275	0.260	0.260	0.260	0
2010-07-30	0.270	0.270	0.260	0.260	0.260	1380000
2010-08-02	0.270	0.270	0.270	0.270	0.270	10000
2010-08-03	0.270	0.270	0.250	0.260	0.260	870000
2010-08-04	0.260	0.260	0.260	0.260	0.260	210000

Figure 37: Example of data corruption: `WPI`.

equity. Several days also print zero volume. The `WPI` symbol was not reassigned until after the 2013 renaming, but these rows cannot be Watson at all: `yfinance` has likely mapped the price history of a *later, unrelated* occupant of the recycled `WPI` ticker back onto the 2010 date range.

A1.3 Example 2: `TIE`, January 2012

```python
import yfinance as yf
from IPython.display import display

raw_price_data = yf.download(
    "TIE",
    start = "2012-01-18",
    end = "2012-01-31",
    auto_adjust = False,
    progress = False,
)

display(raw_price_data)
```

Python Script 3: Downloading raw `TIE` price data via `yfinance`

Price	Adj Close	Close	High	Low	Open	Volume
Ticker	TIE	TIE	TIE	TIE	TIE	TIE
Date						
2012-01-18	7300.000000	7300.000000	7400.00	7200.00	7200.00	4600
2012-01-19	7500.000000	7500.000000	7500.00	7300.00	7400.00	3160
2012-01-20	7800.000000	7800.000000	7800.00	7800.00	7800.00	35080
2012-01-23	1.400000	1.400000	1.40	1.40	1.40	0
2012-01-24	1.400000	1.400000	1.40	1.40	1.40	0
2012-01-25	16.120001	16.120001	16.23	15.85	16.01	1686802
2012-01-26	16.010000	16.010000	16.32	15.92	16.25	1816465
2012-01-27	1.400000	1.400000	1.40	1.40	1.40	0
2012-01-30	8100.000000	8100.000000	8100.00	7900.00	7900.00	14900

Figure 38: Example of data corruption: `TIE`.

What `TIE` was in January 2012: Titanium Metals Corporation (TIMET), listed on the NYSE under `TIE`. The ticker stayed with TIMET until Precision Castparts completed its acquisition: the merger closed on 7 January 2013, after which TIMET common stock ceased to trade on the NYSE [9]. Throughout the January 2012 window, then, `TIE` was unambiguously the live NYSE listing for TIMET, a large-cap S&P 500 member, and the acquisition a year later valued TIMET shares at $16.50 each [9].

What `yfinance` returns: The data downloaded from Python Script 3, presented in Figure 38, shows that across the trading days in the window `2012-01-18–2012-01-31`, the rows fall into three distinct and mutually inconsistent groups under the single `TIE` label:

Days	Close	Volume
Jan 18–20, 30	7,300–8,100	3k–35k
Jan 23, 24, 27	1.40	0
Jan 25–26	16.01–16.12	1.6 M – 1.7 M

Only the two rows in the ∼ $16 range, on volumes of order a million shares, are consistent with the live large-cap NYSE listing; they sit right at the $16.50 figure the acquisition would later place on TIMET. The other two groups are each implausible for that listing and inconsistent with one another. The ∼ $7,300–$8,100 rows, on thin volumes of a few thousand to a few tens of thousands, look like a different instrument sharing the symbol. The $1.40 rows carry zero volume. Whatever their individual origins, the three groups cannot all be the same security, which strongly suggest an instrument swap under one ticker label.

A1.4 The two filter rules we use

Both rules are applied once to the full price panel at the start of the backtest. A ticker that fails either rule is dropped from the panel entirely (across all dates).

Rule 1: Zero-volume rule

Any ticker with at least one zero-volume print is dropped. S&P 500 names effectively never print zero volume on a NYSE trading day, so a zero-volume row is almost always a data-quality issue.

Rule 2: Extreme single-day return rule

Any ticker with at least one single-day return outside the interval $[-90\%, +500\%]$ on **adjusted** close prices is dropped. Adjusted close corrects for splits and dividends, so legitimate corporate actions do not trigger the rule. The bounds are wide, so that a violation is a strong indicator of data glitch, more than a real return.

The `WPI` corruption from A.1 would have been detected by Rule 1, via its zero-volume prints. The `TIE` corruption from A.1 would have been detected by both rules: Rule 1, from the zero-volume $1.40 rows; and Rule 2, because the jumps between the three price groups produce single-day returns far outside the allowed band: for instance, the ∼ $8,000 → $1.40 transition is a return of about −99.98%, and $1.40 → $16 is roughly +1,040%, both outside $[-90\%, +500\%]$.

A practical note on Rule 1

Dropping a ticker entirely on the basis of a *single* zero-volume print is deliberately conservative. This choice prioritizes the integrity of the surviving panel and may result in the loss of some otherwise valid history. We accept this trade-off because the cost of allowing a corrupted series to enter the investable universe at any rebalance date (thereby distorting both the trading signals of our strategy and the subsequent mark-to-market valuation of the position) outweighs the cost of excluding a small number of names from a historical universe of 955 constituents since 2008 (A.5).

A1.5 Implementation output

In our implementation, the function `remove_bad_tickers` (from `modules/universe_construction.py`) returns:

- the cleaned price panel, with bad tickers dropped;
- a bad-ticker report DataFrame, indexed by ticker, with two boolean columns (`zero_volume` and `extreme_return`) flagging which rule(s) triggered for each excluded name.

The number of dropped tickers, and the count under each rule, are logged at `INFO` level. As an example, on our import of price data for the period `2008-01-02` to `2026-02-27`, the filter drops 82 tickers out of 955 historical members.